

Climate Policy Uncertainty and Financial Markets: Evidence from Stock Prices, Return Volatility, and Trading Activity

By VICTORIA GODIO, BAPTISTE MESSANT, AND RAPHAËL PERILHOU*

Supervisor: Pr. Angie Acquatella†

March 2026

Abstract

This study investigates the impact of Climate Policy Uncertainty (CPU) on stock returns, return volatility, and trading volume. We construct a novel text-based CPU index measuring the semantic divergence between U.S. climate-related campaign materials and legislative texts, and apply a stacked event-study for the 30 companies of the Dow Jones Industrial Average over the period 2017-2025. We document three relevant findings: (1) stock returns exhibit a significant increase in the premium in the month preceding the spike, followed by a decline at the spike date; (2) realized volatility falls significantly and persistently at and after CPU spikes, consistent with a wait-and-see mechanism; (3) the first log difference of the trading volume rises significantly before and after the spike and the abnormal trading volume decreases at and after the spike.

Keywords: Climate policy uncertainty, Event Study, NLP, Stock Returns, Realized Volatility, Trading Volume.

JEL Codes: G14, G18, Q54, C58.

*Toulouse School of Economics, Master's students in Economics of Global Risks (victoria.godio@gmail.com, bmessant2@gmail.com, perilhou.raphael@gmail.com)

†Toulouse School of Economics, Assistant Professor (angelique.acquatella@tse-fr.eu)

Contents

1 Introduction **1**

2 Literature Discussion **3**

3 Data presentation **7**
3.1 Climate Policy Uncertainty index 7
3.1.1 Policy intent documents (campaing materials) 7
3.1.2 Legislative Texts 9
3.1.3 Construction of the Climate Policy Uncertainty (CPU) Index . . . 9
3.2 Financial data 11

4 Descriptive statistics **14**
4.1 Climate Policy Uncertainty Index 14
4.1.1 Text Volume Around CPU Spike Months 14
4.1.2 Climate intensity 15
4.1.3 Top-20 Climate Terms at CPU Spike Months 17
4.1.4 Term Frequency: Spike Months vs. Non-Spike Months 18
4.2 Financial data 21

5 Empirical Strategy **25**
5.1 Identification Challenge 25
5.2 Stacked Event-Study Design 25
5.3 Regression Specification 26
5.3.1 Outcome Variables 26
5.4 Spike Detection 27
5.4.1 Method 1: Rolling Threshold 27
5.4.2 Method 2: Local Maxima (Preferred) 28
5.4.3 Spike Visualization 28
5.4.4 Event Study on CPU 30
5.5 Endogeneity Analysis: Is VIX a "Bad Control"? 32
5.5.1 The Problem 32
5.5.2 Granger Causality and VAR Diagnostics 33
5.5.3 Cross-Correlation Analysis 33
5.5.4 The Truly Predetermined Control: VIX$(t-3)$ 35
5.5.5 Conclusion on endogeneity analysis 36

6 Results **36**
6.1 Arithmetic returns . . . 37
6.1.1 Hypothesis . . . 37
6.1.2 Results . . . 37
6.2 Realized Volatility . . . 39
6.2.1 Hypothesis . . . 39
6.2.2 Results . . . 39
6.3 Abnormal Trading Volume . . . 42
6.3.1 Hypothesis . . . 42
6.3.2 Results . . . 42
6.4 First-Difference Log Trading Volume . . . 44
6.4.1 Results . . . 44

7 Conclusion **46**

8 AI Acknowledgment

9 Appendix
9.1 Introduction . . .
9.2 Campaign Materials Documents . . .
9.3 Legislative texts . . .
9.4 Climate Dictionary Term Intensity at CPU Spike Months . . .
9.5 Arithmetic Returns . . .
9.5.1 Rolling Threshold Spike Detection . . .
9.5.2 Local Maxima Spike Detection . . .
9.6 Realized Volatility . . .
9.6.1 Rolling Threshold Spike Detection . . .
9.6.2 Local Maxima Spike Detection . . .
9.7 Abnormal Trading Volume . . .
9.7.1 Rolling Threshold Spike Detection . . .
9.7.2 Local Maxima Spike Detection . . .
9.8 First-Difference Log Trading Volume . . .
9.8.1 Rolling Threshold Spike Detection . . .
9.8.2 Local Maxima Spike Detection . . .

1 Introduction

Climate risk has emerged as one of the most significant challenges of the past decade. It refers to two main categories of risk: physical risk, which arises from climate and weather-related events as well as from human activities directly affecting the environment, and transition risk, which stems from the adjustment process toward a low-carbon economy[1]. Climate risk is no longer only an environmental issue; it also has major economic and social consequences as demonstrated by IPCC (2022). In response, public authorities across the world have increasingly adopted policy measures to address climate change. However, the future trajectory of climate policy remains highly uncertain. Hence, over the last few years, several major environmental policy reversals and shifts have been observed across advanced economies. The fuel tax has been repealed in France after the Yellow Vests movement, Australia recently abandoned its coal power phase-out plan[2] and the U.S. withdraws the Paris accords in 2017 under Trump's administration. Beyond these policy reversals, many other events may be considered as generating Climate Policy Uncertainty (CPU) in the United States, as illustrated in Figure 27.

Such uncertainties related to climate policies can significantly impact the overall economy and, more precisely, the corporate strategies, Gavriilidis et al. (2026) demonstrate that climate policy uncertainty leads to a decline in firms' investment, R&D expenditures and their productivity over longer horizons. Indeed, following the option theory, Dixit and Pindyck (1994) show that if the investments are irreversible and subject to uncertainty, the optimal response for firms is to delay investment until the benefits of immediate action outweigh the expected value of waiting. As a consequence, Huang (2026) finds that a higher level of climate policy uncertainty is associated with lower corporate investment rate, operating performance, and stock returns. Indeed, according to the Present Value Theory, stock prices should be affected by a climate policy uncertainty shock, as stock prices represent the sum of the expected cash-flows discounted. Therefore, a climate policy uncertainty shock should also be reflected in firms' market valuation and consequently its returns. Moreover, according to the Efficient Market Hypothesis, Fama (1970), asset prices in efficient markets should *fully reflect* all available information including climate policy uncertainty. While the existing literature remains limited concerning the impact of the CPU index on the realized volatility and the trading volume, Lv (2023) and Lasisi et al. (2024) find that the CPU index has a significant effect on stock market volatility and possesses out-of-sample predictive power for realized volatility.

[1] Laurent Clerc, *Should Central Banks Care About Climate and Nature Related Risks?*, presentation delivered in the Global Risks course, Toulouse School of Economics, 19 November 2025.

[2] Discovery Alert

By analysing how spikes in climate policy uncertainty affect stock returns, return volatility, and trading activity for large U.S. firms, this paper fills this gap and provides an overall understanding of how climate policy uncertainty affects asset prices and investors' behavior on the U.S market. Our analysis focuses on the 30 companies composing the Dow Jones Industrial Average Index over the period 2017–2025, providing a diversified sectoral representation of the U.S. economy while relying on a widely recognised benchmark of the U.S. stock market. Capturing the CPU through the creation of our own index, based on semantic misalignment between campaign and legislative texts using GloVe word embeddings and cosine similarity, rather than using news-based indices, is another contribution of our paper. Concerning our financial data, we obtained them from the London Stock Exchange Group (LSEG), Yahoo Finance and the FRED databases. We then identify CPU spike episodes using two complementary detection methods ; a Rolling Threshold approach and a Local Maxima approach, the latter being our preferred method and estimate a stacked event-study design with a symmetric event window of ± 2 months around each spike. In addition, to the two different methods for spikes detection, we have added three different panels : the Baseline, The Raw VIX and the VIX$(t-3)$. Indeed, a key methodological contribution of this paper is the treatment of the VIX as a control variable. Our tests show that contemporaneous VIX is a bad control because it is correlated with CPU within the same month. We therefore use VIX$(t-3)$, the earliest fully predetermined lag implied by the construction window of the CPU index, as our preferred control.

The empirical analysis reveals three distinct and robust patterns. First, arithmetic stock returns exhibit a sharp reversal around CPU spikes: returns are significantly positive in the month preceding the spike, consistent with investors demanding a higher risk premium in anticipation of the forthcoming uncertainty shock, before declining sharply and significantly in the spike month itself relative to the pre-spike baseline. Second, and more surprisingly, realized volatility falls significantly at the spike date and remains below its pre-spike level for at least two months thereafter. These findings are consistent with the wait-and-see mechanism of Bloom (2009). Third, the first difference in log trading volume increases significantly one month before and one month after the spike, consistent with short-run anticipation and portfolio adjustment effects. This finding is in line with Peng and Yang (2024), who show that spikes in the CPU index increase short-selling activity because of heightened uncertainty and concerns about the potential overvaluation of assets for which climate-related risks may not yet be fully incorporated into prices. Conversely, the abnormal volume trading variable indicates below-normal trading activity at and after the spike, suggesting that aggregate uncertainty induces a net reduction in trading relative to the volume traded of the last twelve months. This outcome is consistent with the wait and see mechanism of Bloom (2009).

The paper will be organized as follows. Section 2 reviews the existing literature on the impact of climate policy uncertainty on investors' decisions and financial markets. Section 3 presents the data, including the construction of Climate Policy Uncertainty index and the financial variables used in the analysis. Section 4 provides an overview of the descriptive statistics. Section 5 presents the empirical strategy and the event-study regressions. Section 6 presents the empirical results, and finally Section 7 concludes.

2 Literature Discussion

Existing literature shows that climate policy uncertainty has a significant impact on the financial markets. For instance, Yan et al. (2025) examined the impact of climate risk, and more specifically climate policy uncertainty, on firm-level stock pricing. Using data on Chinese A-share listed firms over the period 2009–2022, the authors have found that an increase in climate policy uncertainty significantly increases stock price delays and reduces pricing efficiency, suggesting that CPU disrupts the incorporation of information into stock prices. More precisely, higher climate policy uncertainty appears to raise investors' awareness of the historical underestimation of climate risk, while delaying the incorporation of this information into stock prices. In this context, Yan et al. (2025) suggest that greater transparency and predictability in climate policies are needed in order to reduce information asymmetries and improve stock market pricing efficiency.

In addition, Treepongkaruna et al. (2023a) have examined whether climate policy uncertainty is priced in individual stocks. They have found empirically that: *"The average risk-adjusted annual future returns of stocks with low exposure to the CPU are 5.5%–6.3% higher than those of stocks with a high CPU exposure"*. Interestingly, firms with a low exposure to the CPU index also tend to have a lower crash risk, and they have a higher exposure when the US president is a Democrat; conversely, firms with a high CPU-beta have a higher crash risk and a higher exposure when the US president is a Republican. This result might be explained by the fact that Republican contributors tend to invest in firms qualified as *"socially irresponsible"*, whereas Democrats tend to invest in firms that are *"more socially responsible"*. Vo et al. (2025) have studied the impact of climate policy uncertainty on stock price crash risk, defined as: *"the probability of an abrupt and extreme decline in stock price"*, by using a panel dataset of US companies from 1989 to 2024. This paper relies on the news-based CPU measure from eight leading US newspapers. The finding of this paper is consistent with Treepongkaruna et al. (2023a), who also revealed that a higher level of the CPU index leads to a higher risk of stock price crash. This result could be explained through two different mechanisms: managers have stronger incentives to withhold bad news, as periods of uncertainty tend to reduce transparency. Additionally, uncertain periods make the performance of

vulnerable firms more volatile, which therefore also incentivizes managers to withhold such information. This accumulation of undisclosed negative news can lead to a higher risk of stock price crashes.

More globally, Bezgin and Gungor (2025) have studied how climate policy uncertainty can have an impact on investors' behavior and on stock indices in Sweden, the United Kingdom, Germany, Norway, the Netherlands, and Finland from 2000 to 2023. The paper reveals that co-movements exist between the CPU index and stock market indices, and the causality test shows that a causal relationship exists between the CPU index and all the market indices except for Norway at different periods. While Modern Portfolio Theory assumes fixed correlations for portfolio diversification, here, interestingly, the authors have found that the correlation between the CPU and the stock market indices is not constant over time. For instance, it has been found that there is a statistically significant relationship between the CPU and the stock index OMXH25: a negative relationship between 2007 and 2010, and a positive one between 2022 and 2023. As a consequence, investors have to adopt new dynamic strategies to mitigate and monitor exogenous sources of risk such as the CPU index.

Li et al. (2024) have studied the impact of climate policy uncertainty on financial markets and, more precisely, on stock price synchronicity. Stock price synchronicity is measured to represent the extent to which a firm's stock returns move together with market and industry returns rather than reflecting firm-specific information. This study relies on all A-share non-financial companies listed on the Shanghai and Shenzhen stock exchanges. Li et al. (2024) have found that an increase in climate policy uncertainty significantly reduces stock price synchronicity, and this negative relationship is even more pronounced for high-polluting industries and after the Paris Agreement. This negative relationship indicates that an increasing CPU index tends to reduce the co-movements of stock prices across firms, and this could be explained by the fact that periods of high CPU raise investors' concerns. Hence, in response, firms tend to reveal more information to boost investors' confidence, which leads to a decrease in stock price synchronicity. In particular, a spike in the CPU index could lead high-polluting firms to face greater risk and may affect firms differently by changing their cash holding or R&D decisions, which tends to reduce stock price synchronicity even further.

Furthermore, some studies have focused on the effect of CPU on market volatility. Chen et al. (2023) investigate the effects of climate policy uncertainty on Chinese stock market volatility. The authors find, based on in-sample evidence, that CPU has a significant effect on financial asset price volatility. Indeed, uncertainty surrounding climate policies creates an additional source of risk for investors in financial markets. It can alter expectations regarding future costs, regulation, sectoral profits, and investor behavior, which in turn affects the volatility of companies' stocks. The authors also relate this mechanism to behavioral finance concepts, arguing that the CPU index affects

investor attention and sentiment, thereby amplifying stock market fluctuations. The same conclusion has been reached by other empirical studies, such as Lasisi et al. (2024), who find that in both the UK and the US, higher values of the US-based climate policy uncertainty (CPU-US) index tend to increase stock market volatility. However, the magnitude of the coefficient is larger in the US than in the UK. This suggests that the country's stock market are likely to respond more to the uncertainty associated with climate policies than the other economy. Moreover, although the effect of CPU is not highly persistent over time, it still affects market volatility over a long horizon, as stock market volatility itself is highly persistent in these both economies. In addition, Chen et al. (2023) show that CPU contains useful information for predicting volatility, as forecasting models combining CPU with 5-min high-frequency data outperform other volatility prediction models out-of-sample. The same conclusion has been observed by Bonato et al. (2022) that climate risks, more generally, have predictive power for state-level realized stock-market volatility at intermediate and long forecast horizons by using a 5-min intraday data form Bloomberg State level stock market indices for 50 US States. This finding is valid for good and bad state level realized volatility. However, Yang et al. (2025) add another dimension to this issue by showing that the impact also depends on the industry considered. Indeed, using a GARCH model and a Chinese climate policy uncertainty index constructed from a large corpus of Chinese newspapers, they find that Chinese climate policy uncertainty has no significant effect on the volatility of firms in the chemical, building materials, and metal industries. By contrast, it has a significant impact on firms in the transportation, steel, and electric power industries. Hence, their results highlight the heterogeneous effect of climate policy uncertainty on volatility across industries. The same idea is also demonstrated by Ren et al. (2023), who have studied the impact of the CPU on the idiosyncratic volatility by relying on a panel data of 2,555 Chinese non-financial listed firms from 2009 to 2019. Empirically, they have shown that uncertainty related to climate policies tend to increase significative the idiosyncratic volatility of firms. Their argument is that when climate policy uncertainty increases, firms face unclear future costs, opportunities or constraints. Some firms may become more exposed to risk, while others may try to take advantage of new opportunities by adjusting their investment decisions and risk taking. Since the reactions among firms are heterogeneous, stock price fluctuations are more firm-specific rather than market movements. This effect is even more important for public companies, small market capitalization and high equity concentration firms.

Finally, the existing literature on the impact of climate policy uncertainty on stock trading volume remains very limited, Wanidwaranan et al. (2025) examine the relationship between climate policy uncertainty and trading behavior in the US equity market, with a particular focus on investor herding. Herding behavior is defined as "*a behavior in which investors neglect their own knowledge and beliefs in their trading deci-*

sions, thus leading to imitative trading behavior". Relying on the Fama–French three-factor model, the study shows that climate policy uncertainty has a significant positive effect on market-wide herding, suggesting that, in such a context, investors are more likely to align their trading decisions with overall market behavior. The authors also find that this phenomenon is even more pronounced during extremely dry days and in the post-Paris Agreement period. Hence, climate policy uncertainty affects investors' trading behavior and not only stock returns or volatility. Furthermore, Peng and Yang (2024) have investigated how climate policy uncertainty affects short-selling activity and market pricing. Their study reveals a positive relationship between climate policy uncertainty and short-selling behavior. More specifically, higher climate policy uncertainty increases uncertainty surrounding future climate-related regulations, making it more difficult for market participants to assess firms' future costs, profitability, and exposure to transition risk. In such a context, short sellers may increase their positions, particularly when they believe that some assets are overvalued or that climate-related risks have not yet been fully incorporated into stock prices. Overall, the study shows that periods of high climate policy uncertainty are associated with higher short-selling activity in firms' stocks. A similar result is reported by Sun et al. (2025), who use data on Chinese A-share listed companies over the period 2007-2022. Their empirical analysis shows that higher climate policy uncertainty leads corporate executives to sell more of their shares and to do so more frequently. The authors identify three main channels through which this effect operates. First, greater climate policy uncertainty increases corporate default risk by weakening firms' financial conditions and raising financing costs. In such a context, executives may anticipate a decline in firm value and therefore sell their shares before stock prices fall. Second, during periods of high climate policy uncertainty, firms may become more cautious and take fewer risks, which can reduce future growth opportunities and weaken expected firm performance. As a result, executives may have stronger incentives to sell their shares. Third, an increase in climate policy uncertainty may reduce the quality of corporate information disclosure, thereby increasing information asymmetry between insiders and outside investors. In this context, executives may use their informational advantage to time their stock sales more strategically. In short, periods of high climate policy uncertainty are associated with increased executive stock selling.

The existing literature has already shown that climate policy uncertainty affects stock pricing efficiency, crash risk, investor behavior, and stock market volatility. However, to the best of our knowledge, no study has yet examined the impact of climate policy uncertainty on returns based on the Volume-Weighted Average Price (VWAP) nor on the monthly volume of shares traded. This paper aims to fill this gap by examining the effect of climate policy uncertainty on stock return, stock volatility, and trading volume

for firms included in the Dow Jones 30 Index. By considering these three dimensions jointly, this study provides a more complete firm-level analysis on how climate policy uncertainty may affect financial markets. Moreover, while the existing literature tends to rely on wide cross-sectional samples of listed firms, this paper focuses specifically on the components of the Dow Jones 30, which comprise the largest and most liquid U.S. companies, while also providing a diversified sectoral representation of the U.S. economy. This allows us to offer evidence on how climate policy uncertainty affects trading dynamics among benchmark stocks. In addition, capturing the CPU through the creation of our own index, based on semantic misalignment between campaign and legislative texts using GloVe word embeddings and cosine similarity, rather than using news-based indices, is another contribution of our paper.

3 Data presentation

3.1 Climate Policy Uncertainty index

The CPU index measures the semantic alignment between climate-related campaign materials (APP) and climate-related legislative texts. The index was constructed in several steps. First, we needed to obtain the textual data and clean it for both campaign materials and legislative texts. Secondly we had to create the CPU index. Both steps are described below.

3.1.1 Policy intent documents (campaing materials)

Policy intent documents are collected from the `American Presidency Project`. The extension of Moncomble (2024)[3] allows us to download all the documents after advanced research in an XML format (The Image 28 in appendix shows how the tool works). We selected our time period (2016Q1–2025Q4) and the relevant types of documents, namely: Executive Orders, Memoranda, Statements, Spoken Addresses and Remarks, Campaign Documents, Party Platforms, Debates, Convention Speeches, Proclamations, National Security Presidential Memoranda, State of the Union Addresses, State of the Union Written Messages, Economic Report of the President, Signing Statements, and Messages. An example of campaign material document can be found in the Appendix (Image 29).

The corpus is limited to climate related documents. The dictionary we used is based on Gavriilidis et al. (2023) and is shown in Table 1

[3] Website for the extension of Moncomble (2024)

Table 1: Climate Dictionary used in the analysis (Gavriilidis et al., 2023)

Dictionary	Level	Dictionary Terms
Climate (Policy)	Trigrams	capture and sequestration; climate and energy; sea level rise; environmental protection agency; energy department; cap and trade; clean power plan; energy efficiency program; fuel economy standard; climate change bill; clean air act; climate stewardship act; clear skies act
	Bigram	climate change; climate crisis; climate disaster; climate science; climate scientist; climate action; global warming; climate warming; temperature rise; global temperature; greenhouse gas; renewable energy; renewable fuel; renewable electricity; renewable power; clean energy; clean power; green energy; energy efficiency; fuel economy; fuel efficiency; fossil fuel; energy transition; climate transition; climate mitigation; climate adaptation; climate accord; climate conference; climate activist; climate demonstration; energy policy; energy legislation; energy law; energy tax; energy committee; climate bill; climate policy; climate legislation; climate rule; carbon tax; carbon price; carbon market; carbon trading; carbon limit; emission trading; emission limit; emission permit; emission standard; ghg rule; ghg limit; ghg permit; ghg standard; fuel standard; waxman markey; kerry lieberman
	Unigram	ghg; co2; carbon; emission; methane; environmental; decarbonization; renewables; epa; doe

The American Presidency Project search engine is subject to character limits (256) and case sensitivity. Thus, we did multiple query, separating our dictionary by chunks, entering each n-grams in both lower case and title case. Then it was very straightforward to remove the duplicata using the name of each .xml file.

Then, we use XML parsing to extract key metadata: author, title, date of publication, full text. The final dataset was filtered to include only documents published from 2016 onward. The resulting dataframe contains the following variables:

- **date_published**: Date of publication in the "YYYY-MM-DD" format.
- **title**: Document title.
- **text**: Full cleaned text of the document.

This corpus corresponds to the CSV file "APP_climate.csv" in the replication folder.

3.1.2 *Legislative Texts*

Legislative texts were collected from the `GovInfo Bulk Data` website. We focus on the primary instruments of policymaking: House bills, Senate bills, and joint resolutions. Concurrent and simple resolutions were excluded as they have no legal force. The dataset covers the 114th to the 119th Congresses, corresponding to the period 2015Q1–2025Q4. An example of legislative text can be found in the appendix (Image 30).

To identify climate-related texts, we applied the same climate dictionary used for the APP corpus (Table 1). Only bills containing at least one climate-related term were retained.

We performed the XML parsing using the `xml2` R package. After parsing and filtering, climaterelated legislative texts were compiled into a single dataframe, including an identifier for the chamber (*House* or *Senate*) and instrument type (*Bill* or *Joint Resolution*) in case we need it for later robustness check. The final dataset was sorted by publication date and restricted to documents published from January 2016 onward. The resulting dataframe contains the following variables:

- **bill_number**: Unique identifier of the legislative text.
- **date_published**: Date of publication (YYYYMM-DD).
- **title**: Short title of the bill.
- **text**: Full cleaned text of the bill.
- **chamber**: Legislative chamber (*House* or *Senate*).
- **instrument**: Instrument type (*Bill* or *Joint Resolution*).

This corpus corresponds to the CSV file "legi_climate_dataset.csv" in the replication folder.

3.1.3 *Construction of the Climate Policy Uncertainty (CPU) Index*

The Climate Policy Uncertainty (CPU) index measures the semantic alignment between climate-related campaign materials (from the APP corpus) and climate-related legislative texts. In other words, the index captures the degree to which the language used by policymakers and legislators on climate issues is similar or diverges over time.

We aggeragated both the APP and legislative corpora at the monthly level. For each month, we concatenated all texts published in that month into a single document. This ensures that the CPU index captures overall climate policy communication per month, rather than individual documents.

We have split each monthly document into individual words (tokens) using a word tokenizer. We removed the words with little semantic content (e.g "or","and", "the"). We constructed a vocabulary of words present in pre-trained word embeddings (GloVe vectors) developed by Pennington et al. (2014)[4] so we could restrict the analysis to words with meaningful vector representations.

Finally, we measure the similarity between campaign materials and legislative texts in the same month using cosine similarity. Cosine similarity compares two vectors and returns a value between -1 and 1, where 1 indicates identical direction (high similarity), -1 indicates opposite direct (high dissimilarity) and 0 indicates no similarity. The following transformation ensures that higher CPU values correspond to greater semantic divergence between campaign materials and legislation.

$$\text{CPU}_{t-i:t} = 1 - \cos(\text{CM}_{t-i:t}, \text{Legi}_t)$$

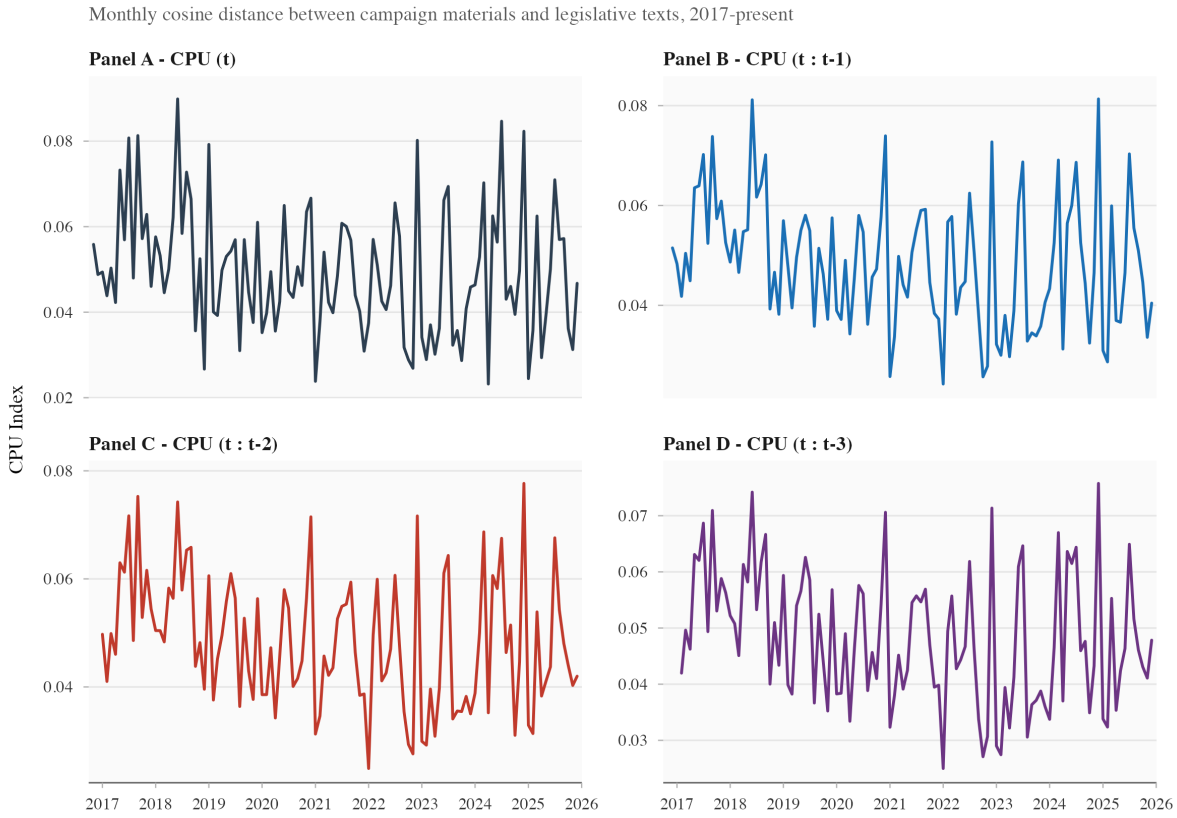
Where CM_t refers to campaign material from month t-i to month t and Legi_t refers to legislative texts at month t. We use a window $\{t-i:t\}$ because of the delay between an issue is discussed by officials and its application on legal texts. In our analysis, we build 4 $\text{CPU}_{t-i:t}$ with $i = \{0,1,2,3\}$ to check how sensitive is the Index to our choice of delay. As displayed on Figure 1, we see the choice of delay does not impact much our index. Hence, we choose a delay of 2 months, such that we take a quarter of campaign materials for the construction of the CPU index. Moreover, it illustrates the monthly evolution of the index. Although overall similarity is high, the series exhibits strong variation from month to month. It highlights the periods where semantic alignment between campaign communications and legislative texts diverges.

If no legislative texts were available in a month, the CPU value was recorded as missing. For some reasons that was the case for August and October 2016. Therefore we restrict our analysis to 2017 onward. Furthermore, our empirical strategy relies on CPU spikes. In the methodological section we will describe them also choose two spike identification methods, and compare them to ensure our results are not sensitive to the method,

As we can see on Table 3, our index varies between roughly 0.025 and 0.078, which reflects relatively high cosine similarity between the campaign materials and the legislative texts over time. This is expected as we take documents that are both climate-related.

[4]Tool available at: `https://nlp.stanford.edu/projects/glove/`

Figure 1: **Climate Policy Uncertainty Index Across Lag Windows**



3.2 Financial data

To analyse the link between climate policy uncertainty and the response of financial markets, we chose to conduct our anylsis with a sample of the 30 companies included in the Dow Jones Industrial Average Index of the 19 January 2026. The Dow Jones Industrial Average Index is a weighted-price measure of 30 US blue-chip companies, including all the industries, except the transportation and utilities[5]. The Dow Jones Industrial Average index composition is based on several factors, including a company's reputation, sustained growth, and broad investor interest. Eligible firms must be headquartered in the United States and derive most of their revenues from the U.S. Additionally, the index composition can change at any time when needed[6]. For this analysis, we focus on companies included in the Dow Jones Industrial Average index as this index is commonly viewed as a benchmark of the U.S. stock market and provides an

[5]S&P Dow Jones Indices (S&P DJI), *Dow Jones Industrial Average®*, available at: https://www.spglobal.com/spdji/en/indices/equity/dow-jones-industrial-average/

[6]S&P Dow Jones Indices, *S&P Dow Jones Indices: Index Mathematics Methodology*, December 2025.

informative picture of broader U.S. economic conditions[7]. As this index includes large and publicly listed companies, according to Badertscher et al. (2014) these companies exhibit a high financial reporting quality (FRQ) due to higher demand for reliable and transparent financial information from the investors. Moreover, we selected the companies from this index to obtain a diversified sectoral composition in our sample, since the Dow Jones Industrial Average selection process explicitly considers sector balance to maintain adequate representation across industries.

Table 2: Company Name List

	Company Name	
GOLDMAN SACHS GP.	CATERPILLAR	MICROSOFT
HOME DEPOT	AMERICAN EXPRESS	SHERWIN
UNITEDHEALTH GROUP	AMGEN	VISA 'A'
JP MORGAN CHASE & CO.	MCDONALDS	INTERNATIONAL BUS. MCHS.
TRAVELERS COS.	APPLE	BOEING
AMAZON.COM	SALESFORCE	HONEYWELL INTL.
JOHNSON & JOHNSON	NVIDIA	3M
CHEVRON	PROCTER & GAMBLE	WALMART
WALT DISNEY	MERCK & COMPANY	CISCO SYSTEMS
COCA COLA	NIKE 'B'	VERIZON COMMUNICATIONS

Afterwards, in order to examine the impact of climate policy uncertainty on financial markets, this empirical analysis relies on three complementary firm-level variables: the monthly arithmetic returns computed from the Volume-Weighted Average Price (VWAP), the realized volatility of returns, and the trading volume of firms' stocks. The selection of these variables is motivated by the fact that according to the Efficient Market Hypothesis, Fama (1970), if the financial markets are qualified as *efficient*, the prices on the markets should always *"fully reflect"* all available information. Therefore, new information and uncertainty shocks such as climate policy uncertainty are expected to be fully incorporated into the asset prices. In this setting, a transaction-based price measure such as the arithmetic returns provide a relevant way to capture how the financial markets react to a climate policy uncertainty shock. Moreover, realized volatility is included in order to capture stock price variability. This choice is supported by Shi et al. (2016), who studied the relationship between public information flows and stock return volatility using news-based sentiment indicators and showed that news sentiment has a significant and positive effect on stock return volatility.

[7]Federal Reserve Bank of St. Louis (FRED), *Dow Jones Industrial Average (DJIA)*, which notes that "the Dow Jones Industrial Average provides a view of the U.S. stock market and economy", available at: https://fred.stlouisfed.org/series/DJIA

Finally, the literature has long emphasized that information arrival is closely related not only to price changes, but also to trading activity. Indeed, as highlighted in the literature reviewed by Karpoff (1987), price changes could be interpreted as the market's evaluation of new information, while changes in trading volume are rather interpreted as the extent to which investors disagree about the meaning of the information. A similar result has been found by Copeland (1976), who constructed a model in which investors receive new information sequentially and randomly and studied its impact on total trading volume. The result was that, while total trading volume is identical when all traders are either optimistic or pessimistic, Copeland (1976) also considers the more realistic case in which traders disagree. In this setting, the simulation shows that trading volume is lowest when disagreement is relatively high and highest under complete unanimity of opinion. Therefore, trading volume appears to be a relevant variable for this study, as it can capture investors' reactions and the degree of disagreement generated by a shock such as climate policy uncertainty.

Having motivated the choice of these variables, the next step is to describe their construction and the data sources used for the empirical analysis. The dataset is built from three main sources and covers the period from 1 January 2017 to 31 December 2025.

Stock Prices. Daily volume-weighted average prices (VWAP) are collected for the 30 companies listed in Table 2 from the LSEG (London Stock Exchange Group) database. The VWAP represents the average transaction price of a stock over a given trading day, weighted by trading volume, and serves as the basis for computing both arithmetic returns and realized volatility, as described in Section 5.3.

Trading Volume. Daily trading volume for each of the 30 companies listed in Table 2 is obtained from Yahoo Finance over the same sample period. This series is used to construct both the abnormal trading volume and the first-difference log volume measures defined in Section 5.3.

VIX Index. The CBOE Volatility Index (VIX) is included as a market-level control variable. The VIX captures market expectations of near-term volatility derived from S&P 500 index option prices and is widely used in the empirical finance literature as a proxy for aggregate market uncertainty (Basaglia et al., 2025). Daily VIX data are obtained from the Federal Reserve Economic Data (FRED) database maintained by the Federal Reserve Bank of St. Louis.

4 Descriptive statistics

4.1 Climate Policy Uncertainty Index

Table 3: Summary Statistics for CPU (t:t-2)

Mean	Median	Sd.Dev	Min	Max
0.0483	0.0471	0.012	0.0249	0.0777

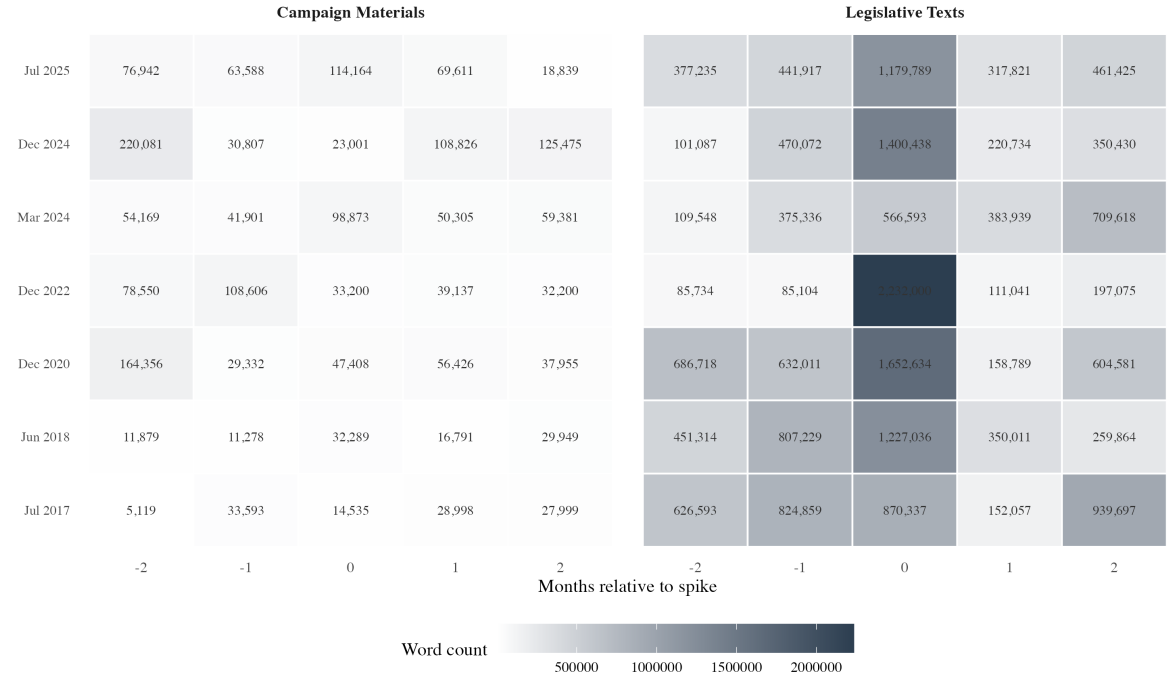
We already mentionned that our CPU reflects an overall high semantic similarity between climate related campaign materials and legislative texts. Now, we want to explore difference within the texts arround spike event to understand better how divergence is captured.

4.1.1 Text Volume Around CPU Spike Months

A potential concern with embedding-based similarity measures is that variation in corpus size may influence the aggregate document embedding, and therefore the measured semantic divergence underlying the CPU index. A month with an unusually large legislative corpus may produce a different average embedding, mechanically affecting the cosine distance from campaign materials regardless of any genuine change in policy content. To assess this, we examine the total word count of each corpus in the two months before and after each spike, for both campaign materials and legislative texts.

Figure 2 displays word counts as a heatmap, where rows correspond to spike episodes, columns to relative time, and colour intensity to word count. Two observations stand out. First, legislative texts are systematically larger than campaign materials by an order of magnitude across all spike episodes and all relative periods. Second, legislative texts display a visible concentration of word volume at t=0, with the spike month consistently exhibiting higher word counts than surrounding periods across most episodes. This pattern suggests that CPU spikes partly coincide with months of unusually high legislative output, which may partly inflate the measured semantic divergence at the spike date. This is a limitation worth keeping in mind when interpreting the CPU index, as it cannot fully disentangle misalignment from the effect of corpus size.

Figure 2: Text volume (word count) around CPU spike episodes



4.1.2 *Climate intensity*

To assess whether CPU spikes are driven by changes in climate topic salience rather than semantic divergence, we examine the evolution of climate intensity in both corpora around spike events. We measure climate intensity for each month and source as the share of words matching the Gavriilidis (2021) climate dictionary[8]. For month t and source s (Campaign vs Legislative texts), we define the climate intensity as follows:

$$CI_T = \frac{\text{Number of dictionary matches in corpus}_{t,s}}{\text{Total word count of corpus}_{t,s}} \tag{1}$$

Table 4 reports descriptive statistics for the monthly climate dictionary intensity of each corpus. On average, campaign materials (APP) and legislative texts (BILLS) allocate 0.36% and 0.59% of their words to climate-related terminology, respectively. Although modest in absolute terms, both series display substantial month-to-month variation (Std. Dev. of 0.15% and 0.13% respectively), confirming that climate intensity is an informative signal rather than a near-constant baseline.

Figure 3 is informative on two points. First, legislative texts exhibit systematically higher climate intensity throughout the sample, suggesting that bills engage more directly with climate terminology than campaign materials on average.

[8]This is the same dictionary used to filter our sample to climate-relevant documents, ensuring consistency between the selection and measurement steps.

Second, and more importantly for the validity of the CPU index, the identified spike dates do not particularly coincide with periods in which both sources peak simultaneously, nor with periods of sharp divergence between the two series. Although this evidence is purely descriptive, the absence of coincidence between spike dates and simultaneous peaks across corpora is reassuring, as it suggests that the CPU index does not simply capture general climate salience rather than semantic divergence.

However, the lack of alignment between spike dates and large gaps in climate intensity across the two series is more ambiguous. On the one hand, such gaps may reflect differences in topic intensity rather than meaningful policy misalignment; in this case, the fact that CPU spikes do not coincide with these periods strengthens the interpretation of the index as capturing policy uncertainty. On the other hand, if divergence in climate salience is interpreted as a disconnect between what officials publicly advocate and what is actually debated or enacted in legislation (which is arguably an important dimension of climate policy uncertainty), then the absence of CPU spikes during these episodes weakens the ability of the index to fully capture this concept.

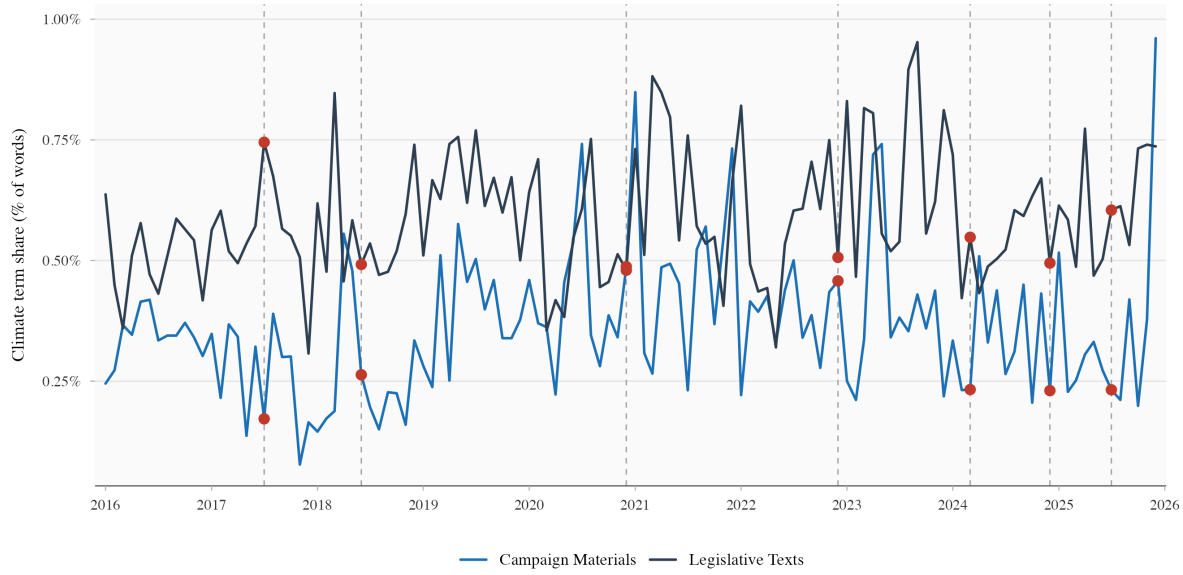
Overall, these considerations highlight an inherent limitation of our approach. While the CPU index captures meaningful variation in the relationship between campaign and legislative discourse, it cannot fully disentangle semantic divergence from differences in topic salience. As such, the index should be interpreted as a partial measure of climate policy uncertainty. Addressing this distinction more explicitly, for instance by jointly modeling topic intensity and semantic content, is a promising avenue for future research.

Table 4: Descriptive Statistics: Monthly Climate Dictionary Intensity

Series	N	Mean	Median	Std. Dev.	Min	Max
Campaign Materials (APP)	120	0.0036	0.0034	0.0015	0.0008	0.0096
Legislative Texts (BILLS)	118	0.0059	0.0057	0.0013	0.0031	0.0095

Notes: Climate intensity = number of Gavriilidis (2023) climate dictionary matches divided by total monthly word count. Sample: monthly observations, 2017–present.

Figure 3: Share of words matching the climate dictionary, by source — local maxima spikes marked



Notes: Climate intensity = matches from the Gavriilidis (2023) climate dictionary / total words.
Red dots = CPU(t:t-2) local-maxima spike months. Dashed lines = same spike dates.

4.1.3 Top-20 Climate Terms at CPU Spike Months

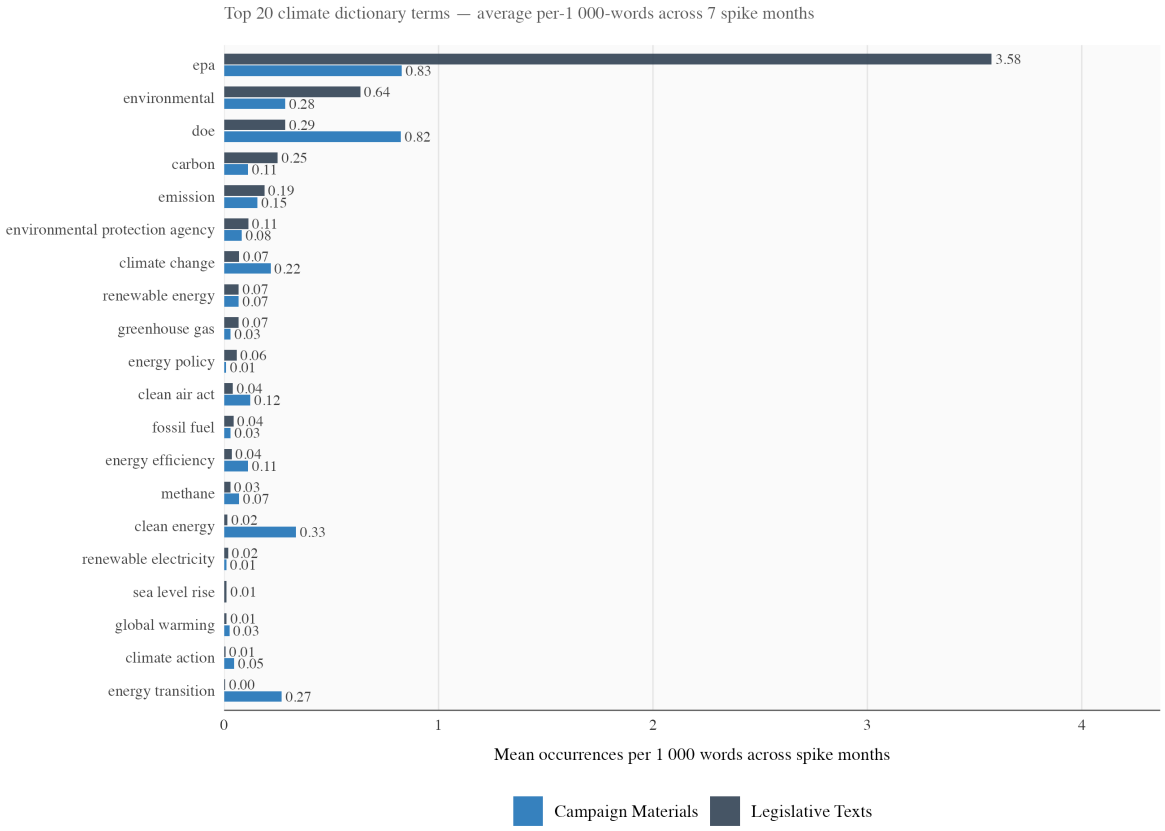
To further characterise the nature of CPU spikes, we examine which climate dictionary terms drive intensity at each spike month. For each spike identified by the local-maxima algorithm, we count the occurrences of each dictionary term in both corpora and normalise by total word count (occurrences per 1,000 words) to ensure comparability across months with different corpus sizes. We report the top-20 terms by average normalised frequency, displayed side by side for campaign materials (APP) and legislative texts (BILLS).

At the aggregate level, pooling counts across all spike months, EPA, Environmental, and DOE emerge as the three most frequent terms in legislative texts, while EPA, DOE, and *clean energy* lead in campaign materials (Figure 4). Two patterns are worth noting. First, the dominance of institutional acronyms (EPA and DOE) across both sources suggests that climate subjects at spike months focus on regulatory agencies rather than broader climate concepts. Second, campaign materials display relatively higher intensity on policy-oriented terms such as *clean energy* and *energy transition*, while legislative texts are more intensive on regulatory vocabulary.

At the individual spike level, the composition of top terms is remarkably stable across episodes separated by several years, suggesting that CPU spikes are not driven

by the emergence of novel climate topics but rather by recurring tensions around the same core vocabulary. Some variation is nonetheless observable across specific episodes, likely reflecting the particular legislative or political context of each spike. The full set of individual spike figures is provided in Appendix 9.4 (Figure 31–Figure 37).

Figure 4: Top-20 Climate Dictionary Terms: Average Intensity Across All CPU Spike Months



The preceding analyses characterise the texts at spike months in terms of volume and climate salience. We now turn to a direct comparison between spike and non-spike months to assess whether CPU spikes are associated with a shift in the climate vocabulary used by each source.

4.1.4 Term Frequency: Spike Months vs. Non-Spike Months

To assess whether the climate vocabulary at spike months is systematically different from non-spike periods, we compare term frequencies across the two groups (spike vs non-spike). Rather than averaging monthly per-1,000-word rates, which is sensitive to variation in corpus size across months, we pool all text within each period and compute term frequency on the pooled corpus. Formally, the pooled frequency for term k, source

s, and period $p \in \{\text{spike}, \text{non-spike}\}$ is:

$$f_{k,s,p} = \frac{\sum_{t \in p} \text{hits}_{k,s,t}}{\sum_{t \in p} \text{words}_{s,t}} \times 100 \tag{2}$$

Where hits correspond to words belonging in the climate dictionary.

Figure 5 reports the pooled term frequency for the top-20 terms, comparing spike and non-spike months side by side for each source. Three terms dominate overwhelmingly: *epa*, *doe*, and *environmental* in legislative texts, where *epa* alone accounts for 0.342% of all words at spike months, roughly four times higher than any other term. Campaign materials are more evenly distributed across terms, with *epa*, *doe*, and *climate change* leading. Crucially, the ordering of terms is nearly identical between spike and non-spike months in both sources, suggesting that the core climate vocabulary is stable across the sample rather than spike-specific.

Figure 5: Top-20 Climate Terms: Pooled Frequency at Spike vs. Non-Spike Months

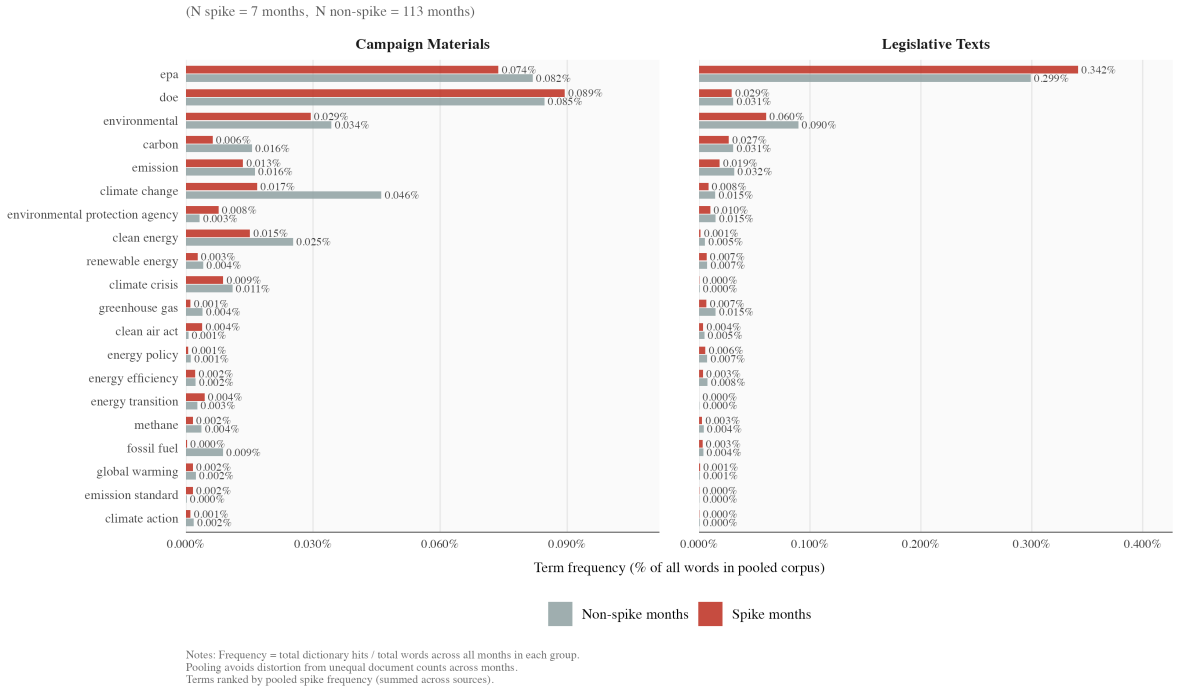


Figure 6 sharpens this comparison by reporting the relative frequency ratio — the ratio of pooled spike frequency to pooled non-spike frequency — for each term and source. A value of 1 indicates identical relative frequency across both periods. Two terms stand out as clear outliers in campaign materials: *clean air act* and *emission standard* both display relative frequencies well above 2×, suggesting they are disproportionately salient at spike months. *Environmental protection agency* also shows modest over-representation. By contrast, legislative texts cluster tightly around the baseline of 1 for nearly all terms, with *global warming* being the only slight exception. This asymmetry is noteworthy:

it suggests that when CPU spikes occur, campaign materials shift somewhat in their specific regulatory vocabulary, while legislative texts remain compositionally stable — consistent with the interpretation that spikes reflect campaign-side framing shifts rather than legislative agenda changes.

Figure 6: Relative Frequency: Spike vs. Non-Spike Pooled Term Frequency

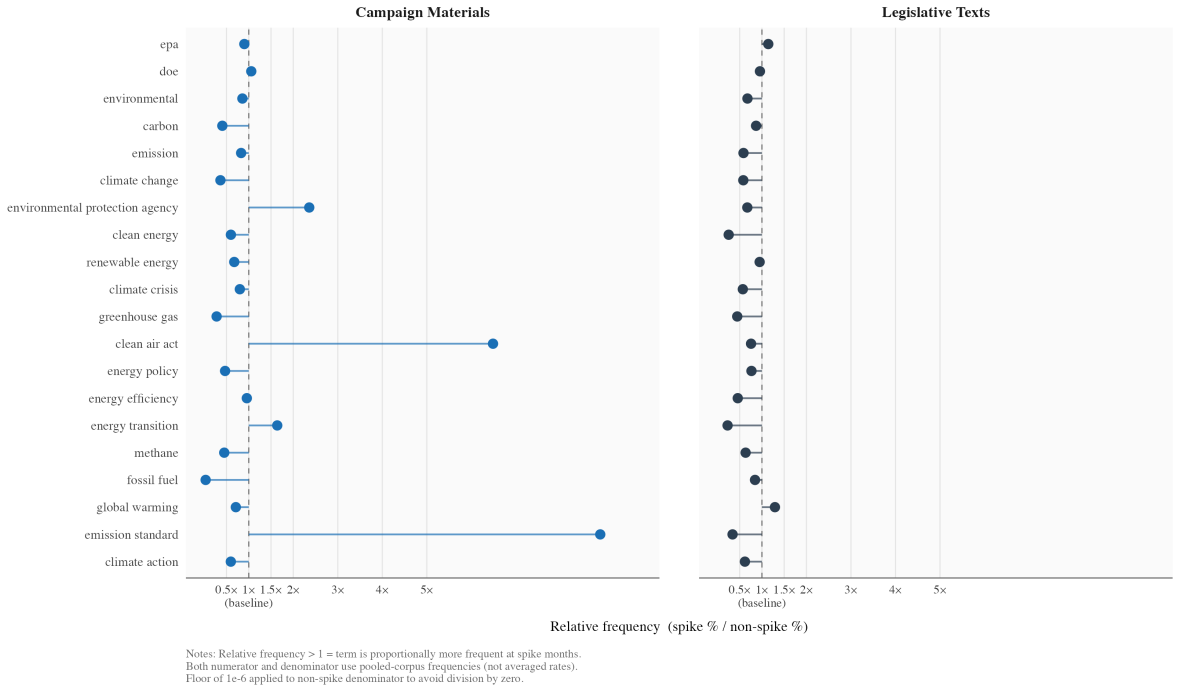
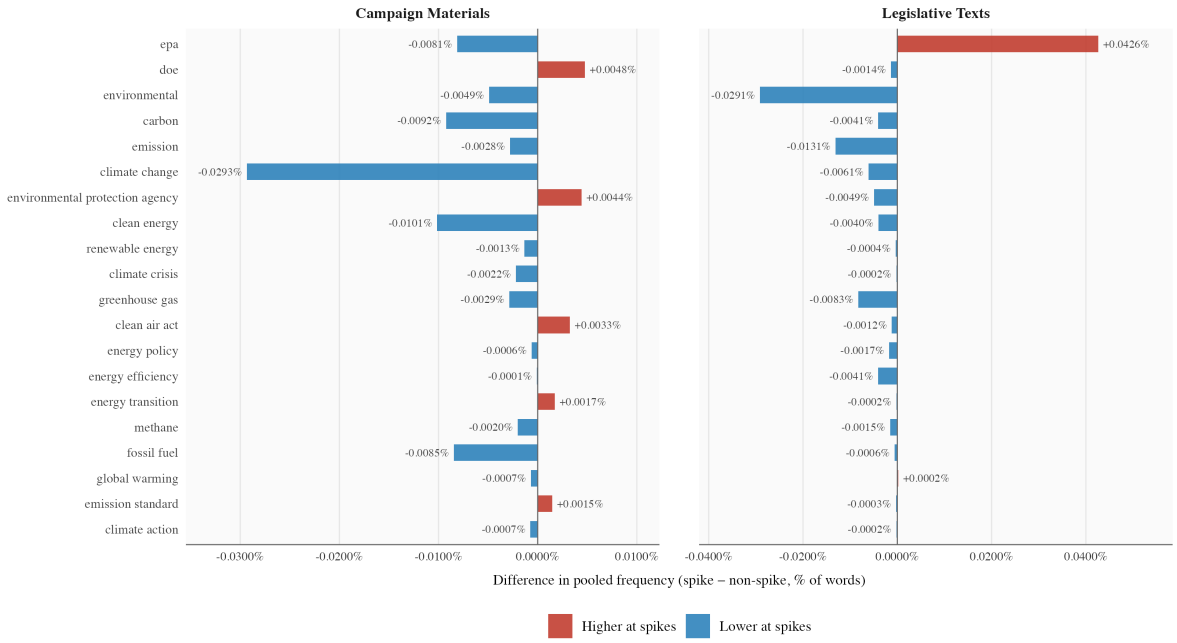


Figure 7 completes the picture by showing the absolute difference in pooled frequency (spike minus non-spike). The dominant feature is the large positive bar for *epa* in legislative texts (+0.0426%), which is the single largest shift in the entire figure. This is consistent with the word volume result in Figure 2: spike months coincide with months of unusually high legislative output, which mechanically inflates EPA-related counts in the BILLS corpus. On the campaign materials side, *climate change* displays the largest negative shift (−0.0293%), meaning it is proportionally *less* frequent at spike months than during normal periods — a counterintuitive finding suggesting that CPU spikes are not moments of heightened broad climate rhetoric in campaign materials. Meanwhile, *doe*, *environmental protection agency*, *clean air act*, and *emission standard* show positive differences in campaign materials, pointing to a shift toward more specific regulatory vocabulary at spike months. Overall, the magnitudes remain small throughout, and no consistent pattern of joint over-representation across both sources emerges, further supporting the conclusion that CPU spikes do not reflect a general surge in climate salience.

Figure 7: Absolute Difference in Pooled Term Frequency: Spike Minus Non-Spike



4.2 Financial data

Table 5: Descriptive statistics Company level

Variable	Mean	Median	SD	Min	P25	P75	Max
Returns VWAP	0.0004	0.0007	0.0203	-0.9961	-0.0062	0.0076	0.3128
Realized Volatility	0.0622	0.0537	0.0348	0.0151	0.0400	0.0748	0.5472
Abnormal Volume	0.0037	-0.0214	0.2448	-0.7132	-0.1446	0.1226	1.8621

Table 5 presents the descriptive statistics for the three outcome variables used in the analysis. VWAP returns are centered around zero (mean 0.0004, median 0.0007), consistent with an efficient market, though the distribution presents some extreme observations in both tails. Realized volatility displays a median (5.37%) below its mean (6.22%), suggesting that high volatility episodes, visible in Figure 8 and Figure 9, pull the mean upward. Abnormal volume is centered close to zero (mean 0.0037, median -0.0214), suggesting that trading activity does not systematically deviate from its 12-month rolling benchmark. The inter-quartile ranges of all three variables are relatively narrow compared to their full ranges, confirming that extreme observations are episodic rather than persistent. No seasonal pattern is detectable from these summary statistics alone.

Figure 8: CHEVRON monthly realized volatility over time

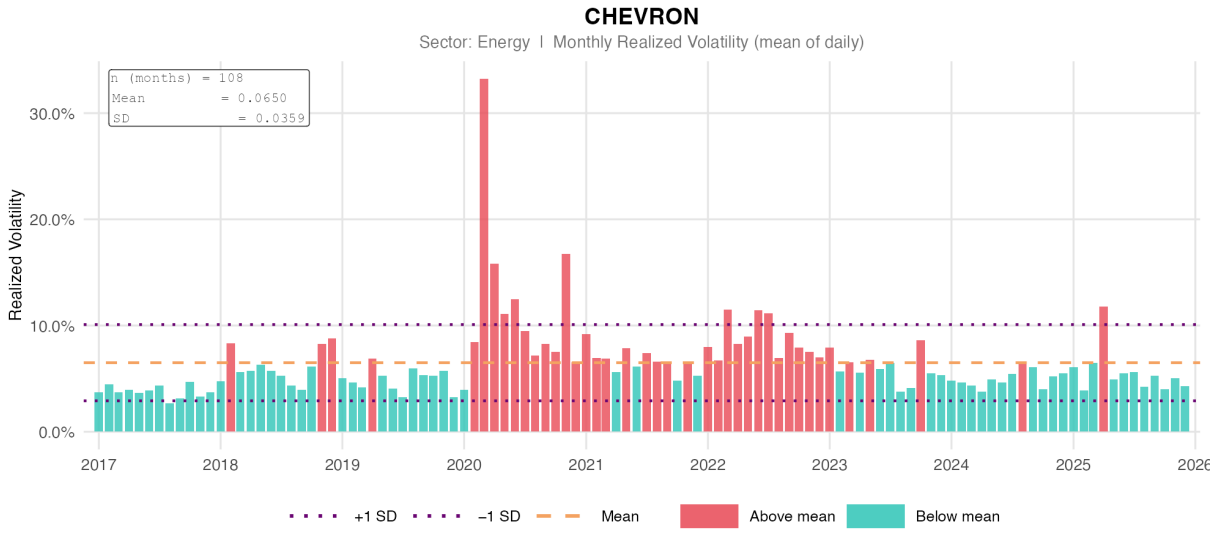


Figure 9: APPLE monthly realized volatility over time

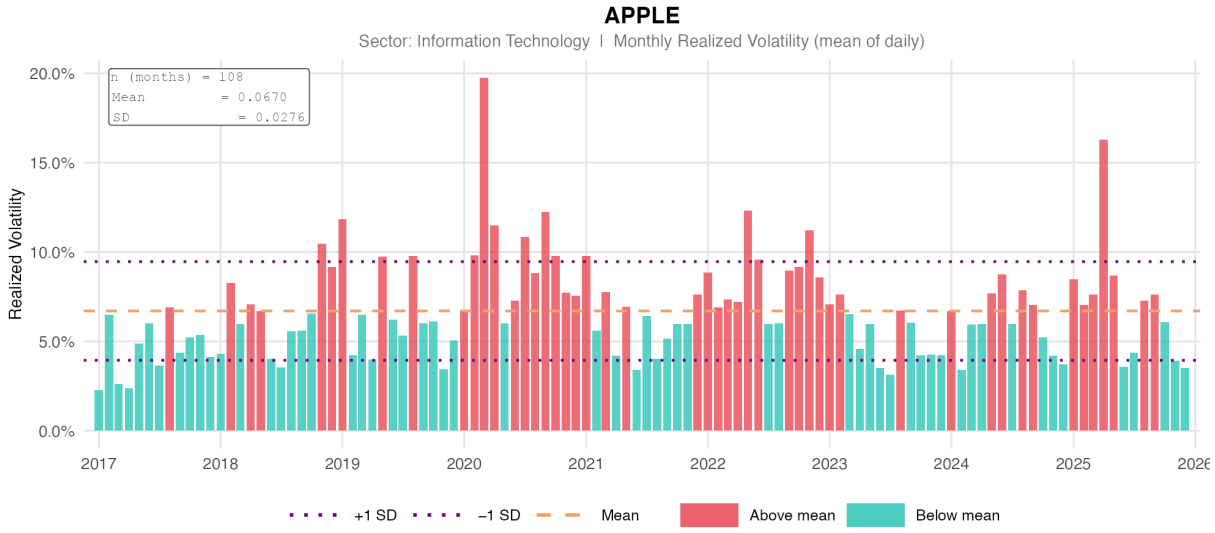


Figure 8 and Figure 9 display the monthly realized volatility for Chevron and Apple respectively. An exceptional volatility spike is visible in 2020 for both firms, far more pronounced for Chevron (∼33%) than Apple (∼20%). Outside 2020, both series remain relatively stable around their respective means. Chevron displays more above-mean months than Apple, suggesting a globally higher risk profile. Volatility appears to revert quickly to its mean after each spike. No seasonal pattern is visually identifiable in either series.

Figure 10: CHEVRON monthly trading volume over time

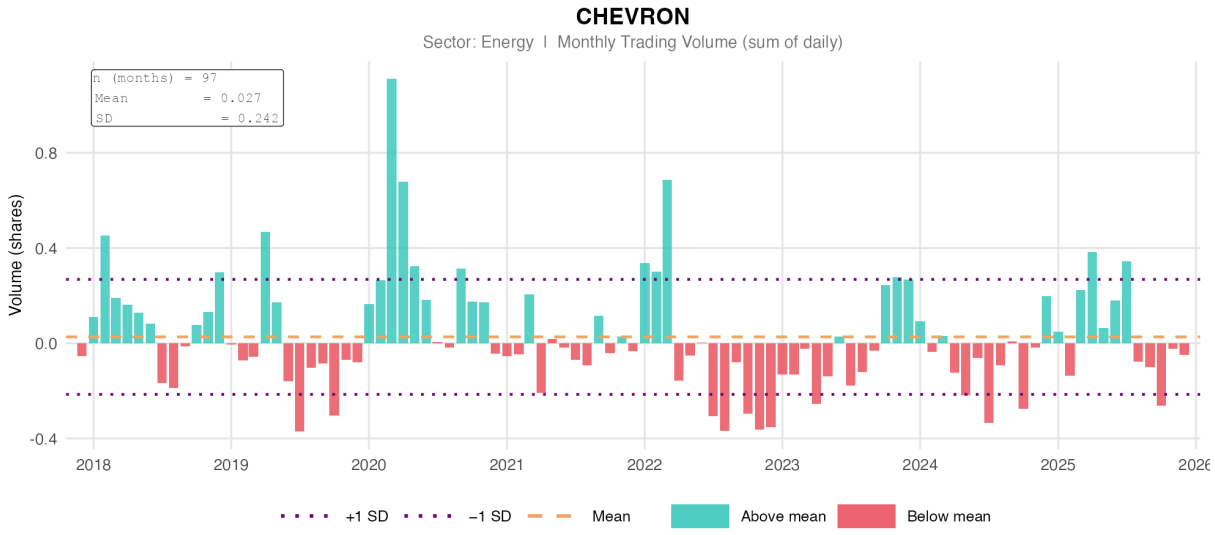


Figure 11: APPLE monthly trading volume over time

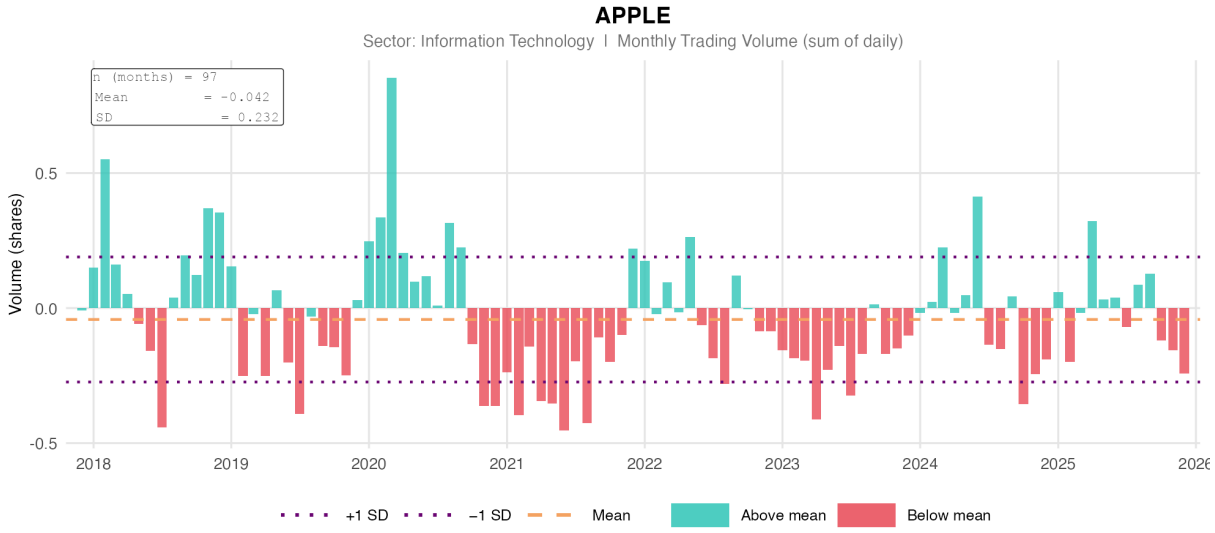


Figure 10 and Figure 11 display the monthly trading volume for Chevron and Apple respectively. Both series oscillate symmetrically around zero with no visible time trend. The 2020 spike is present for both firms but proportionally larger for Chevron. Apple shows a slightly negative mean (-0.042) versus positive for Chevron (0.027), though this is not visually striking. Bar dispersion appears relatively homogeneous throughout the period outside 2020. No seasonal pattern is visually detectable in either series.

Figure 12: CHEVRON monthly VWAP returns over time

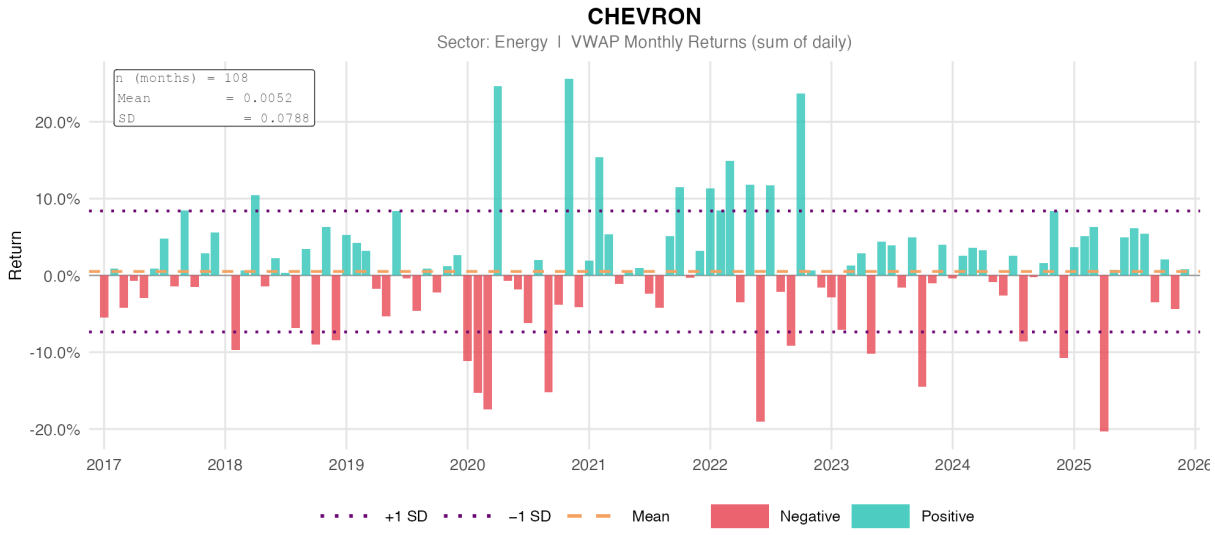


Figure 13: APPLE monthly VWAP returns over time

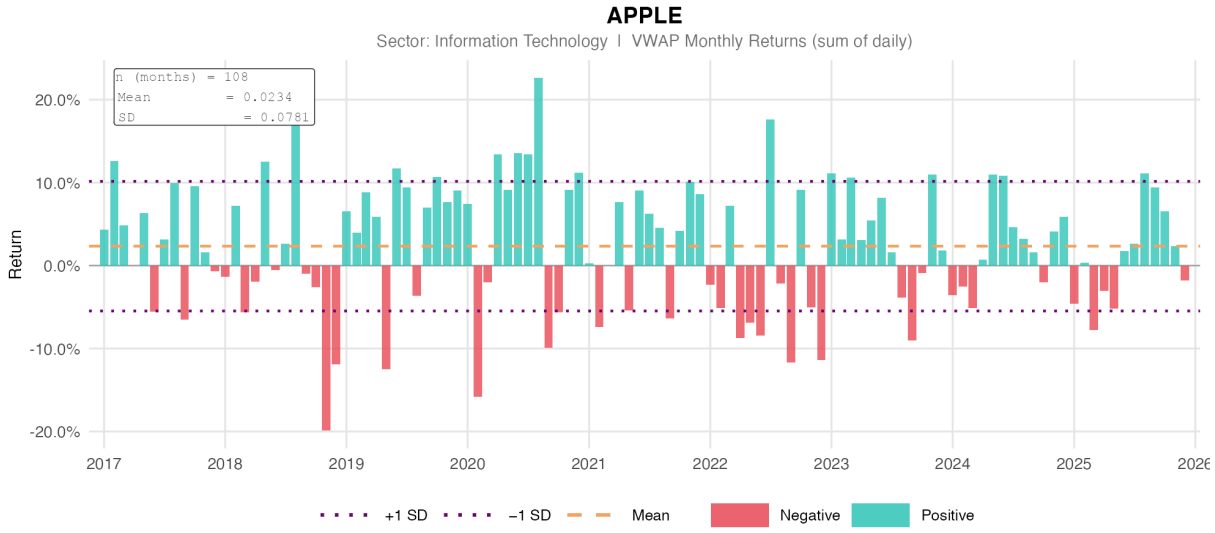


Figure 12 and Figure 13 display the monthly VWAP returns for Chevron and Apple respectively. Apple displays a visually slightly more positive trend than Chevron over the full period. Both series exhibit sharp drops in 2020, more severe for Chevron. Return dispersion appears comparable across both firms despite their different sectors. Positive and negative returns alternate without apparent temporal structure, consistent with the absence of trend in log returns. No seasonal pattern is visually identifiable in either series.

5 Empirical Strategy

5.1 Identification Challenge

Our goal is to estimate the causal effect of climate policy uncertainty shocks on firm-level financial outcomes. The key identification challenge is that the CPU index is an aggregate time-series shock: all firms in our sample are simultaneously exposed to the same CPU realisation in each month, which precludes the standard difference-in-differences strategy that relies on comparing treated and untreated units exposed to the same shock at different times.

We address this challenge by exploiting the *time-series variation* in the CPU index. Rather than comparing firms exposed to a policy change against unexposed ones, we ask: do firm-level financial outcomes behave differently in the months surrounding CPU spike episodes compared to non-spike months? The identifying assumption is that, absent the CPU spike, firm-level outcomes would have followed a parallel trend across episodes. We assess this assumption empirically by testing whether pre-spike coefficients are statistically indistinguishable from zero.

A second challenge concerns confounding by aggregate market conditions. Months of high CPU may coincide with elevated general market uncertainty, captured by the VIX index. Including contemporaneous VIX as a control is tempting but problematic: as we document in Section 5.5, CPU and VIX are contemporaneously correlated within the same calendar month, so conditioning on raw VIX absorbs part of the CPU effect rather than isolating it. We address this by using $\text{VIX}(t-3)$, the first VIX lag with no temporal overlap with any of the three months entering the $\text{CPU}(t{:}t-2)$ construction window.

5.2 Stacked Event-Study Design

To estimate the dynamic effects of CPU spikes on firm-level outcomes, we implement a stacked event-study design following Baker et al. (2022) and Wing et al. (2024). This approach has two advantages over a standard two-way fixed effects (TWFE) estimator. First, TWFE estimators are known to produce biased coefficients in settings with heterogeneous treatment effects across units or time (De Chaisemartin and d'Haultfoeuille, 2020; Goodman-Bacon, 2021). Second, by explicitly stacking episodes, the estimator uses each spike as an independent local experiment, reducing the likelihood that results are driven by a single confounding event.

Let $\mathcal{E}$ denote the set of identified CPU spike episodes and $\mathcal{I}$ the set of firms in our sample. For each episode $e \in \mathcal{E}$ with spike date t_e, we define a symmetric event window $h \in \{-2, -1, 0, +1, +2\}$ months around t_e, where h denotes time relative to the spike. The window of ± 2 months is chosen to balance two considerations: it is wide enough to capture anticipation effects one month before the spike and the persistence of the

shock two months after, while remaining narrow enough to minimise contamination from other macroeconomic events.

For each episode e, we extract the firm-month observations falling within the event window, yielding a panel indexed by firm i, episode e, and relative time h. These episode-specific panels are then stacked into a single estimation dataset. Firm fixed effects are included within each stack to absorb time-invariant heterogeneity across firms.

5.3 Regression Specification

For each outcome variable $y_{i,e,h}$, we estimate the following stacked event-study regression:

$$y_{i,e,h} = \alpha_i + \sum_{h \neq -2} \beta_h \, \mathbf{1}[\text{rel_time}_e = h] + \gamma X_{i,e,h} + \varepsilon_{i,e,h} \tag{3}$$

where α_i are firm fixed effects absorbing all time-invariant heterogeneity, $\mathbf{1}[\text{rel_time}_e = h]$ are relative-time dummies equal to one when observation (i, e) falls h months relative to spike e, and $X_{i,e,h}$ is the set of control variables described in Section 5.5. The period $h = -2$ is the omitted baseline category, so all β_h coefficients measure the deviation in the outcome variable relative to two months before each spike. Standard errors are clustered at the firm level to account for serial correlation within firms across episodes.

The coefficients of interest are β_{-1}, β_0, β_{+1}, and β_{+2}. A statistically insignificant β_{-1} supports the parallel-trends assumption: outcomes were not already trending prior to the spike, which would suggest that the spike is the source of the subsequent change rather than a pre-existing drift. A significant β_0 identifies the contemporaneous effect of the shock, while significant β_{+1} and β_{+2} identify its persistence.

5.3.1 Outcome Variables

This study examines four outcome variables capturing distinct dimensions of financial market activity: stock returns, realized volatility, abnormal trading volume, and first-difference log trading volume. Each variable is constructed at the firm-month level as described below.

Arithmetic Returns. The monthly arithmetic return for firm i in month t is computed as the month-over-month change in volume-weighted average price (VWAP):

$$r_{i,t} = \frac{P_{i,t} - P_{i,t-1}}{P_{i,t-1}}$$

where $P_{i,t}$ denotes the VWAP of firm i in month t.

Realized Volatility. For each firm i and calendar month t, realized volatility is computed as the standard deviation of daily VWAP returns over all trading days within month t, scaled by $\sqrt{21}$ to reflect the average number of trading days per month:

$$\sigma_{i,t} = \sqrt{21} \cdot \widehat{\text{sd}} \left(\frac{P_{i,\tau} - P_{i,\tau-1}}{P_{i,\tau-1}} \right)_{\tau \in t}$$

where $P_{i,\tau}$ is the VWAP of firm i on day τ.

Abnormal Trading Volume. Abnormal trading volume captures the deviation of log average monthly trading volume from its historical benchmark. Specifically, it is defined as:

$$\text{AbnormalVol}_{i,t} = \text{LogVol}_{i,t} - \overline{\text{LogVol}}^{(12)}_{i,t-1}$$

where $\text{LogVol}_{i,t} = \log\left(\frac{1}{D_{i,t}} \sum_{d \in t} \text{Vol}_{i,d}\right)$ is the log average daily trading volume in month t, $D_{i,t}$ is the number of trading days for firm i in month t, and $\overline{\text{LogVol}}^{(12)}_{i,t-1} = \frac{1}{12} \sum_{j=1}^{12} \text{LogVol}_{i,t-j}$ is the 12-month rolling mean of log volume, lagged by one month.

First-Difference Log Volume. To remove firm-level trends in trading activity and capture month-to-month changes in volume, the first difference of log trading volume is defined as:

$$\Delta \log(\text{Volume}_{i,t}) = \log(\text{Volume}_{i,t}) - \log(\text{Volume}_{i,t-1})$$

where $\text{Volume}_{i,t}$ denotes the total trading volume of firm i in month t.

5.4 Spike Detection

We identify months of unusually high CPU ("spikes") using two complementary methods. In both cases, we enforce a minimum 5-month gap between consecutive spikes to ensure non-overlapping event windows.

5.4.1 Method 1: Rolling Threshold

A spike is identified when the CPU index exceeds a time-varying threshold defined by the trailing 12-month rolling mean plus 1.5 rolling standard deviations:

$$\text{Spike}_t = \mathbb{1}\left\{\text{CPU}_t > \overline{\text{CPU}}_{t-12:t-1} + 1.5 \times \sigma_{t-12:t-1}\right\} \tag{4}$$

When multiple consecutive months are flagged, we retain only the month with the highest CPU value within each cluster.

5.4.2 *Method 2: Local Maxima (Preferred)*

A spike is a month in which the CPU is strictly higher than both the preceding and following months (a local maximum), and additionally exceeds the global mean plus 1.5 global standard deviations:

$$\text{Spike}_t = \mathbb{1}\left\{\text{CPU}_t > \text{CPU}_{t-1} \ \wedge \ \text{CPU}_t > \text{CPU}_{t+1} \ \wedge \ \text{CPU}_t > \overline{\text{CPU}} + 1.5\sigma\right\} \quad (5)$$

This is our preferred method because it identifies true peaks rather than elevated plateaus, is less sensitive to the rolling window length, and is more intuitive: a spike corresponds to a genuine peak in the time series.

5.4.3 *Spike Visualization*

Figure 14 and Figure 15 display the identified CPU spike episodes under the rolling threshold and local maxima methods respectively, overlaid on the $\text{CPU}(t-i:t)$ time series across all four lag windows. Several observations are worth noting. First, the two methods identify largely overlapping sets of episodes, which provides early reassurance that our results are not artefacts of the spike detection procedure. Both methods flag episodes concentrated in two broad periods: the first Trump administration (2017–2020), where the divergence between campaign rhetoric and legislative output on climate was particularly pronounced, and the 2023–2025 period, which reflects renewed policy uncertainty around the energy transition agenda.

Second, the rolling threshold (Figure 14) tends to identify slightly more episodes than the local maxima method, as any month exceeding the trailing 12-month mean by 1.5 standard deviations qualifies, even if CPU does not constitute a local peak in the time series. The local maxima method (Figure 15) is more selective: it requires that the flagged month be strictly higher than both its neighbours, ensuring that only genuine turning points in the CPU series are retained. For this reason, we adopt the local maxima method as our preferred identification strategy throughout the empirical analysis, while using the rolling threshold as a robustness check. Reassuringly, the pattern of spikes is qualitatively consistent across the four lag windows (Panels A–D), confirming that the choice of campaign materials aggregation window does not materially affect spike identification.

Figure 14: CPU index with rolling threshold spike detection. Red dots indicate identified spike episodes. Threshold: trailing 12-month mean + 1.5 SD.

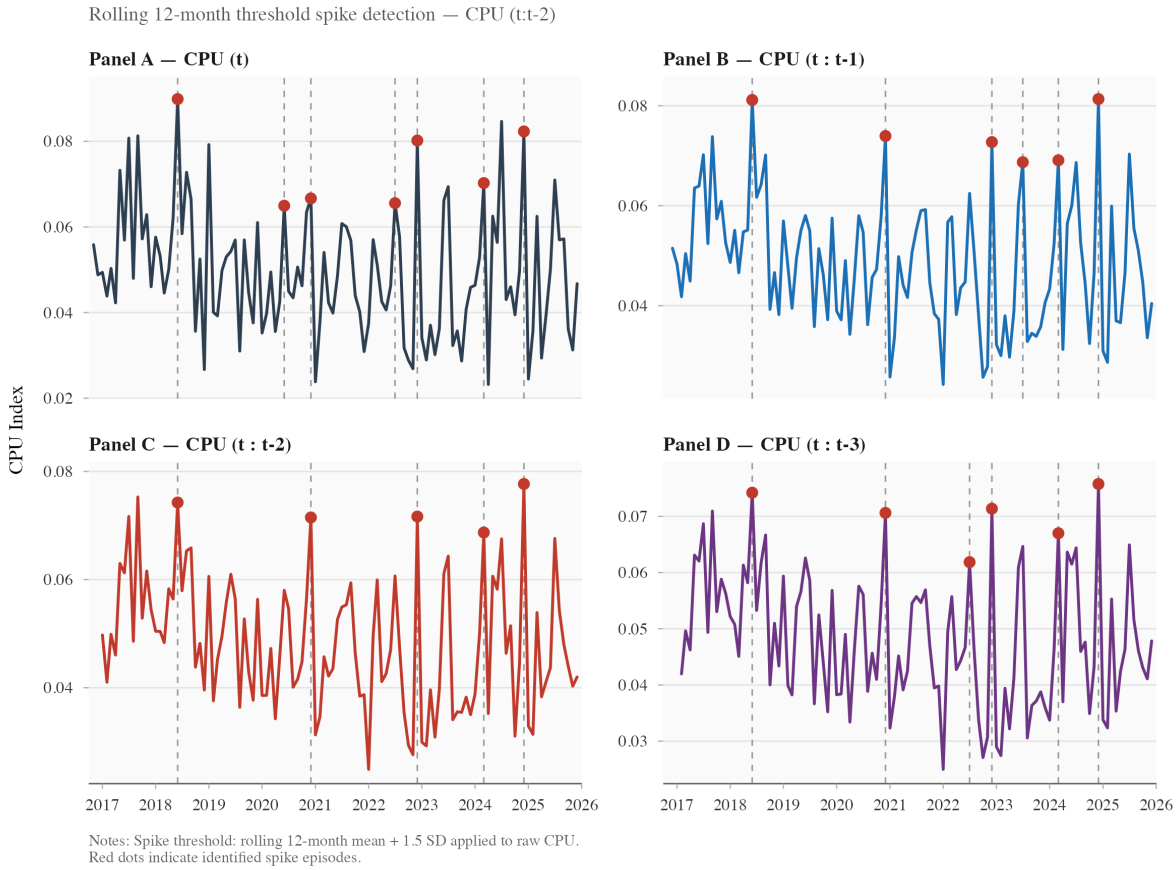
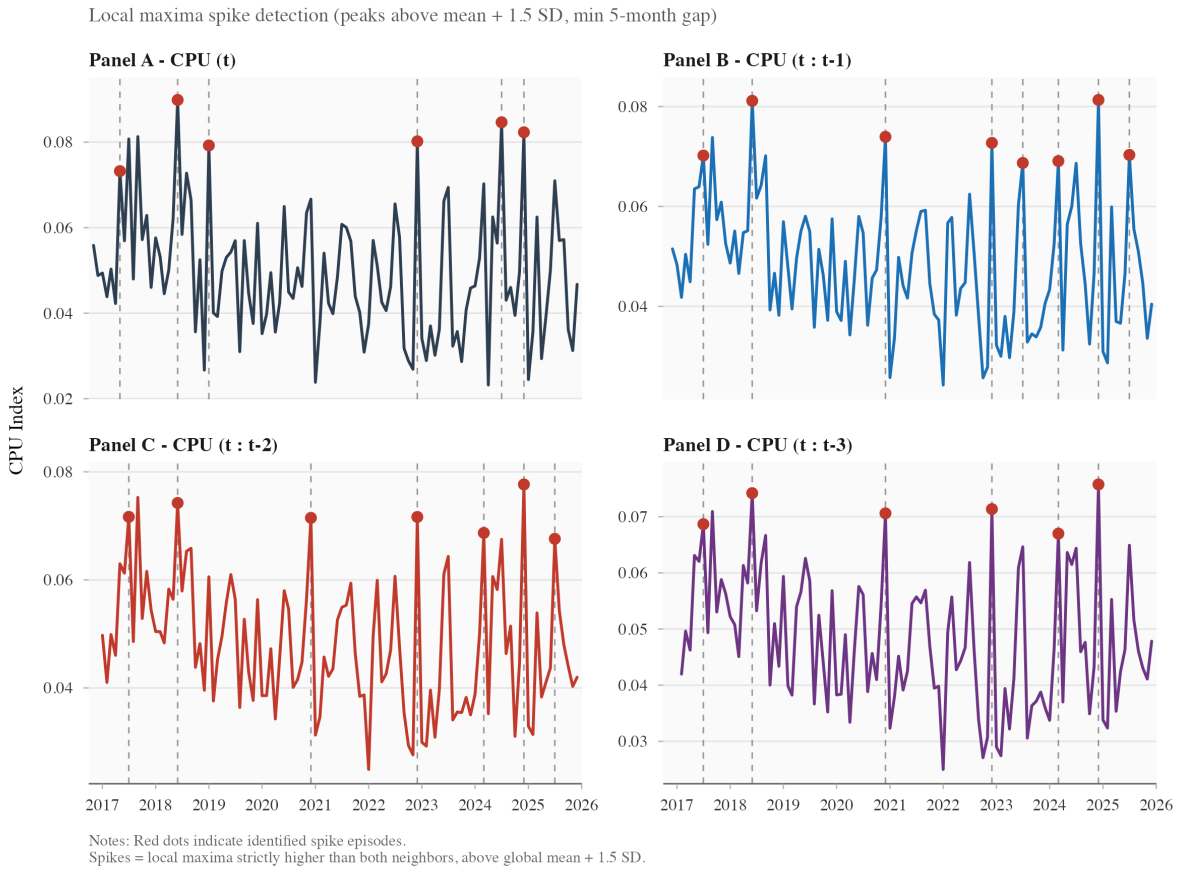


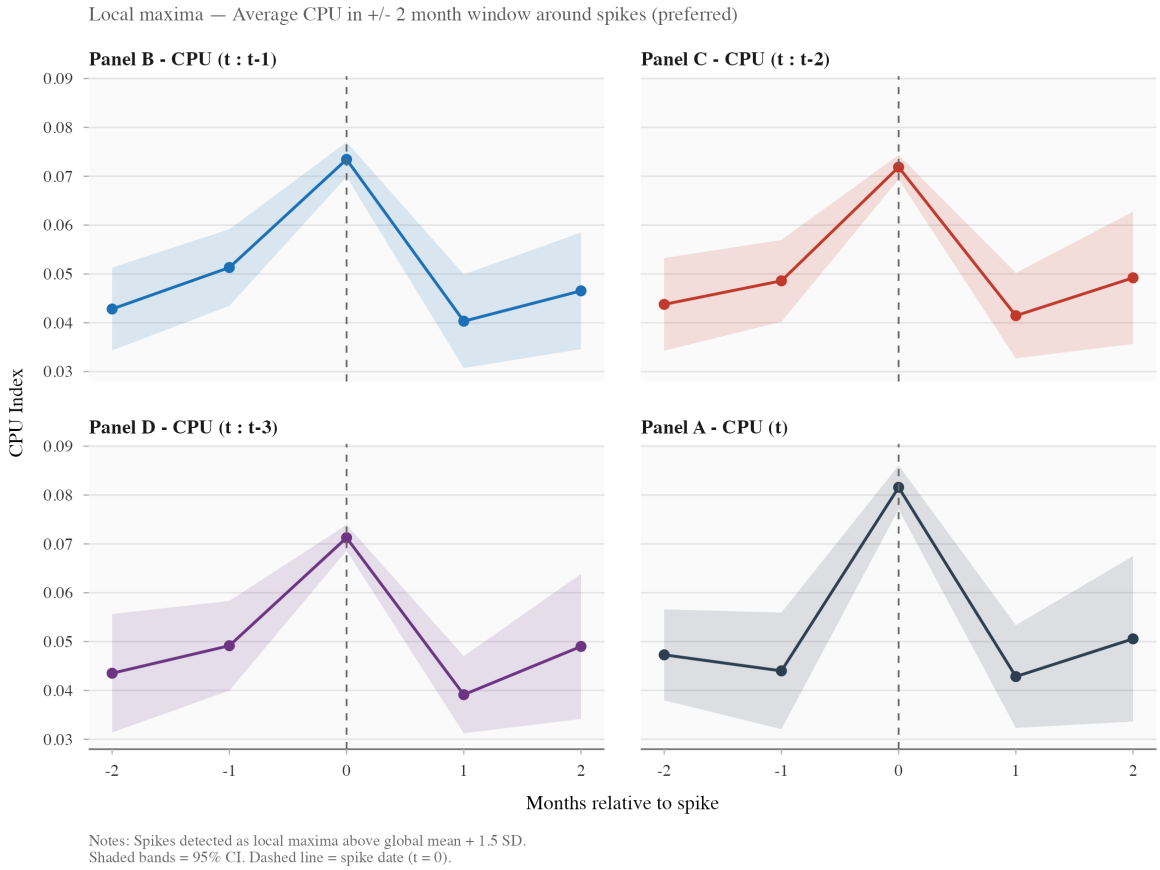
Figure 15: CPU index with local maxima spike detection (preferred). Red dots indicate identified spike episodes. Spikes are local maxima above the global mean + 1.5 SD, with a minimum 5-month separation.



5.4.4 *Event Study on CPU*

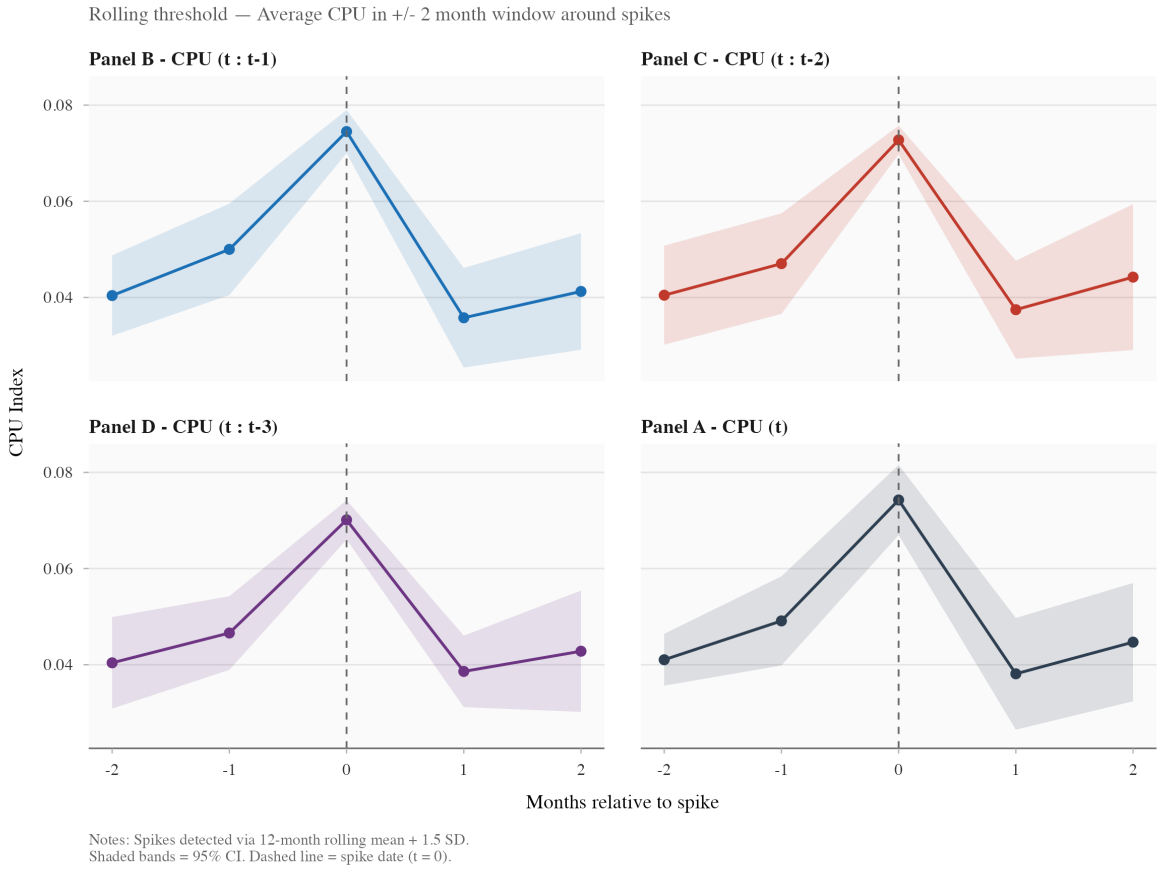
Figure 16 and Figure 17 present the event-study estimates applied to the CPU index itself under the local maxima and rolling threshold methods respectively. Both methods yield a consistent inverted-V profile: CPU rises in the two months preceding the spike, peaks at $t = 0$, and reverts thereafter. This confirms that the identified episodes correspond to genuine, transitory peaks in climate policy uncertainty rather than persistent level shifts, and that results are to both the detection method and the choice of delay.

Figure 16: Event-study estimates across lag windows, ±2 months around spike (local max)



Notes: Spikes detected as local maxima above global mean + 1.5 SD.
Shaded bands = 95% CI. Dashed line = spike date (t = 0).

Figure 17: Event-study estimates across lag windows, ±2 months around spike (rolling)



5.5 Endogeneity Analysis: Is VIX a "Bad Control"?

5.5.1 The Problem

A key concern in our event-study framework is whether the VIX index constitutes a valid control variable. First VIX may be a common driver of both CPU and the different outcome variables, a genuine confounder that should be controlled for. Second, CPU spikes may be contemporaneously correlated with VIX, in which case conditioning on VIX constitutes a "bad control" that removes part of the structural effect of interest (Angrist and Pischke, 2009). A further and more fundamental constraint arises from the construction of CPU$(t{:}t-2)$ itself: because campaign materials from months t, t-1, and t-2 all enter the index, any VIX realization from those months overlaps with a CPU input and is therefore ineligible as a predetermined control. We address both concerns through a sequence of diagnostic tests before deriving the appropriate control lag.

5.5.2 *Granger Causality and VAR Diagnostics*

We begin with bivariate Granger causality tests to assess whether past values of CPU predict future VIX (and vice versa), using three monthly lags. The results are reported in Table 6. Neither direction yields a statistically significant result: the F-statistic for CPU $\rightarrow$ VIX is $F(3, 98) = 1.817$ ($p = 0.149$), and for VIX $\rightarrow$ CPU it is $F(3, 98) = 1.516$ ($p = 0.215$). These results suggest that CPU and VIX do not Granger-cause each other at the monthly frequency, lagged values of one series carry no incremental predictive power for the other.

Table 6: Granger Causality Tests: CPU and VIX (3 lags)

Direction	F-statistic	df	p-value	Conclusion
CPU $\rightarrow$ VIX	1.817	$(3, 98)$	0.149	Fail to reject H_0
VIX $\rightarrow$ CPU	1.516	$(3, 98)$	0.215	Fail to reject H_0

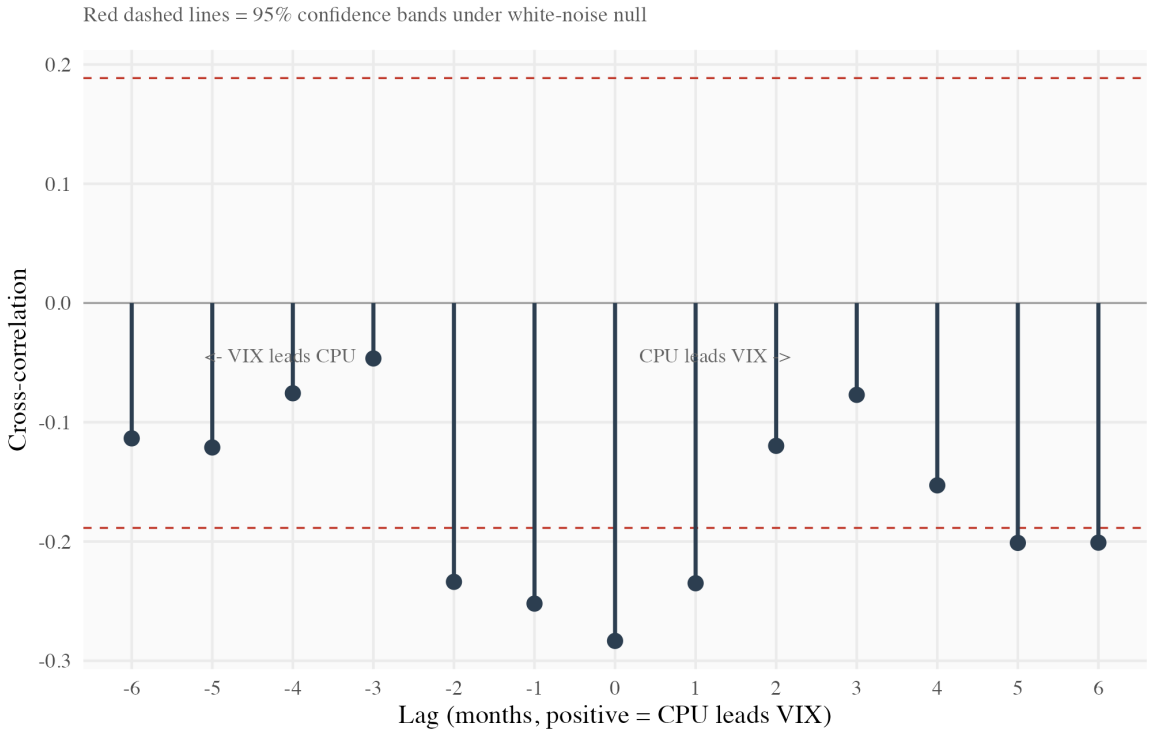
Notes: H_0: the cause variable does not Granger-cause the outcome. Tests use 3 monthly lags. Neither direction is significant at conventional levels.

We complement these tests with a Vector Autoregression (VAR) estimated on the joint CPU–VIX system, with optimal lag length selected by AIC. The VAR block-exogeneity test confirms the absence of Granger causality from CPU to VIX ($F = 0.581$, $p = 0.447$). However, the test for *instantaneous* causality (whether CPU and VIX move together within the same month) yields $\chi^2(1) = 2.757$ ($p = 0.097$), marginally significant at the 10% level. This finding is critical: it indicates that CPU and VIX are *contemporaneously correlated* within the same calendar month, even though neither leads the other in a temporal sense. It is precisely this within-month co-movement that causes raw VIX to absorb the CPU effect when included as a contemporaneous control.

5.5.3 *Cross-Correlation Analysis*

Figure 18 presents the cross-correlation function (CCF) between CPU$(t{:}t-2)$ and VIX at lags ranging from -6 to $+6$ months. Positive lags indicate that CPU leads VIX; negative lags indicate that VIX leads CPU.

Figure 18: Cross-Correlation Function: CPU$(t{:}t-2)$ and VIX.



Notes: Positive lags denote CPU leading VIX; negative lags denote VIX leading CPU. Red dashed lines indicate 95% confidence bands under the white-noise null ($\pm 1.96/\sqrt{n}$). Bars extending beyond the bands are statistically significant.

Two features of the plot are noteworthy. First, the contemporaneous correlation (lag $= 0$) is the strongest in absolute value (≈ -0.30), confirming the within-month co-movement identified by the VAR instantaneous causality test. The negative sign implies that months with elevated CPU tend to coincide with *lower* VIX readings, consistent with the interpretation that political climate uncertainty and market-implied volatility capture distinct dimensions of risk: the former reflects divergence between political promises and legislative output, while the latter reflects forward-looking option-based risk premia.

Second, correlations at both positive and negative lags are substantially smaller in magnitude and, with the exception of a few lags at horizons of four to six months, largely fall within the 95% confidence bands. This is consistent with the Granger causality results: neither series systematically leads the other over time. The significant negative correlations at longer positive lags (CPU leading VIX by four to six months) are suggestive but should be interpreted with caution given the multiple-testing environment.

Taken together, the Granger and CCF evidence points to a specific form of endogeneity: *simultaneous correlation* rather than temporal causality. This motivates replacing contemporaneous VIX with a truly predetermined lag, whose admissible range is derived directly from the CPU construction window.

5.5.4 *The Truly Predetermined Control: VIX$(t-3)$*

The choice of lag is not arbitrary but constrained by the construction of the index. CPU$(t{:}t-2)$ uses campaign materials over months t, $t-1$, and $t-2$, and the legislative text embedding at month t. The inputs to the index therefore span three calendar months. Any VIX realisation from months t, $t-1$, or $t-2$ is contemporaneous with at least one input entering the index and is ineligible as a predetermined control. The earliest admissible lag is VIX$(t-3)$: the most recent month whose VIX realisation has *no overlap* with any month contributing to CPU$(t{:}t-2)$. This mapping is summarised in Table 7.

Table 7: CPU$(t{:}t-2)$ Construction Window and VIX Admissibility

Month	Role in CPU$(t{:}t-2)$	VIX admissible as control?
t	APP text + BILLS text enter	No — contemporaneous
$t-1$	APP text enters	No — overlapping
$t-2$	APP text enters	No — overlapping
$t-3$	Not used	**Yes — fully predetermined**

Notes: CPU$(t{:}t-2)$ averages campaign material (APP) embeddings over months t, $t-1$, $t-2$ and uses legislative (BILLS) text at month t. VIX$(t-3)$ is the first lag with no overlap with any input month and is therefore fully predetermined by construction.

It is important to note that VIX(t-3) is used here as a *predetermined control variable*, not as an instrumental variable.

The IV exclusion restriction is unlikely to hold for lagged VIX. The reason is well-established in the financial econometrics literature: volatility is highly persistent. Engle (1982) and Bollerslev (1986) document that periods of high volatility tend to be followed by further periods of high volatility, and periods of low volatility by further low volatility. This phenomenon is known as volatility clustering. As a direct consequence of this persistence, VIX(t-3) carries predictive power for realized volatility at t. The exclusion restriction is therefore violated by construction and using VIX(t-3) as an instrument would thus produce biased and inconsistent estimates of the causal effect of contemporaneous VIX.

Our approach is more modest and does not require the exclusion restriction to hold. Rather than attempting to isolate the causal effect of contemporaneous VIX on realized volatility, we simply condition on the pre-existing volatility environment before any month contributing to CPU(t-2:t) is realized. As established in Table 7, VIX(t-3) is the earliest VIX realisation with no temporal overlap with any of the three months entering the CPU index construction window.

5.5.5 *Conclusion on endogeneity analysis*

The joint evidence from the Granger tests, VAR instantaneous causality, CCF analysis, and the VIX$(t-3)$ regressions supports a clear conclusion. Contemporaneous VIX is a bad control in this setting: it does not confound the relationship between CPU spikes and outcome variables, but absorbs part of the CPU effect through within-month co-movement. The correct predetermined control is VIX$(t-3)$, which is the first VIX lag with no overlap with any input month of CPU$(t{:}t-2)$, making it fully predetermined by construction and without requiring additional timing assumptions. We therefore adopt VIX$(t-3)$ as our preferred control specification in all subsequent analyses.

6 Results

This section presents the event-study estimates for the four outcome variables: arithmetic returns, realized volatility, abnormal trading volume, and first-difference log trading volume. For each variable, we report results under both spike detection methods (rolling threshold and local maxima) and across three different specifications (Baseline, Raw VIX, VIX$(t-3)$). Full tables and figures for all specifications are collected in Section 9.

The results reveal three distinct channels through which CPU spikes affect financial markets. First, arithmetic returns exhibit a sharp reversal pattern: a positive pre-spike appreciation followed by a significant decline at the spike date, consistent with a pricing-in then correction mechanism. Second, realized volatility falls significantly and persistently around CPU spikes, a counterintuitive finding that we interpret as reflecting investor inaction rather than heightened trading under uncertainty. Third, trading volume displays a lagged response: the first-difference of log volume rises significantly in the month preceding and following the spike, while the levels-based abnormal volume measure indicates below-normal activity in the spike month itself, suggesting that aggregate uncertainty induces a net reduction in trading relative to historical benchmarks even as heterogeneous agents trade actively against one another in the short run. All three patterns are robust to the choice of spike detection method and to the VIX$(t-3)$ endogeneity correction.

For some outcome variables, the pre-trend coefficient β_{-1} is statistically significant, indicating that the parallel trends assumption is not fully satisfied. Two interpretations are possible. First, this may reflect genuine anticipation effects: if market participants observe the divergence between campaign and legislative climate discourse before it reaches its local maximum, firm-level outcomes may begin adjusting prior to the identified spike date. This would be consistent with the efficient markets hypothesis and would not invalidate the event study design, but would suggest that our estimates

capture a lower bound of the total effect. Second, it may reflect a violation of the parallel trend assumption. We cannot fully disentangle these two explanations, and results for affected outcome variables should therefore be interpreted with caution

6.1 Arithmetic returns

6.1.1 Hypothesis

As mentioned earlier Treepongkaruna et al. (2023b) demonstrate that climate policy uncertainty (CPU) is a priced risk factor in equity markets. They document a statistically significant negative relationship between a firm's CPU exposure and its subsequent stock returns. Stocks with low CPU exposure generate annualised risk-adjusted returns that are 5.5–6.3% higher than those with high exposure, suggesting that investors systematically adjust valuations in response to climate policy uncertainty. Looking at those results studying how VWAP and the VWAP arithmetic returns move with CPU can add additional information on the way climate uncertainty impact markets. Hence, as our sample contains several sectors with high exposure to the CPU index (e.g. energy, transport and manufacturing), we can expect the following result :

Hypothesis 1 : The arithmetic returns are negatively impacted by the spikes in the CPU.

6.1.2 Results

Local Maxima Following the results reported in Table 12 and Figure 19 displaying the event-study coefficients for the three different approaches (Baseline, Raw VIX, VIX t(-3)). The estimated coefficients are highly similar across the three specifications. The coefficients are significant and positive at 1% level for the month preceding the spike (t=-1), indicating that stock returns are higher relative to the reference period of two months before the event. In contrast, the coefficients turn significantly and negative at 1% level in the spike month (t=0), pointing to an immediate market reaction once the CPU shock occurs. For instance, according to the baseline model, arithmetic returns increase by 2.25 percentage points one month before the spike relative to returns two months before the shock. Conversely, at t=0, the returns decrease by 2.59 percentage points relative to the returns two months before the shock. Additionally, at month t = 2, across the three panels, the coefficients are negative but are significant at only 5% and 10% level. Overall, these findings suggest that CPU spikes have short-term effects on the stock returns, with a positive effect before the event followed by a decline when the shock occurs.

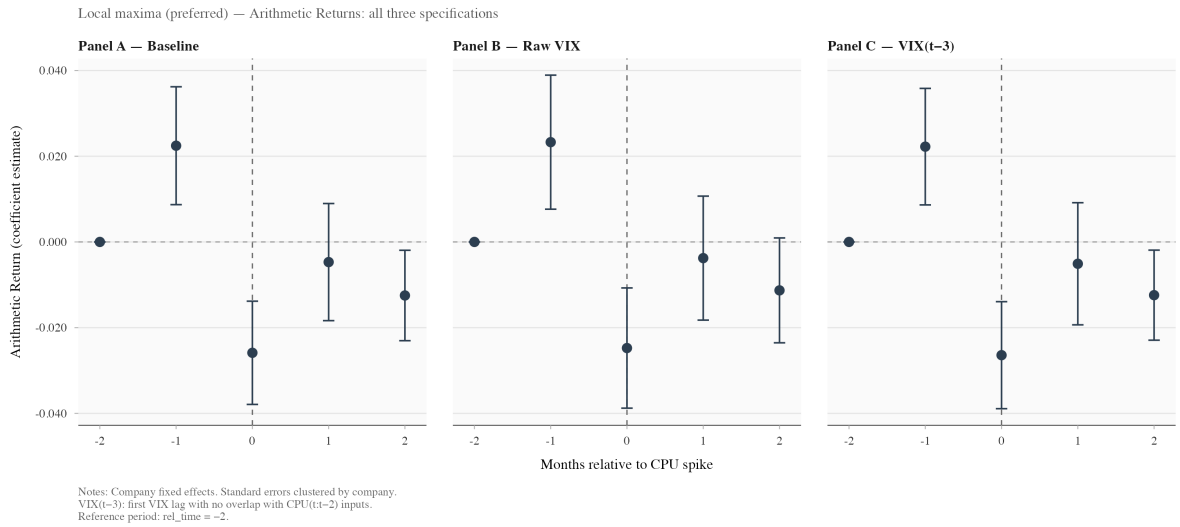


Figure 19: Arithmetic returns — Local maxima: all three specifications (5% level of significance).

Rolling Threshold Turning to the rolling-threshold approach, Table 8 and Figure 20 indicate a pattern that is very similar to the one obtained with the local-maxima method. In all three specifications, the coefficient at (t=-1) is positive and statistically significant, whereas the coefficient at (t=0) is negative and significant both at 1% level. These results suggest that stock returns increase in the month preceding the CPU spike and then fall in the month in which the spike occurs.

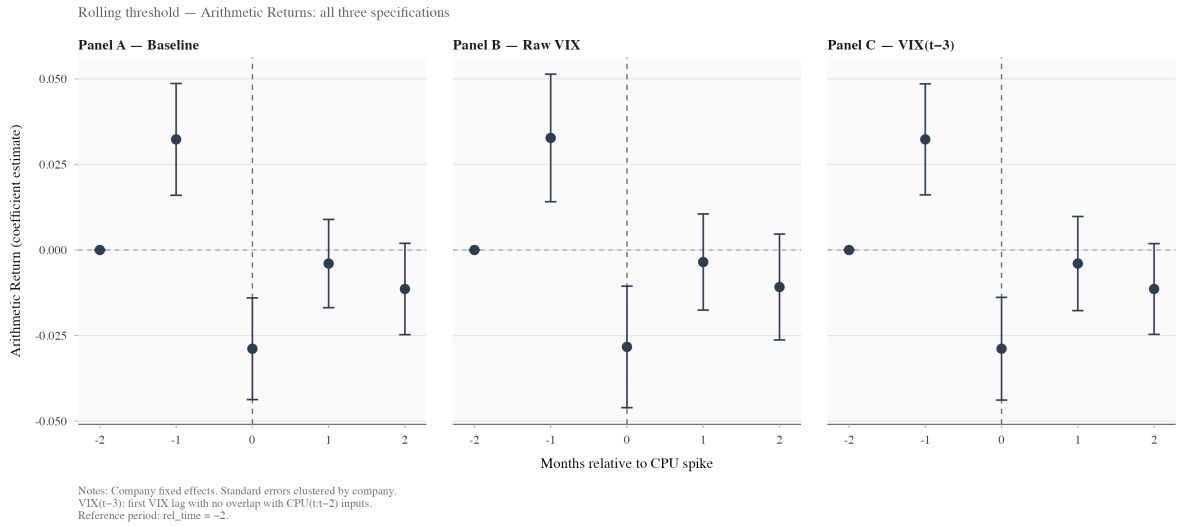


Figure 20: Arithmetic returns — Rolling threshold: all three specifications (5% level of significance).

Overall Results and Discussion Overall, the results obtained for arithmetic returns reveal a strong pattern around CPU spikes. Under both the local-maxima and the rolling-threshold approaches and across the three model specifications (Baseline, Raw

VIX, and VIX($t-3$)), the coefficients are significantly positive in the month preceding the spike and significantly negative in the spike month itself, suggesting that stock returns are higher before the uncertainty shock but lower once the CPU spike occurs relative to the returns of the second month before the spike. The finding is robust to the spike detection methodology, with both rolling-threshold and local-maxima approaches yielding coefficients of similar sign, magnitude, and significance. The positive and significant coefficient at t-1 may be explained by anticipation effects in financial markets: investors may observe early signals of a forthcoming increase in policy uncertainty and begin adjusting their expectations by asking a higher risk premium before the spike is fully captured by the CPU index, generating a brief period of above-normal returns prior to the recorded event. The negative and significant coefficient at t=0 is aligned with Hypothesis 1 and consistent with Treepongkaruna et al. (2023a), who show that firms with high exposure to CPU exhibit a negative relationship between the CPU index and their returns. This result is further supported by Vo et al. (2025), who find that higher levels of the CPU index increase the risk of stock price crashes due to greater volatility in firm performance and reduced transparency, consistent with a downward pressure on equity returns when uncertainty peaks.

6.2 Realized Volatility

6.2.1 Hypothesis

Bonato et al. (2022), demonstrate that climate-related risks possess predictive capabilities with regard to realized stock-market volatility over time horizons of one to two months, as evidenced by their analysis of high-frequency intraday data across all 50 U.S. states. Their findings, which apply to both upside and downside volatility, suggest that climate uncertainty encompassing physical and transition risks influences the dynamics of equity market volatility. In light of these results, we are interested in studying the impact of CPU on the realized volatility of Dow Jones Average Index firms. Moreover, given the results of Bonato et al. (2022), we expect realized volatility to move with CPU, but it is unclear in which direction.

Hypothesis 2: The realized volatility of stock returns is expected to be affected by spikes in the CPU over time horizons of one to two months.

6.2.2 Results

Local Maxima Table 20 and Figure 21 report the event-study estimates for realized volatility under the local-maxima approach. The pre-spike coefficient at t-1 is small and statistically insignificant in both the Baseline and the preferred VIX($t-3$) specifications. In the spike month (t=0), realized volatility drops and significantly at 1% level. At t=0 the CPU spike decreases the realized volatility by 1.18 percentage points (Baseline) and

by 1.62 percentage points (VIX($t-3$)) relative to the realized volatility at the second month before the CPU spike. This volatility-dampening effect persists and strengthens in the post-spike period: at $t+1$, the estimates are $\hat{\beta}_1 = -0.0044$ ($p < 0.01$) and $\hat{\beta}_1 = -0.0075$ ($p < 0.01$), and at $t+2$ both specifications deliver negative coefficients significant at the 1% level, confirming that the effect extends at least two months beyond the shock. The Raw VIX specification produces attenuated and sign-reversed post-spike coefficients compared to the two other models, consistent with a potential bias induced by a bad control, and is therefore not discussed further.

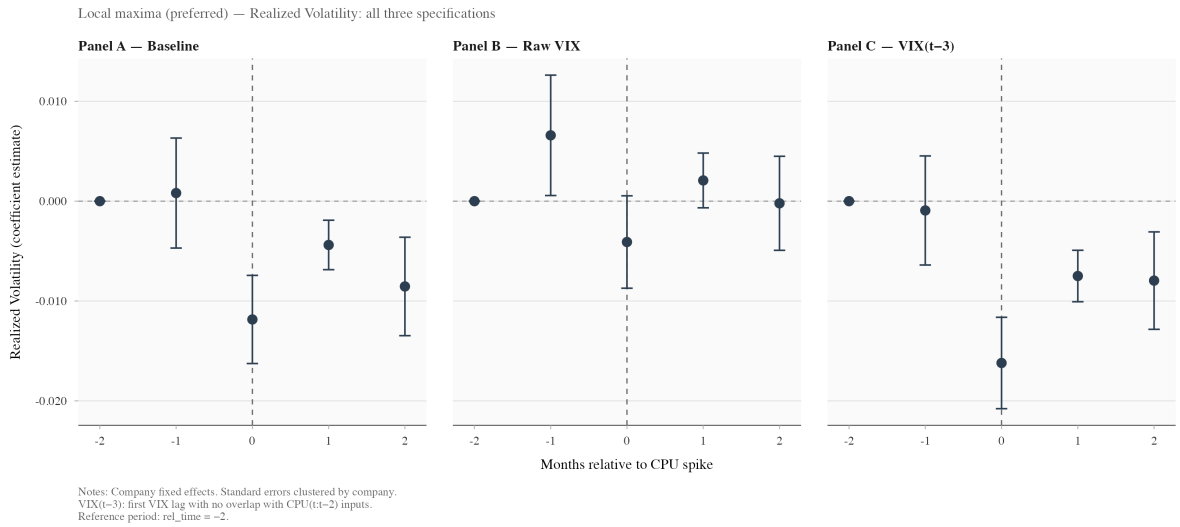


Figure 21: Realized volatility — Local maxima: all three specifications (5% level of significance).

Rolling Threshold The rolling-threshold estimates reported in Table 16 and Figure 22 are similar to the the local-maxima findings for the period t=-1 and t=0 for the Baseline and VIX($t-3$) models. There is a insignificant coefficient for t=-1 and a negative coefficient for t=0 both significant at 1% level as the Local Maxima specification. This negative effect persists at $t+2$ with negative and significant coefficient at 5% level conversely to 1% level with the Local Maxima for both the Baseline and VIX($t-3$) specifications, suggesting that the volatility-dampening effect of CPU spikes extends at least two months beyond the shock. As with the Local Maxima model, the Raw VIX specification produces distorted estimates and is not discussed further.

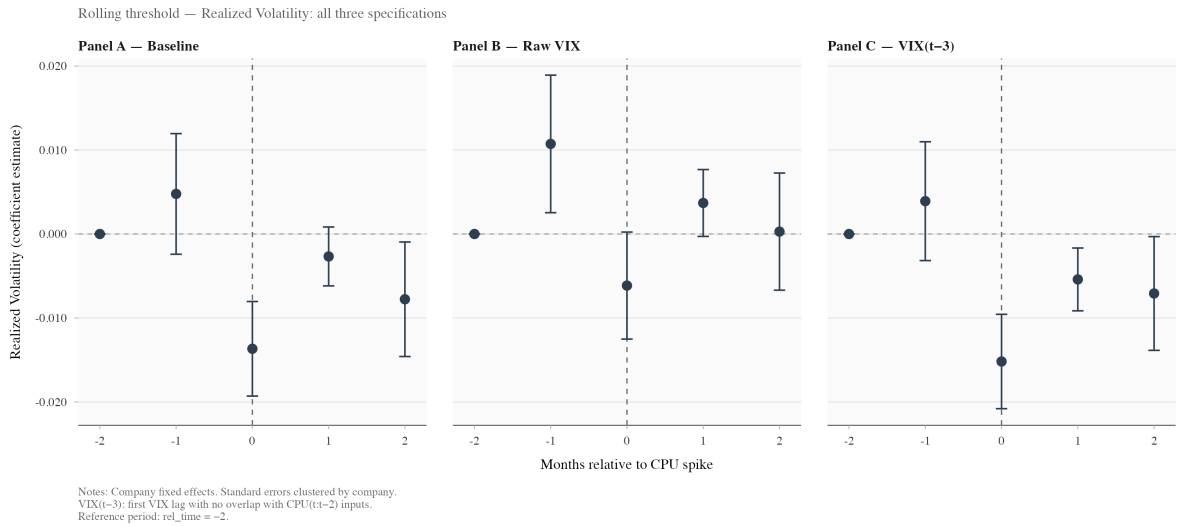


Figure 22: Realized volatility — Rolling threshold: all three specifications (5% level of significance).

Overall Results and Discussion Across the three different panels and the two specifications (Local Maxima and Rolling Threshold), the results for realized volatility are quite similar. CPU spikes are not preceded by any significant change in realized volatility at t-1 and therefore provides evidence against anticipation effects. These results are consistent with a wait-and-see mechanism. As showed by Bloom (2009) higher uncertainty causes firms to temporarily pause their investment and hiring. Moreover, Bernanke (1983) emphasize higher uncertainty increases the value of waiting for new information and therefore retards the current rate of investments. In our case we interpret it as follows: when climate policy uncertainty rises, as measured by the divergence between campaign rhetoric and enacted legislation, investors reduce trading activity rather than rebalancing portfolios. This reduction in trading activity mechanically compresses the dispersion of daily returns within the month, lowering realized volatility. The finding is robust to the spike detection methodology, with both rolling-threshold and local-maxima approaches yielding coefficients of similar sign, magnitude, and significance. Additionally, our results demonstrate that this decrease persists for at least two months following the shock. This outcome is consistent with Lasisi et al. (2024), who find that although the effect of CPU is not highly persistent over time, it still affects market volatility over a longer horizon, as stock market volatility itself is highly persistent in both economies. This volatility-dampening effect is not driven by market-wide volatility conditions, as we obtained the same result with the VIX($t-3$) specification. The outcomes found are aligned with Hypothesis 2. In addition, these findings are aligned with the results of Chen et al. (2023) and Bonato et al. (2022), demonstrating that the CPU has a significant impact on the realized volatility, as uncertainty surrounding climate policies creates an additional source of risk for investors in financial markets. This

uncertainty could affect future costs, regulation, sectoral profits, and investor behavior, which in turn affects the volatility of companies' stocks.

6.3 Abnormal Trading Volume

6.3.1 Hypothesis

Climate policy uncertainty (CPU) can significantly impact financial markets by influencing firm decisions and investor expectations. Using a novel CPU index constructed from three major US newspapers over the period 1990–2019, Basaglia et al. (2025), show that increases in CPU lead to higher stock price volatility and lower returns for firms operating in CO2 intensive sectors. These effects are primarily driven by uncertainty indicating potential policy weakening (CPU) rather than potential tightening (CPU+), suggesting that investors are particularly sensitive to the prospect of climate policy being rolled back. Such portfolio adjustments, driven by shifts in risk perception and expectations about future regulatory environments, are likely to affect trading activity as well as prices and volatility. If investors collectively revise their beliefs about the likelihood of climate policy changes, one would expect a corresponding increase in trading volume as heterogeneous agents rebalance their portfolios. Therefore, studying the relationship between CPU and trading volume can provide valuable insights into how policy-related uncertainty shapes market liquidity and investor behavior.

Hypothesis 3 : The trading volume is positively impacted by the spikes in the CPU.

6.3.2 Results

Local Maxima Now, looking at the local maxima method in Figure 23 and Table 28, a clearer trend is observable. Firstly, no significant reaction is observed for any of the three models at (t=-1). Conversely, when the shock occurs at (t=0), all three models react by decreasing trading activity by between 3.1 and 5.17 percentage points, with the coefficients being significant at the 5% level for the baseline and Raw VIX models, and at the 10% level for the VIX(t-3) model. At the (t = 1) period, only the Raw VIX model has a significant reaction at the 10% confidence level, decreasing trading activity by 4.6 percentage points. Finally, at the (t = 2) period, all three models react with a significant coefficient between the 10% and 5% levels, decreasing trading activity by between 4.6 and 6 percentage points.

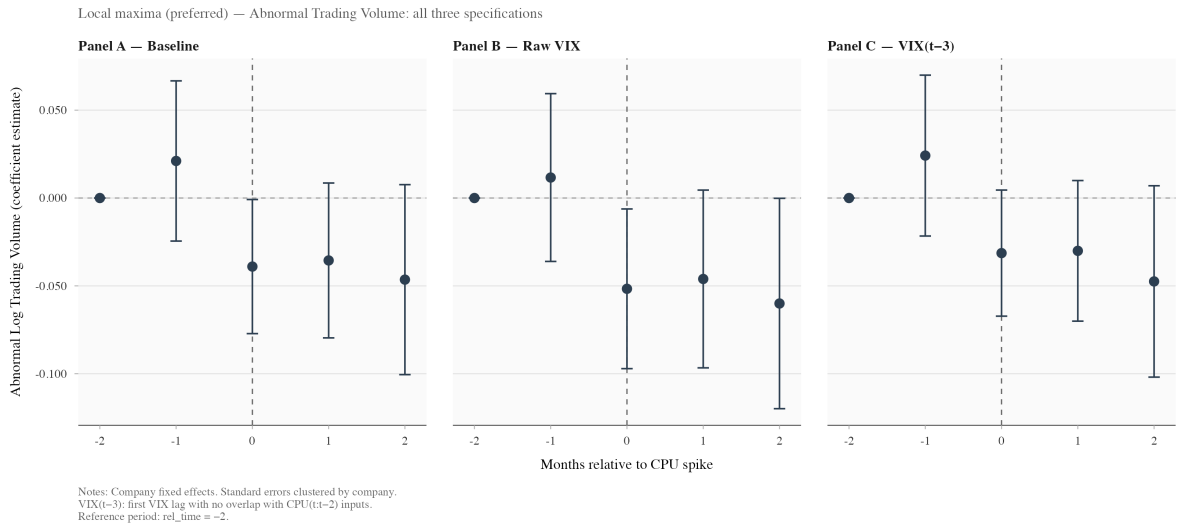


Figure 23: Abnormal trading volume — Local maxima: all three specifications (5% level of significance).

Rolling Threshold Looking at the results in Figure 24 and Table 24, it can be seen that the results differ across different model specifications. Under the baseline model, the abnormal trading volume is unaffected by the CPU shock in all periods before and after the shock, suggesting that trading activity remains stable before and after the shock. Conversely, under the Raw VIX model specification, trading activity is affected by the shock in two different periods: firstly, trading activity decreases by 7.16 percentage points at t=0, and secondly, there is another decrease of 9.91 percentage points at (t=2). These coefficients are both significant at the 10% and 5% levels. Finally, under the VIX(t-3) model, trading activity only reacts at t=2, decreasing by 6.69 percentage points, with a coefficient that is significant at the 10% level.

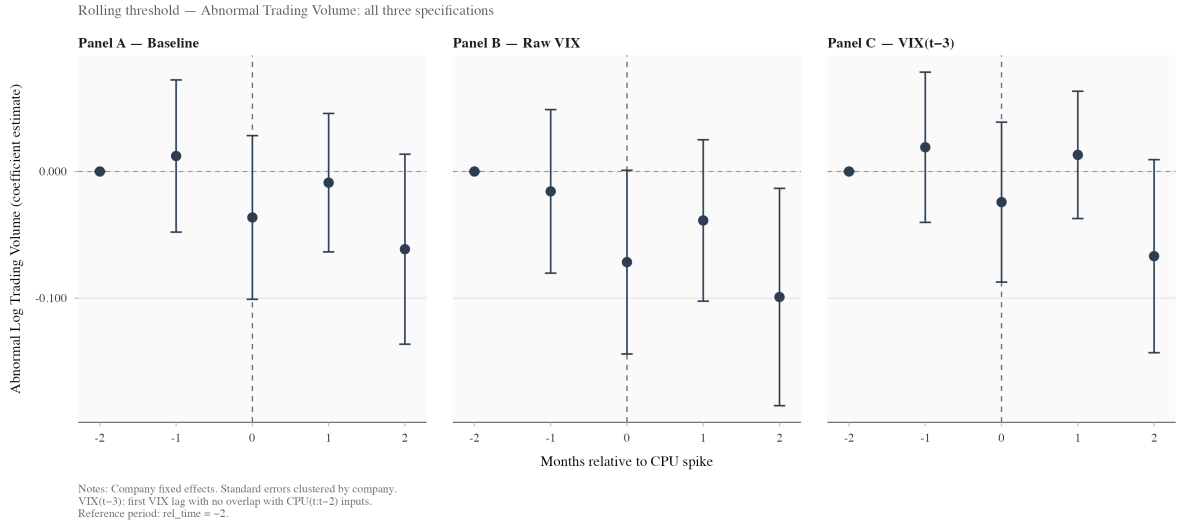


Figure 24: Abnormal trading volume — Rolling threshold: all three specifications (5% level of significance).

Overall Results and Discussion Overall, it is not possible to determine a clear trend among the different models, since they are significant at different periods and to different magnitudes. The results of these two cases reveal one key fact: trading activity appears to remain stable prior to the shock. This suggests that traders do not modify their trading activity before a CPU shock occurs. However, these results may be biased by a trend, since the abnormal trading volume is calculated on an annual basis. This means that the observed deviation could be cancelled out by periods without shocks. To test this theory, the same models are being recomputed using the first difference to reveal the deviation using more stable and accurate data.

6.4 First-Difference Log Trading Volume

6.4.1 Results

Local Maxima Following the results reported in Figure 25 and Table 36, which show the coefficients of the three different approaches, it can be seen that the results are similar for all the different model specifications. Firstly, all the coefficients at (t = -1) and (t = 1) are significant at the 1% level, and all the coefficients at (t = 2) are significant at the 5% level. During the (t=-1) period, the first difference in trading volume increased by approximately 10 percentage points for all model specifications, indicating a short-run anticipation of the CPU shock. Additionally, at (t=1), trading activity increases by 8 percentage points for all models, and finally, at (t=2), trading increases by approximately 6.5 percentage points for all models. These increases, which occur one and two periods after the shock, show persistence in the response of trading activity after the CPU shock.

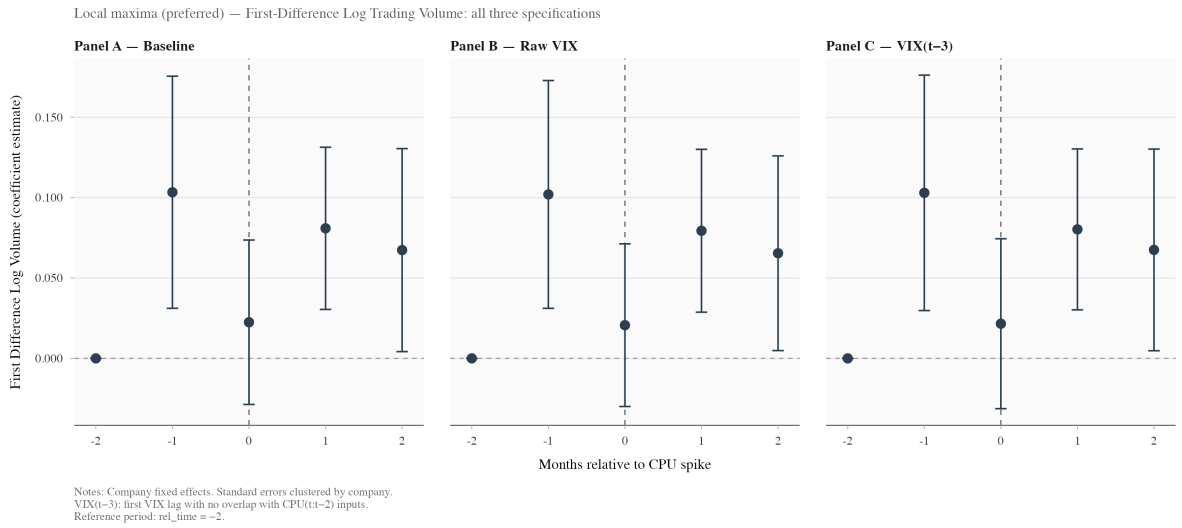


Figure 25: First-difference log volume — Local maxima: all three specifications (5% level of significance).

Rolling Threshold Looking at the results in Figure 26 and Table 32, it can be seen that only the coefficients at (t = 1) are significant at the 1% level for all three models. The impact of the CPU shock at (t = 1) consists of an increase in trading activity of approximately 8.5 percentage points for all models.

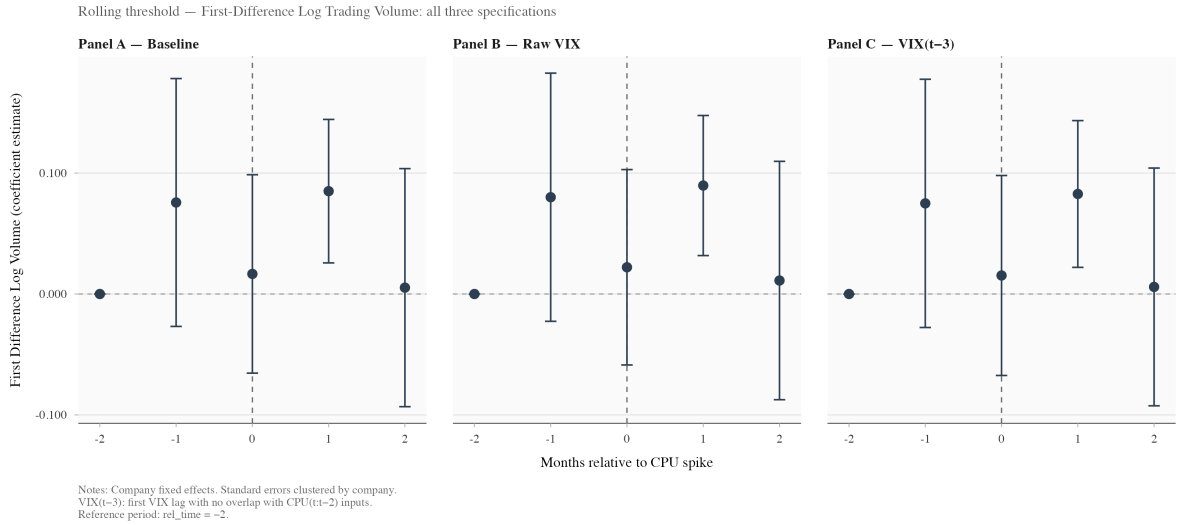


Figure 26: First-difference log volume — Rolling threshold: all three specifications (5% level of significance).

Overall Results and Discussion The study of the impact of first-difference trading activity reveals different results. It can be seen that the CPU shock has a positive impact on the first-difference trading volume, with an impact occurring before the shock, which suggests that traders anticipated the shock before it happened. Before making models of the trading activity, the hypothesis was that the CPU had a positive impact on trading activity. Examining the results, this hypothesis was found to be true for the first difference in trading volume, but false for abnormal trading volume. As mentioned by (Basaglia et al., 2025), higher CPU levels lead to higher stock price volatility and lower returns for firms in CO intensive sectors. Consequently, a CPU shock may lead to an increase in trading activity to hedge against potential losses due to the shock. This is what we observe with the first difference in trading volume, with an increase in trading activity before and after the shock. This could correspond to traders anticipating the shock in order to hedge and limit potential losses, or to further arranging their portfolios after the CPU shock in order to optimize them and/or best hedge against the shock's impact. However, the results for abnormal trading volume are different, and the hypothesis does not hold true. Results based on abnormal trading volume show the difference in trading activity compared to the yearly mean. The negative coefficients show that, after the impact of the CPU shock, the trading volume is lower than the yearly mean.

7 Conclusion

This study examines the impact of Climate Policy Uncertainty (CPU) on three dimensions of financial market activity : stock returns, realized volatility, and trading volume for the 30 firms composing the Dow Jones Industrial Average over the period 2017–2025. By constructing a text-based CPU index based on the semantic misalignment between campaign and legislative texts using GloVe embeddings and cosine similarity, and by implementing stacked event-study regressions, this paper provides a firm-level analysis of how climate policy uncertainty affects U.S. financial markets. The empirical analysis yields three main findings. First, across the three different panels and with the two different spike detection methods, arithmetic returns increase in the month preceding the spike. This is consistent with investors demanding a higher risk premium in anticipation of the future uncertainty shock, before declining significantly at the spike itself. This decline is supported by Vo et al. (2025). Second, realized volatility falls significantly and persistently at and after the CPU spike. This outcome is consistent with the wait-and-see mechanism (Bloom, 2009; Bernanke, 1983): when policy uncertainty rises, investors reduce trading activity rather than rebalancing their portfolios, mechanically compressing the dispersion of daily returns. Third, the two trading volume measures (abnormal trading volume and first log difference of trading volume) reveal complementary dynamics. On one hand, first log difference of trading volume increases significantly in the month preceding and following the spike. This result could be explained by the portfolio rebalancing mechanism (Basaglia et al., 2025): a higher CPU index leads to an increase in trading volume as investors hedge against potential losses. On the other hand, abnormal trading volume decreases at and after the spike, indicating that aggregate activity falls below its last 12-month benchmark. Although those two results seem paradoxical, we note that these two measures represent different time horizons. The first-difference measure identifies month-to-month changes, while abnormal volume compares levels against a 12-month rolling benchmark. Therefore, we could interpret these results as follows: CPU spikes provokes an increase in trading as some investors rebalance their portfolios, while overall trading activity remains below its previous 12-month period. However, this study has limitations, for the CPU index we observe that during months with unusually high legislative activity, the measured increase in text dissimilarity may partly reflect the larger number of documents rather than a genuine rise in climate policy uncertainty. In addition, we have some concerns about the parallel-trends assumption since the pre-trend coefficients are significant for some outcome variables. Moreover, our sample is restricted to the 30 firms of the Dow Jones Industrial Average. This limits the external validity of the findings, particularly for smaller firms potentially exposed differently to climate policy uncertainty.

8 AI Acknowledgment

- **Coding :** We use CLAUDE Code to help us program.
- **Research assistance :** Help with finding relevant references
- **Writing assistance :** Help with the clarity and structure of written sections
- **README Organisation :** We use AI tools in order to help us organise the README.txt file associated to each R script.

9 Appendix

9.1 Introduction

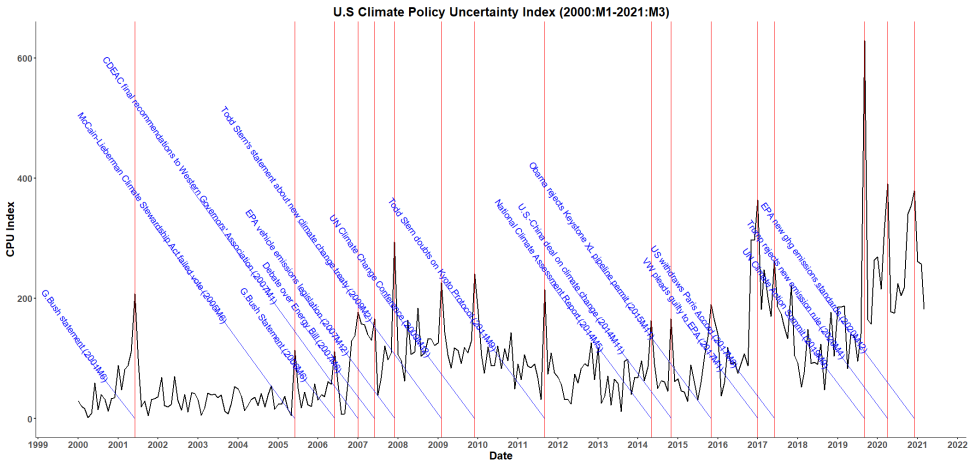


Figure 27: Climate Policy Uncertainty Events.

Source: www.policyuncertainty.com

9.2 Campaign Materials Documents

Below find the different chunks of query for recovering the climate related Campaign materials documents. **Chunk 1:** "Climate Change" "climate change" "Global Warming" "global warming" "Climate Crisis" "climate crisis" "Climate Disaster" "climate disaster" "Sea Level Rise" "sea level rise"
Chunk 2: "Greenhouse Gas" "greenhouse gas" "Carbon Emission" "carbon emission" "Carbon Tax" "carbon tax" "Carbon Price" "carbon price" "Carbon Market" "carbon market"
Chunk 3: "Renewable Energy" "renewable energy" "Clean Energy" "clean energy" "Renewable Power" "renewable power" "Clean Power" "clean power" "Energy Transition" "energy transition"
Chunk 4: "Energy Efficiency" "energy efficiency" "Fuel Economy" "fuel economy" "Fuel Efficiency" "fuel efficiency" "Fossil Fuel" "fossil fuel" "Renewables" "renewables"
Chunk 5: "Climate Policy" "climate policy" "Climate Legislation" "climate legislation" "Climate Bill" "climate bill" "Energy Policy" "energy policy" "Energy Legislation" "energy legislation"
Chunk 6: "Clean Air Act" "clean air act" "Clean Power Plan" "clean power plan" "Climate Change Bill" "climate change bill" "Fuel Economy Standard" "fuel economy standard"
Chunk 7: "Environmental Protection Agency" "environmental protection agency" "Energy Department" "energy department" "EPA" "epa" "DOE" "doe"
Chunk 8: "Cap and Trade" "cap and trade" "Emission Trading" "emission trading" "Emission Permit" "emission permit" "Emission Standard" "emission standard"
Chunk 9: "GHG" "ghg" "CO2" "co2" "Methane" "methane" "Decarbonization" "decarbonization" "Carbon Trading" "carbon trading"
Chunk 10: "Waxman Markey" "waxman markey" "Kerry Lieberman" "kerry lieberman" "Clear Skies Act" "clear skies act" "Climate Stewardship Act" "climate stewardship act"
Chunk 11: "Capture and Sequestration" "capture and sequestration" "Climate and Energy" "climate and energy" "Sea Level Rise" "sea level rise" "Energy Efficiency Program" "energy efficiency program"
Chunk 12: "Climate Transition" "climate transition" "Climate Mitigation" "climate mitigation" "Climate Adaptation" "climate adaptation"
Chunk 13: "Climate Accord" "climate accord" "Climate Conference" "climate conference" "Climate Activist" "climate activist"
Chunk 14: "Climate Demonstration" "climate demonstration" "Energy Committee" "energy committee" "Climate Rule" "climate rule"
Chunk 15: "Carbon" "carbon" "Emission" "emission" "Environmental" "environmental"

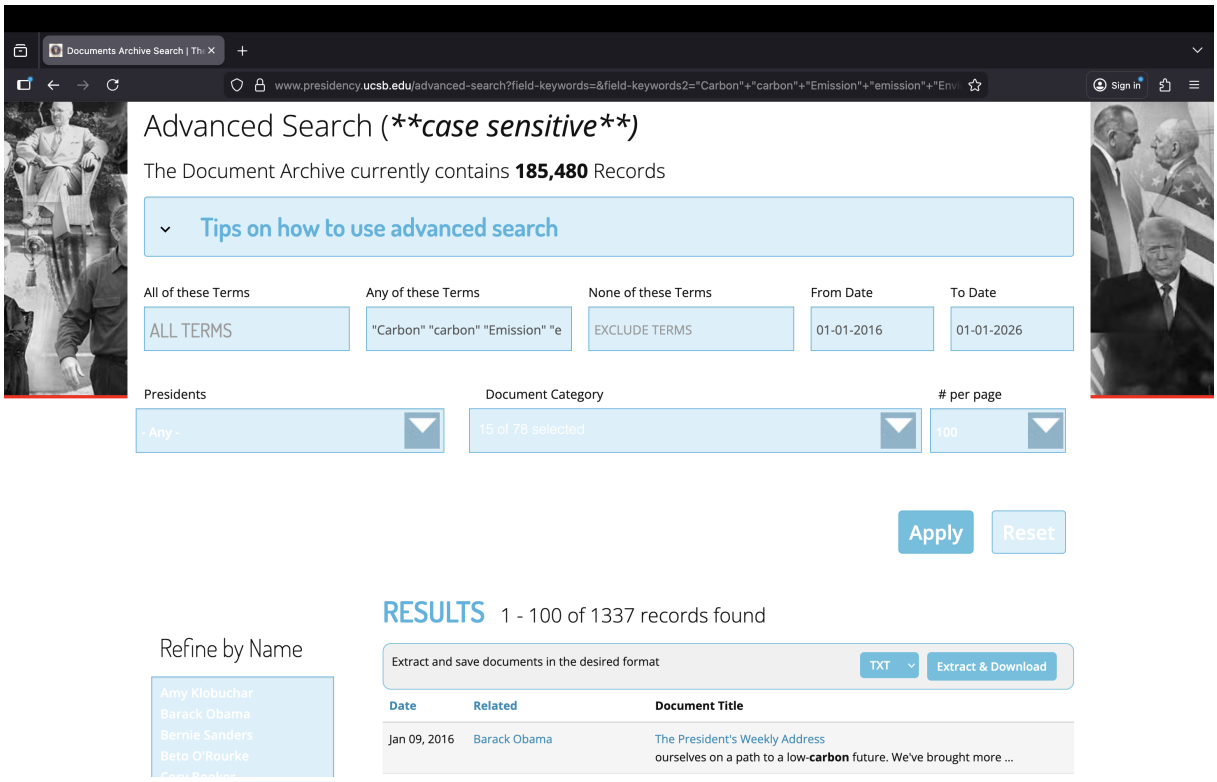


Figure 28: Advanced Research Tool for American Presidency Project for XML downloading



BARACK OBAMA

44th President of the United States: 2009 - 2017

The President's Weekly Address

January 09, 2016

Hi, everybody. Seven years ago, the American auto industry was on the brink of collapse. Plants were closing. Hundreds of thousands of workers were getting laid off from jobs that had been their ticket to a middle class life. And as the pain spread across the country, another 1 million Americans would have lost their jobs in the middle of the worst economic crisis of our lifetimes.

Some said it was too late to turn things around. But I refused to turn my back on so many of the workers that I had met. Instead, I placed my bet on American workers. I placed my bet on American manufacturing. In exchange for help, we demanded responsibility. We said the auto industry would have to truly change, not just pretend that it did. We got labor and management to settle their differences. We got the industry to retool and restructure. Everyone had some skin in the game.

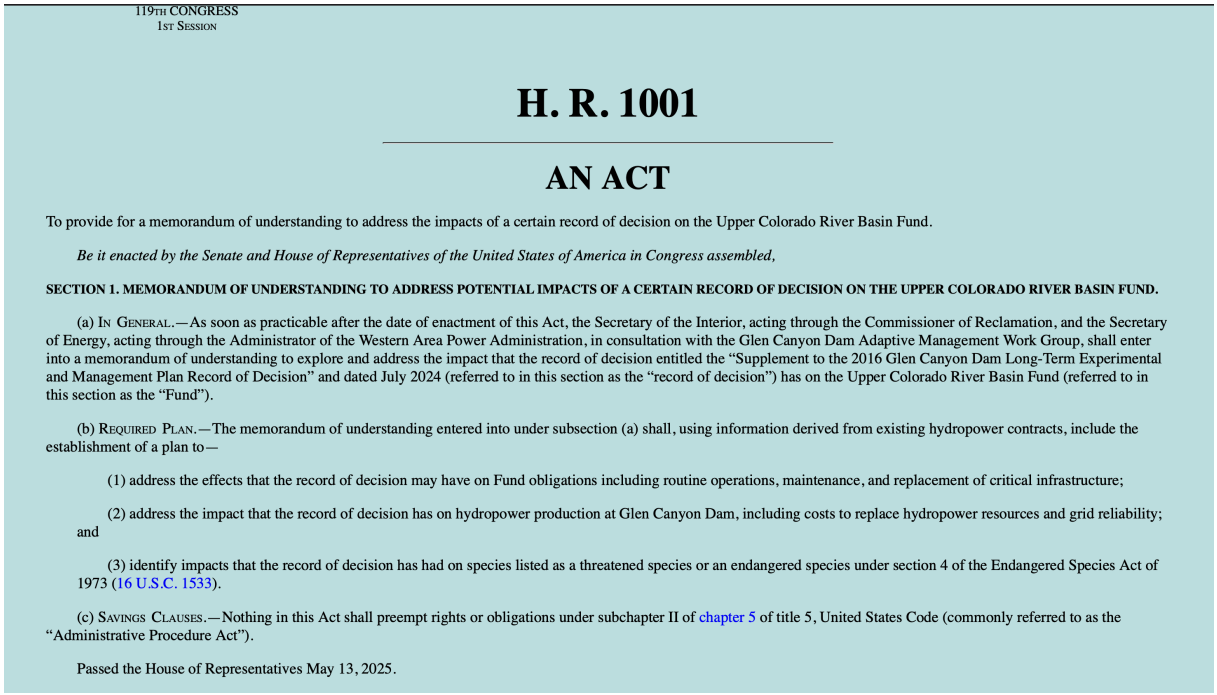
Our plan wasn't popular. Critics said it was a "road to socialism," or a "disaster" waiting to happen. But I'd make that bet again any day of the week. Because today, the American auto industry is back. Since our plan went into effect, our automakers have added more than 640,000 new jobs. We've cut the Detroit-area unemployment rate by more than half. The Big Three automakers are raising wages. Seven years ago, auto sales hit a 27-year low. Last year, they hit an alltime high. Later this month, I'll visit the Detroit Auto Show to see this progress firsthand. Because I believe that every American should be proud of what our most iconic industry has done.

It's not unlike what America overall has done these past 7 years. Our businesses are now on a 70-month streak of job creation, with more than 14 million new jobs in all. We've revamped our schools and the way we pay for college. We've made historic investments in clean energy and put ourselves on a path to a low-carbon future.

Figure 29: Example of Campaign Material Text

9.3 Legislative texts

Figure 30: Example of Legislative Text



119TH CONGRESS
1ST SESSION

H. R. 1001

AN ACT

To provide for a memorandum of understanding to address the impacts of a certain record of decision on the Upper Colorado River Basin Fund.

Be it enacted by the Senate and House of Representatives of the United States of America in Congress assembled,

SECTION 1. MEMORANDUM OF UNDERSTANDING TO ADDRESS POTENTIAL IMPACTS OF A CERTAIN RECORD OF DECISION ON THE UPPER COLORADO RIVER BASIN FUND.

(a) IN GENERAL.—As soon as practicable after the date of enactment of this Act, the Secretary of the Interior, acting through the Commissioner of Reclamation, and the Secretary of Energy, acting through the Administrator of the Western Area Power Administration, in consultation with the Glen Canyon Dam Adaptive Management Work Group, shall enter into a memorandum of understanding to explore and address the impact that the record of decision entitled the "Supplement to the 2016 Glen Canyon Dam Long-Term Experimental and Management Plan Record of Decision" and dated July 2024 (referred to in this section as the "record of decision") has on the Upper Colorado River Basin Fund (referred to in this section as the "Fund").

(b) REQUIRED PLAN.—The memorandum of understanding entered into under subsection (a) shall, using information derived from existing hydropower contracts, include the establishment of a plan to—

(1) address the effects that the record of decision may have on Fund obligations including routine operations, maintenance, and replacement of critical infrastructure;

(2) address the impact that the record of decision has on hydropower production at Glen Canyon Dam, including costs to replace hydropower resources and grid reliability; and

(3) identify impacts that the record of decision has had on species listed as a threatened species or an endangered species under section 4 of the Endangered Species Act of 1973 (16 U.S.C. 1533).

(c) SAVINGS CLAUSES.—Nothing in this Act shall preempt rights or obligations under subchapter II of chapter 5 of title 5, United States Code (commonly referred to as the "Administrative Procedure Act").

Passed the House of Representatives May 13, 2025.

9.4 Climate Dictionary Term Intensity at CPU Spike Months

This appendix documents the most frequently occurring climate dictionary terms at each CPU spike month identified by the local-maxima detection algorithm. For each spike, we report the top-20 terms by total occurrence across both corpora, normalized by total word count (occurrences per 1,000 words) to ensure comparability across months with different corpus sizes. Campaign materials (APP) and legislative texts (BILLS) are displayed side by side to highlight which source drives term salience at each episode. The final figure aggregates counts across all spike months to identify the terms that systematically dominate climate discourse at moments of heightened policy uncertainty. Bars show occurrences per 1,000 words of each climate dictionary term (Gavriilidis, 2023) in campaign materials (blue) and legislative texts (dark).

Figure 31: Top-20 Climate Dictionary Terms at CPU Spike: July 2017

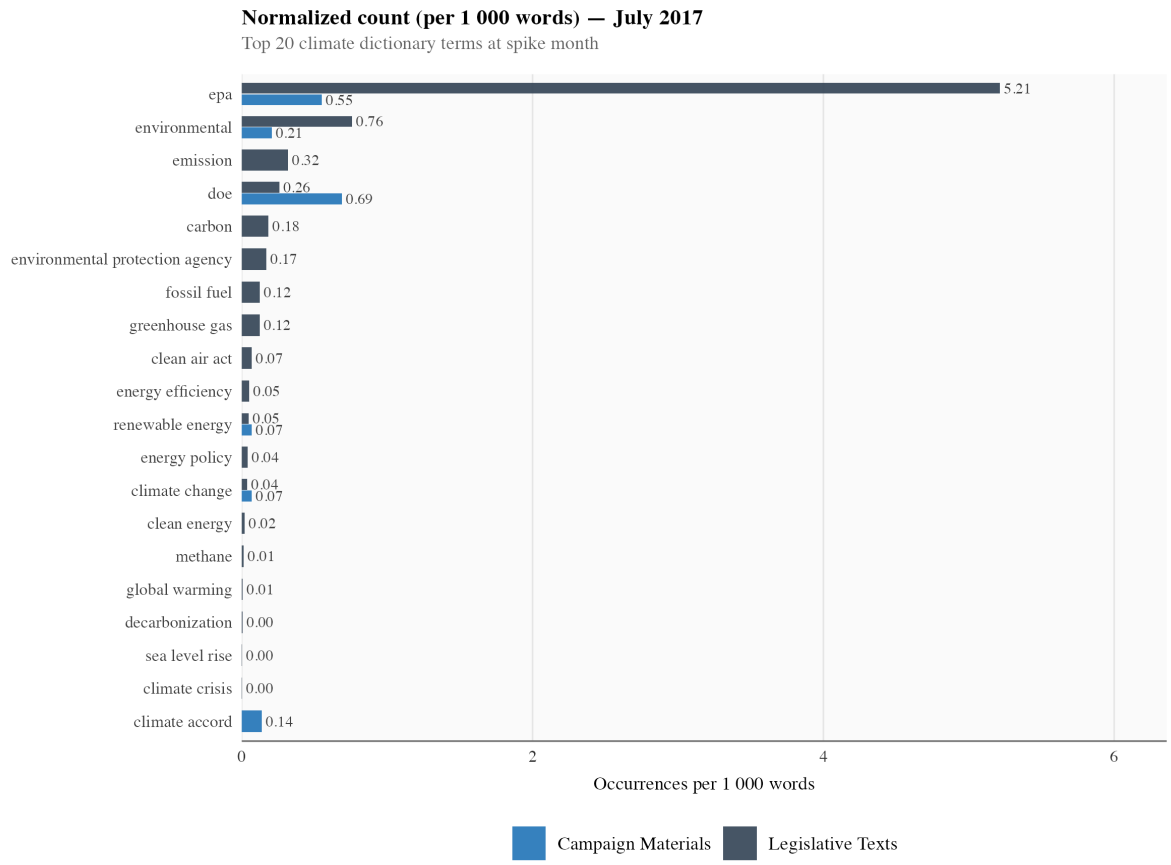


Figure 32: Top-20 Climate Dictionary Terms at CPU Spike: June 2018

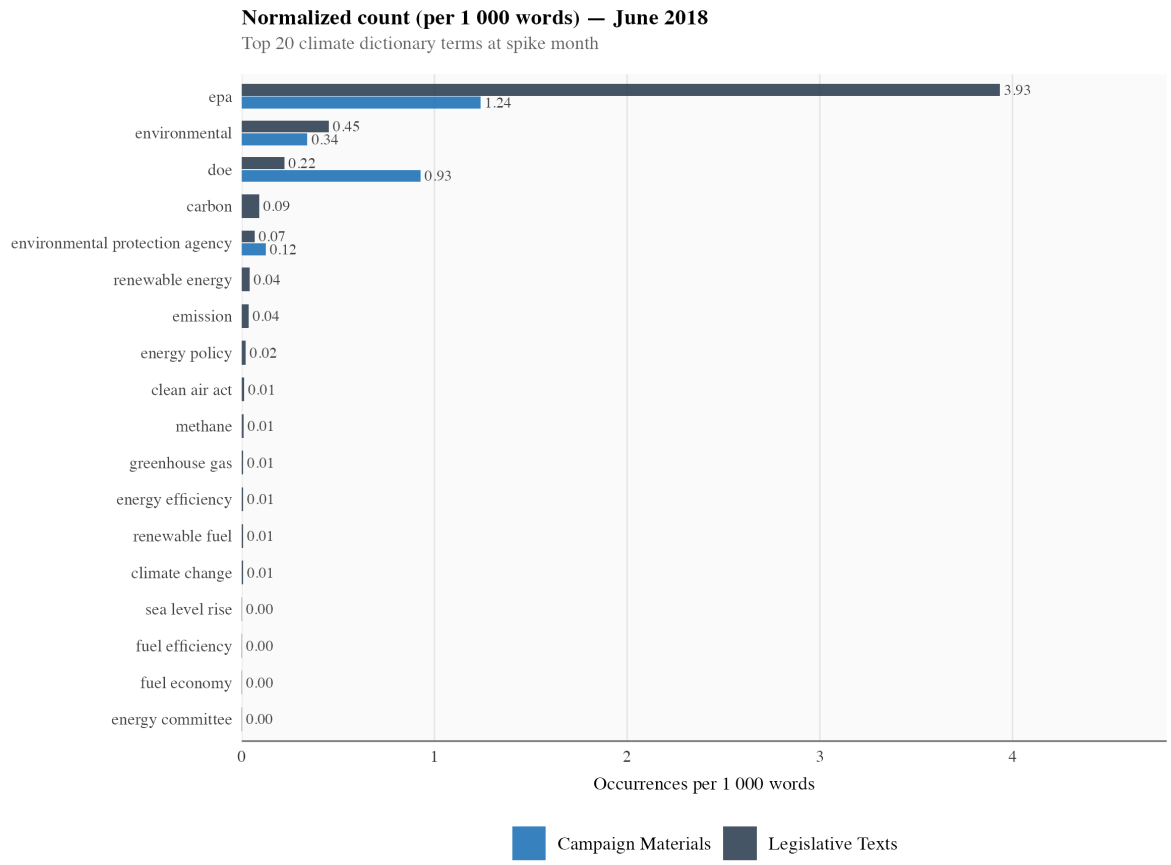


Figure 33: Top-20 Climate Dictionary Terms at CPU Spike: December 2020

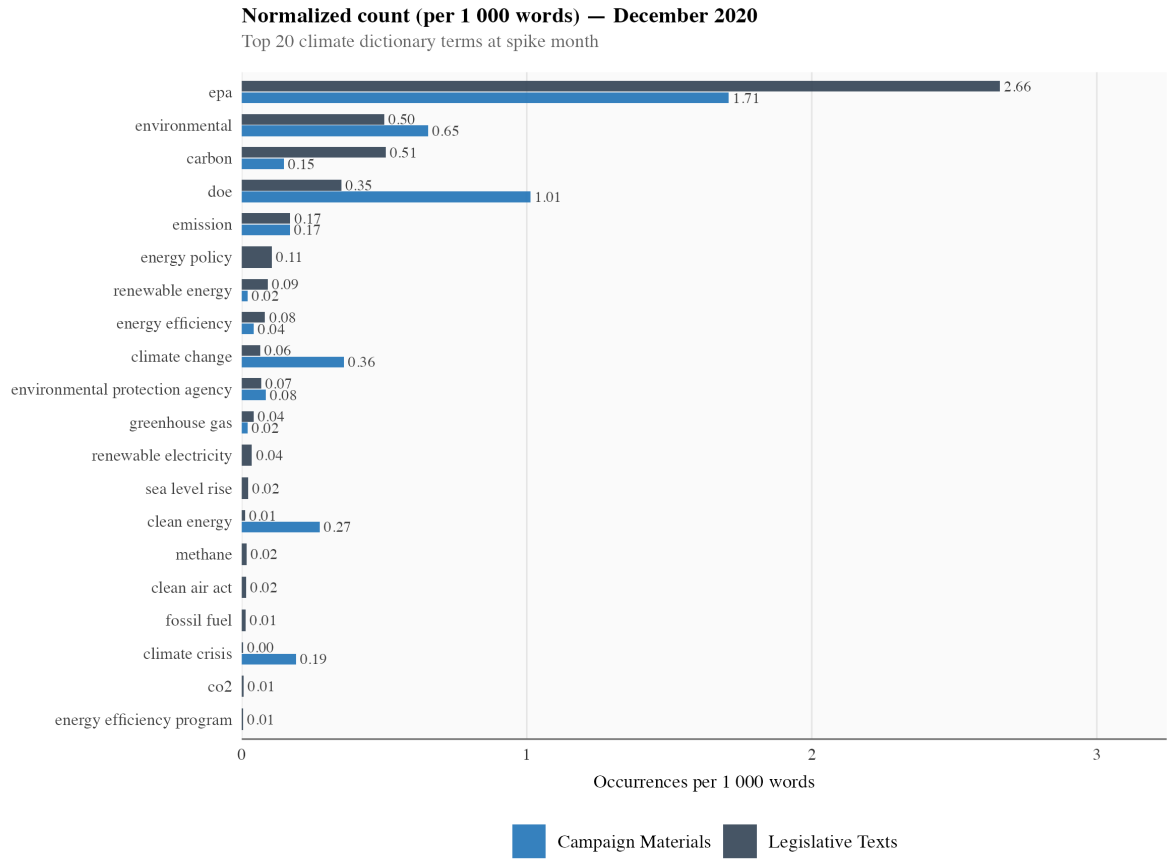


Figure 34: Top-20 Climate Dictionary Terms at CPU Spike: December 2022

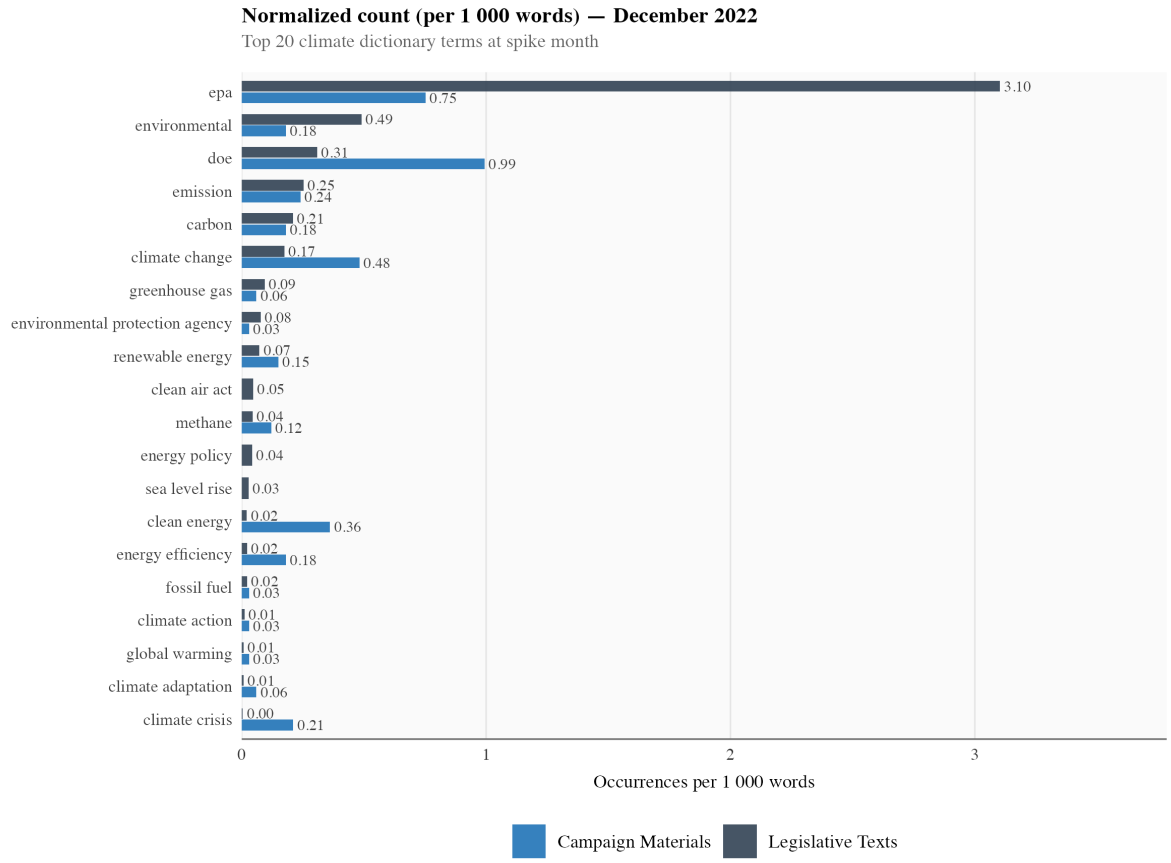


Figure 35: Top-20 Climate Dictionary Terms at CPU Spike: March 2024

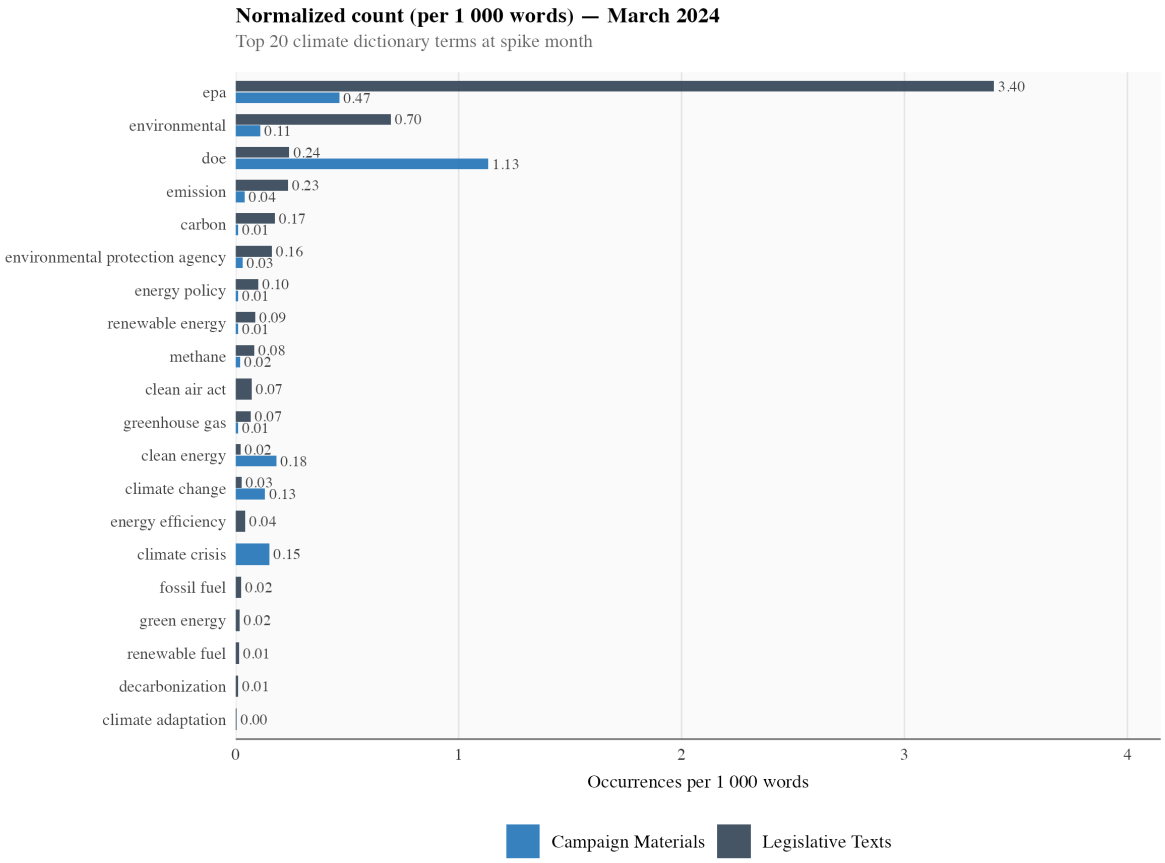


Figure 36: Top-20 Climate Dictionary Terms at CPU Spike: December 2024

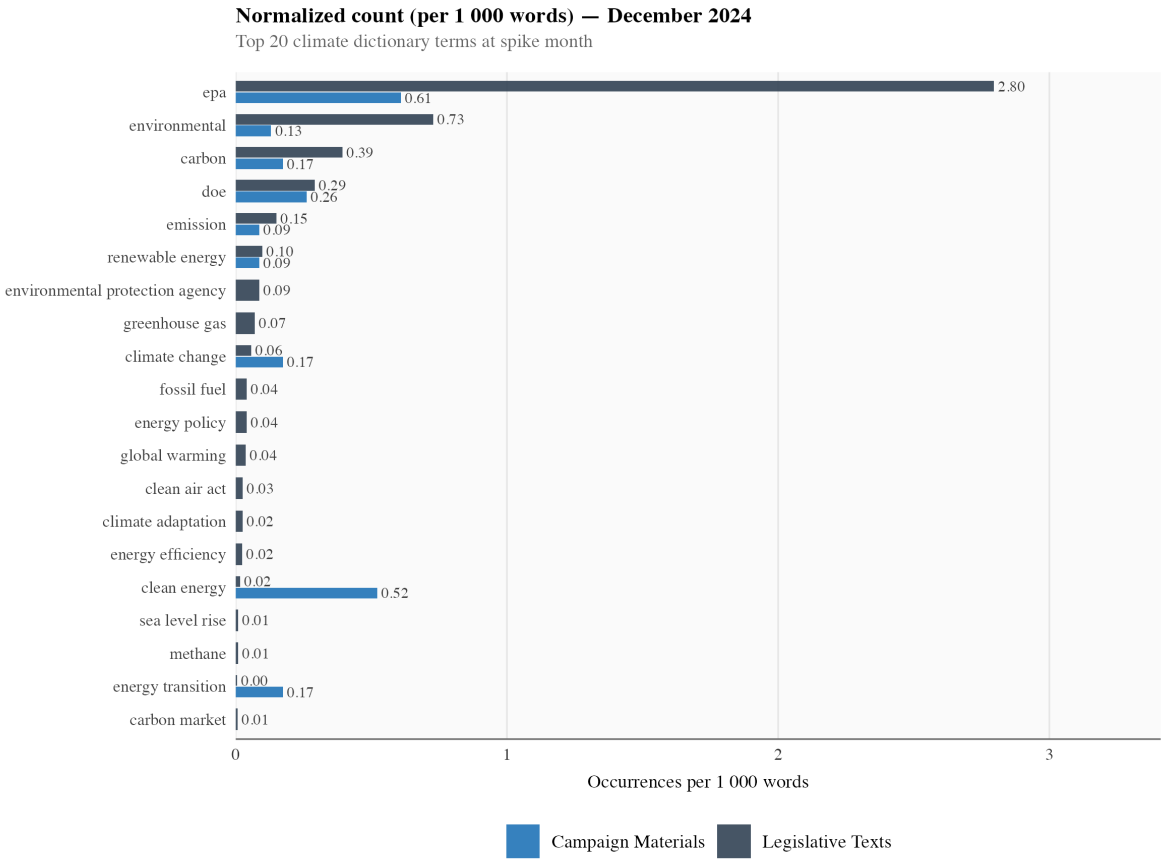
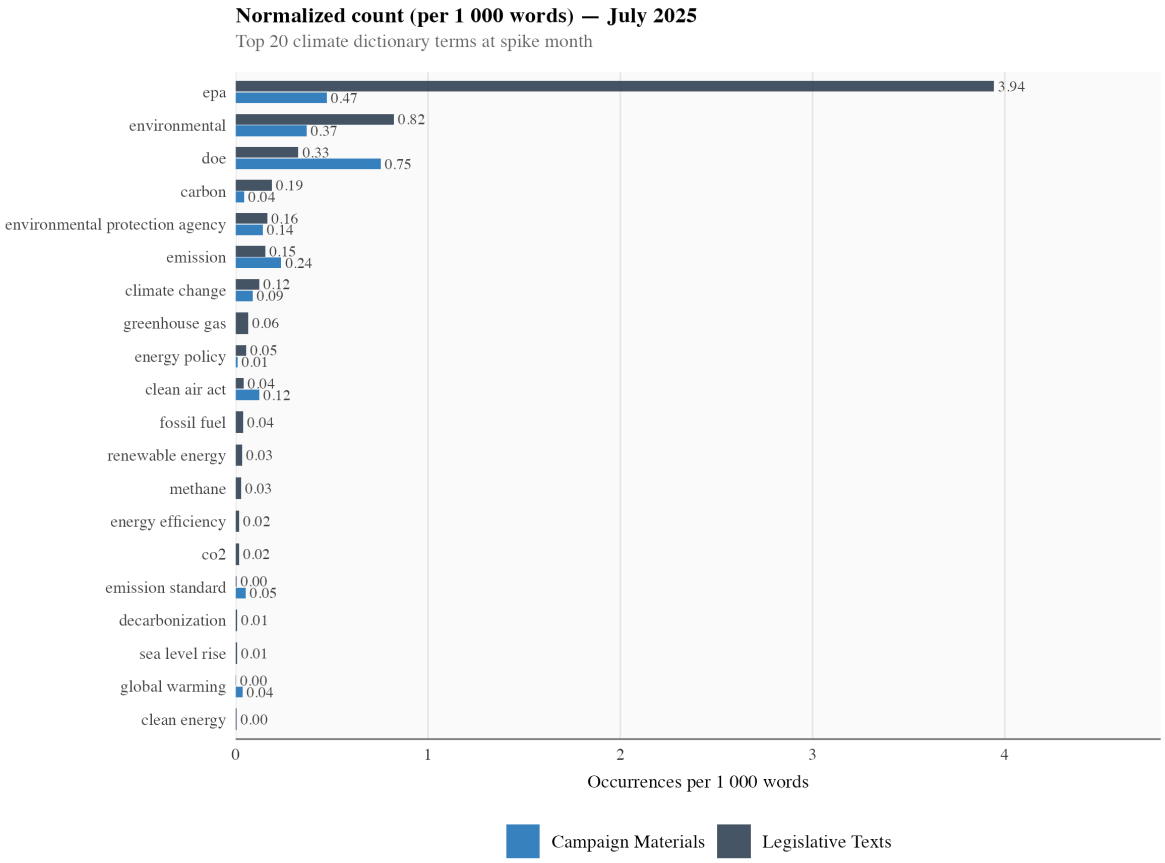


Figure 37: Top-20 Climate Dictionary Terms at CPU Spike: July 2025



Reading guide. Each subsection below covers one outcome variable. Within each subsection, results are organised by spike detection method (*Rolling threshold* then *Local maxima*). For each method, the combined table and figure summarise all three specifications side by side, followed by the three individual results. The reference period throughout is $t-2$ (two months prior to the spike), so all coefficients measure deviations from the pre-spike baseline. Standard errors are clustered by company.

Specifications.

- **Baseline:** $y_{it} = \sum_k \beta_k \mathbf{1}[\text{rel_time} = k] + \alpha_i + \varepsilon_{it}$, no market controls.
- **Raw VIX:** adds contemporaneous VIX as a control (shown to be a *bad control* due to within-month simultaneity with CPU; see Section 5.5).
- **VIX**$(t-3)$**:** adds VIX lagged three months, the first VIX realisation with no overlap with any input month of CPU$(t{:}t-2)$; fully predetermined by construction.

9.5 Arithmetic Returns

Arithmetic returns are defined as $r_{i,t} = \left(P_{i,t}^{\text{last}} - P_{i,t}^{\text{first}}\right) / P_{i,t}^{\text{first}}$, where $P_{i,t}$ is the VWAP and *first*/*last* refer to the first and last trading day of calendar month t.

9.5.1 Rolling Threshold Spike Detection

Table 8: Arithmetic returns — Rolling threshold: all three specifications. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Dependent Variable:	ar_returns		
	Baseline	Raw VIX	VIX(t-3)
Model:	(1)	(2)	(3)
Variables			
rel_time-1	0.0323***	0.0328***	0.0323***
	(0.0080)	(0.0091)	(0.0079)
rel_time0	-0.0289***	-0.0283***	-0.0288***
	(0.0073)	(0.0087)	(0.0073)
rel_time1	-0.0040	-0.0035	-0.0040
	(0.0063)	(0.0069)	(0.0067)
rel_time2	-0.0114*	-0.0108	-0.0114*
	(0.0065)	(0.0076)	(0.0065)
Fixed-effects			
company_name	Yes	Yes	Yes
Fit statistics			
Observations	750	750	750
R^2	0.12350	0.12355	0.12350
Within R^2	0.07523	0.07528	0.07523

Clustered (company_name) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 9: Arithmetic returns — Rolling threshold, Baseline.

Dependent Variable:	ar_returns
Model:	(1)
Variables	
rel_time-1	0.0323***
	(0.0080)
rel_time0	-0.0289***
	(0.0073)
rel_time1	-0.0040
	(0.0063)
rel_time2	-0.0114*
	(0.0065)
Fixed-effects	
company_name	Yes
Fit statistics	
Observations	750
R^2	0.12350
Within R^2	0.07523

Clustered (company_name) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

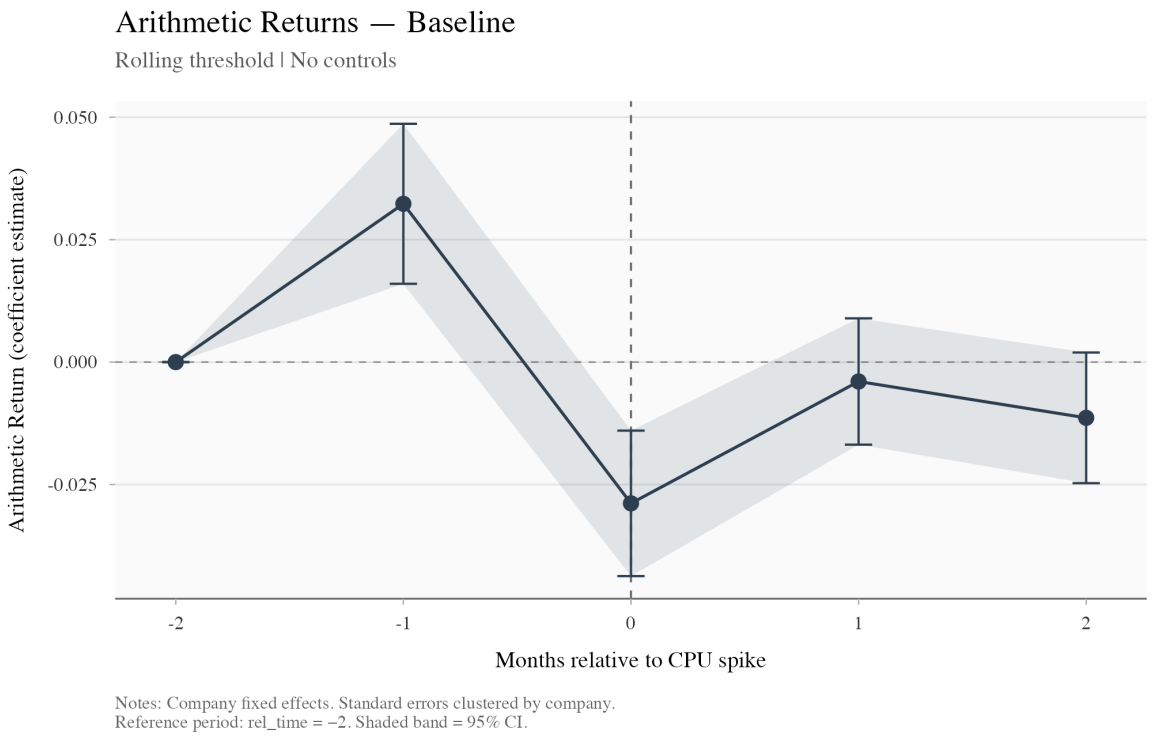


Figure 38: Arithmetic returns — Rolling threshold, Baseline.

Table 10: Arithmetic returns — Rolling threshold, Raw VIX.

Dependent Variable:	ar_returns
Model:	(1)
Variables	
rel_time-1	0.0328***
	(0.0091)
rel_time0	-0.0283***
	(0.0087)
rel_time1	-0.0035
	(0.0069)
rel_time2	-0.0108
	(0.0076)
Fixed-effects	
company_name	Yes
Fit statistics	
Observations	750
R^2	0.12355
Within R^2	0.07528

Clustered (company_name) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

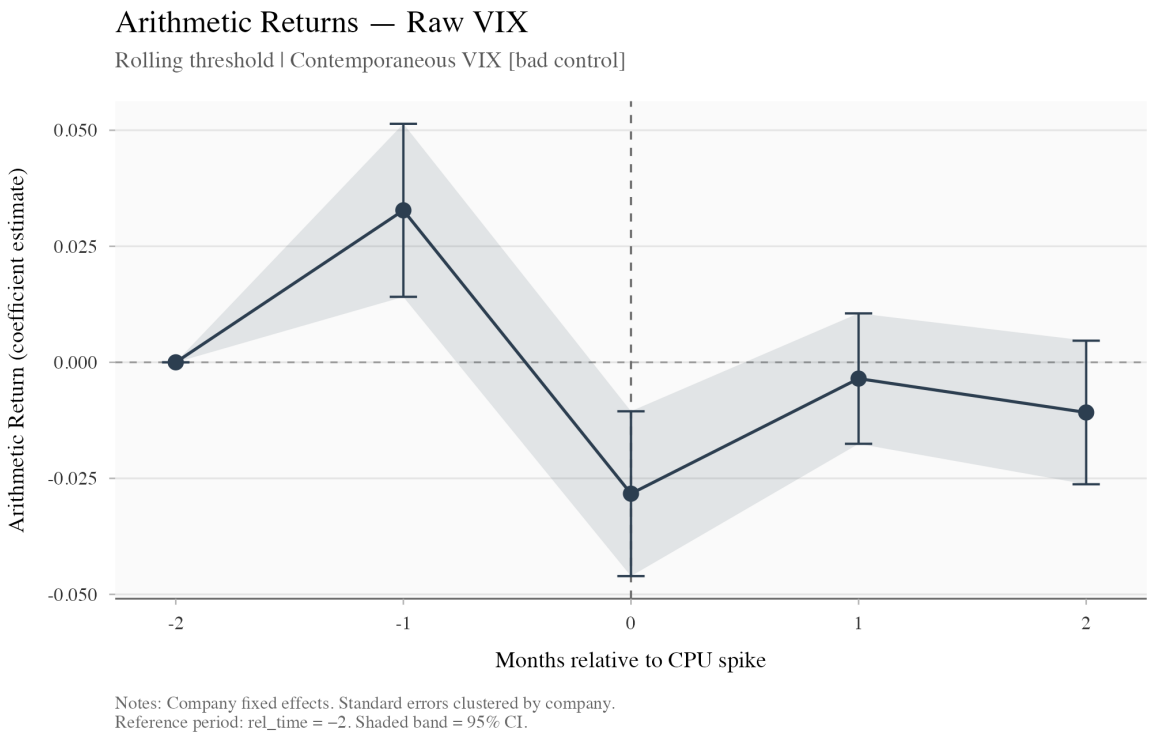


Figure 39: Arithmetic returns — Rolling threshold, Raw VIX.

Table 11: Arithmetic returns — Rolling threshold, VIX$(t-3)$.

Dependent Variable:	ar_returns
Model:	(1)
Variables	
rel_time-1	0.0323***
	(0.0079)
rel_time0	-0.0288***
	(0.0073)
rel_time1	-0.0040
	(0.0067)
rel_time2	-0.0114*
	(0.0065)
Fixed-effects	
company_name	Yes
Fit statistics	
Observations	750
R^2	0.12350
Within R^2	0.07523

Clustered (company_name) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

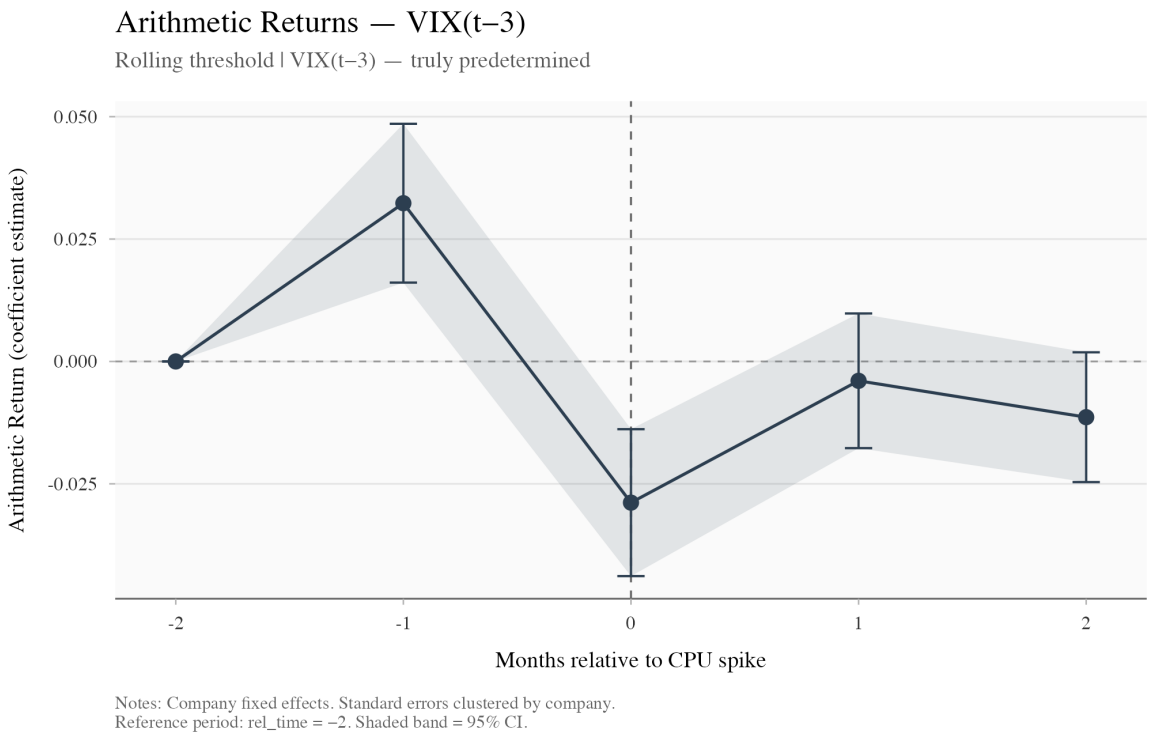


Figure 40: Arithmetic returns — Rolling threshold, VIX$(t-3)$.

9.5.2 *Local Maxima Spike Detection*

Table 12: Arithmetic returns — Local maxima: all three specifications. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Dependent Variable:	ar_returns		
Model:	Baseline (1)	Raw VIX (2)	VIX(t-3) (3)
Variables			
rel_time-1	0.0225***	0.0233***	0.0222***
	(0.0067)	(0.0076)	(0.0066)
rel_time0	-0.0259***	-0.0248***	-0.0264***
	(0.0059)	(0.0069)	(0.0061)
rel_time1	-0.0047	-0.0038	-0.0051
	(0.0067)	(0.0071)	(0.0070)
rel_time2	-0.0125**	-0.0113*	-0.0124**
	(0.0052)	(0.0060)	(0.0051)
Fixed-effects			
company_name	Yes	Yes	Yes
Fit statistics			
Observations	1,050	1,050	1,050
R^2	0.11391	0.11427	0.11407
Within R^2	0.05250	0.05288	0.05267

Clustered (company_name) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 13: Arithmetic returns — Local maxima, Baseline.

Dependent Variable:	ar_returns
Model:	(1)
Variables	
rel_time-1	0.0225***
	(0.0067)
rel_time0	-0.0259***
	(0.0059)
rel_time1	-0.0047
	(0.0067)
rel_time2	-0.0125**
	(0.0052)
Fixed-effects	
company_name	Yes
Fit statistics	
Observations	1,050
R^2	0.11391
Within R^2	0.05250

Clustered (company_name) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

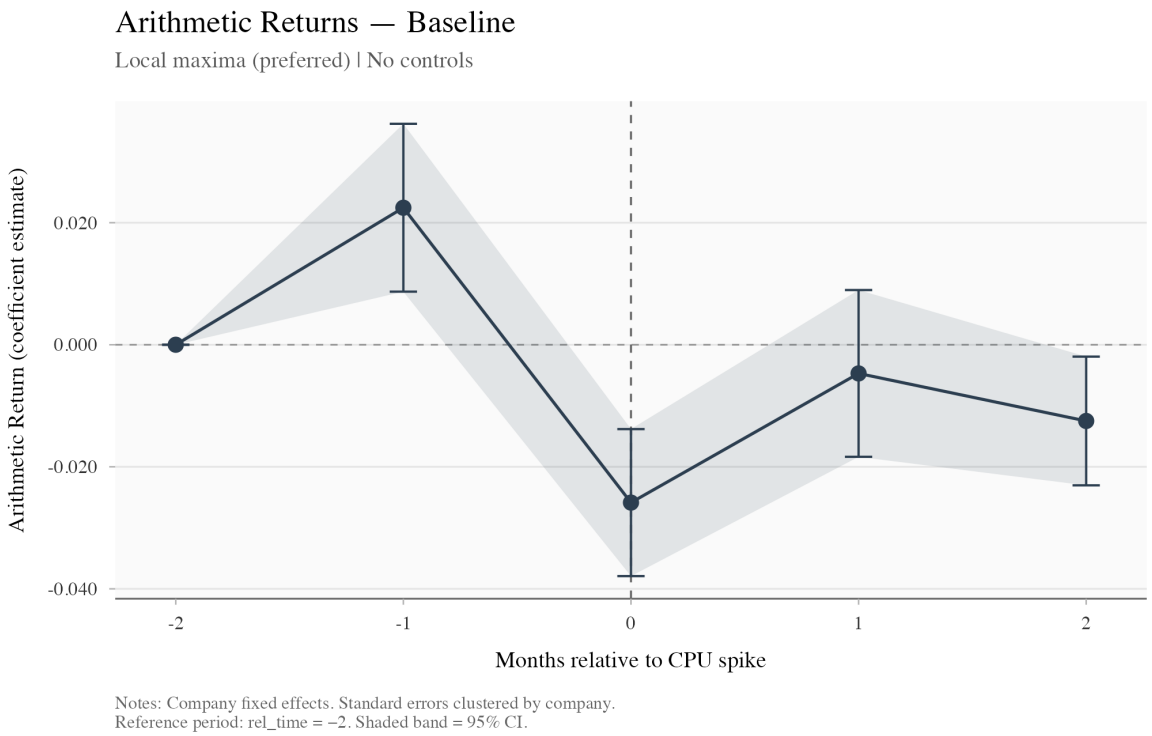


Figure 41: Arithmetic returns — Local maxima, Baseline.

Table 14: Arithmetic returns — Local maxima, Raw VIX.

Dependent Variable:	ar_returns
Model:	(1)
Variables	
rel_time-1	0.0233***
	(0.0076)
rel_time0	-0.0248***
	(0.0069)
rel_time1	-0.0038
	(0.0071)
rel_time2	-0.0113*
	(0.0060)
Fixed-effects	
company_name	Yes
Fit statistics	
Observations	1,050
R^2	0.11427
Within R^2	0.05288

Clustered (company_name) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

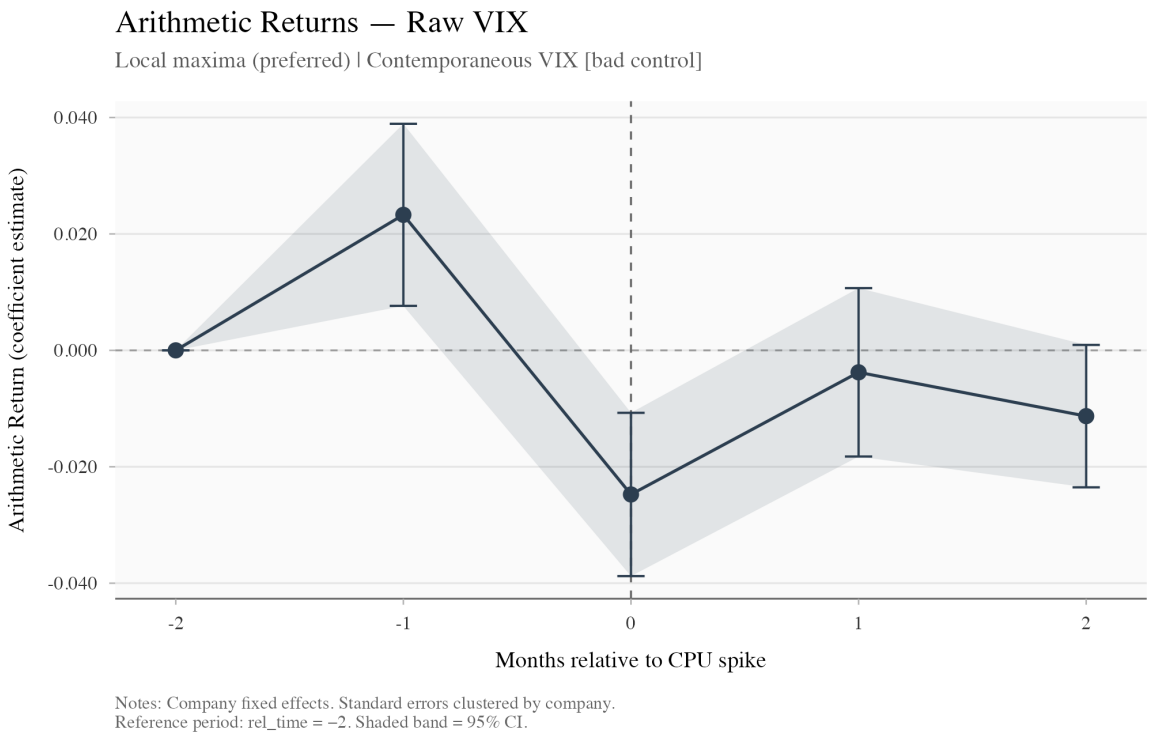


Figure 42: Arithmetic returns — Local maxima, Raw VIX.

Table 15: Arithmetic returns — Local maxima, VIX$(t-3)$.

Dependent Variable:	ar_returns
Model:	(1)
Variables	
rel_time-1	0.0222***
	(0.0066)
rel_time0	-0.0264***
	(0.0061)
rel_time1	-0.0051
	(0.0070)
rel_time2	-0.0124**
	(0.0051)
Fixed-effects	
company_name	Yes
Fit statistics	
Observations	1,050
R^2	0.11407
Within R^2	0.05267

Clustered (company_name) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

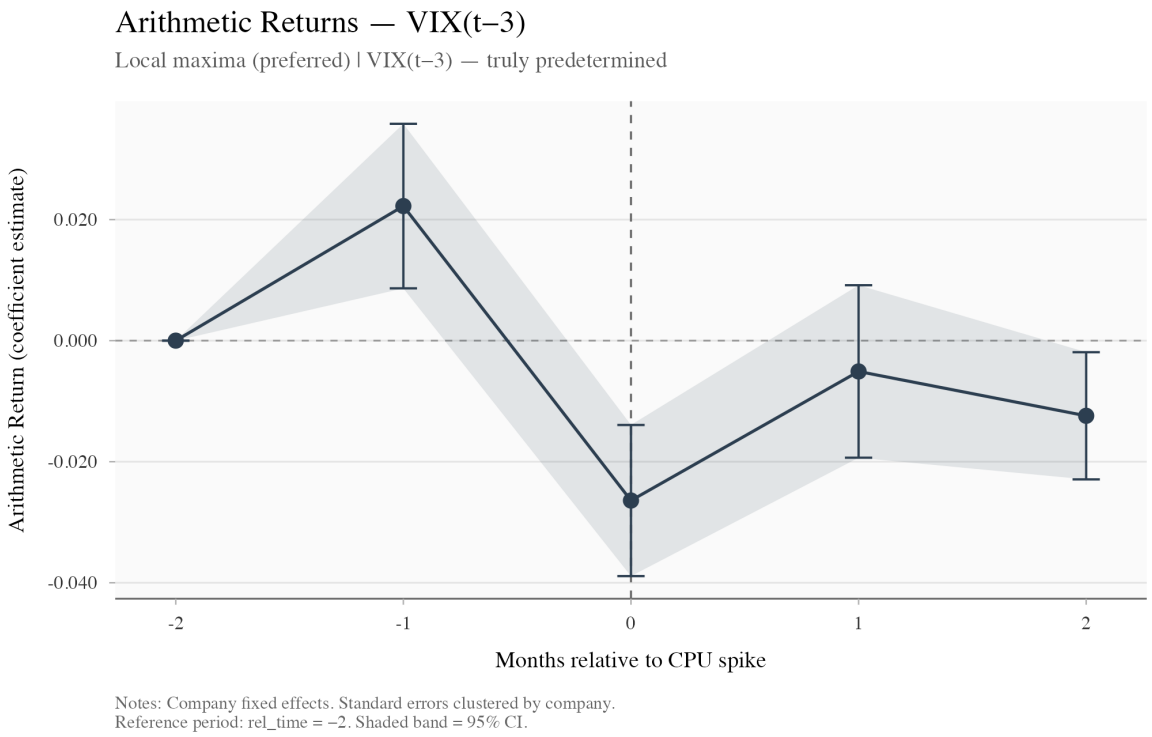


Figure 43: Arithmetic returns — Local maxima, VIX$(t-3)$.

9.6 Realized Volatility

Realized volatility is computed as the standard deviation of daily VWAP returns within each calendar month, scaled by $\sqrt{21}$. A negative coefficient indicates that realized volatility is lower in that event-window month relative to the pre-spike baseline at $t-2$.

9.6.1 Rolling Threshold Spike Detection

Table 16: Realized volatility — Rolling threshold: all three specifications. Columns (1)–(3): Baseline, Raw VIX, VIX$(t-3)$. Clustered (company) s.e. in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Dependent Variable:	realized_vol		
	Baseline	Raw VIX	VIX(t-3)
Model:	(1)	(2)	(3)
Variables			
rel_time-1	0.0048	0.0107**	0.0039
	(0.0035)	(0.0040)	(0.0035)
rel_time0	-0.0137***	-0.0061*	-0.0152***
	(0.0028)	(0.0031)	(0.0027)
rel_time1	-0.0027	0.0037*	-0.0054***
	(0.0017)	(0.0019)	(0.0018)
rel_time2	-0.0078**	0.0003	-0.0071**
	(0.0033)	(0.0034)	(0.0033)
Fixed-effects			
company_name	Yes	Yes	Yes
Fit statistics			
Observations	750	750	750
R^2	0.35073	0.41756	0.37948
Within R^2	0.06810	0.16402	0.10936

Clustered (company_name) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 17: Realized volatility — Rolling threshold, Baseline. No market controls.

Dependent Variable:	realized_vol
Model:	(1)
Variables	
rel_time-1	0.0048
	(0.0035)
rel_time0	-0.0137***
	(0.0028)
rel_time1	-0.0027
	(0.0017)
rel_time2	-0.0078**
	(0.0033)
Fixed-effects	
company_name	Yes
Fit statistics	
Observations	750
R^2	0.35073
Within R^2	0.06810

Clustered (company_name) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

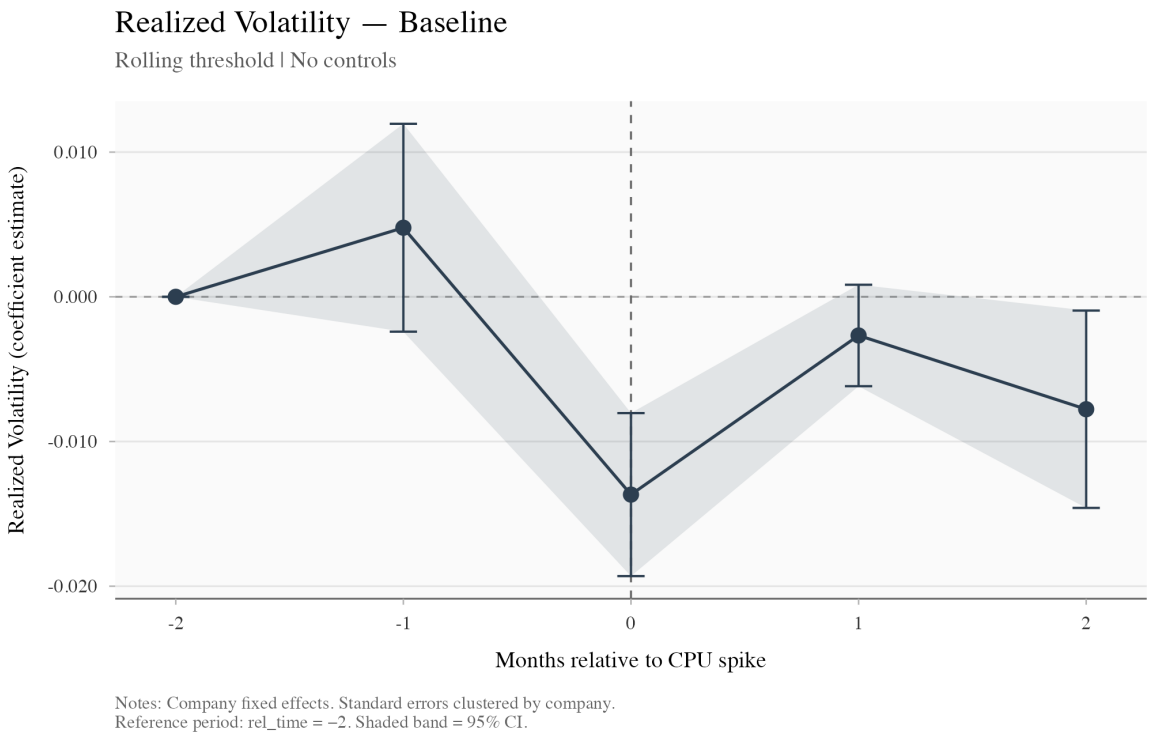


Figure 44: Realized volatility — Rolling threshold, Baseline.

Table 18: Realized volatility — Rolling threshold, Raw VIX. Within-month simultaneity with CPU makes this a bad control.

Dependent Variable:	realized_vol
Model:	(1)
Variables	
rel_time-1	0.0107**
	(0.0040)
rel_time0	-0.0061*
	(0.0031)
rel_time1	0.0037*
	(0.0019)
rel_time2	0.0003
	(0.0034)
Fixed-effects	
company_name	Yes
Fit statistics	
Observations	750
R^2	0.41756
Within R^2	0.16402

Clustered (company_name) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

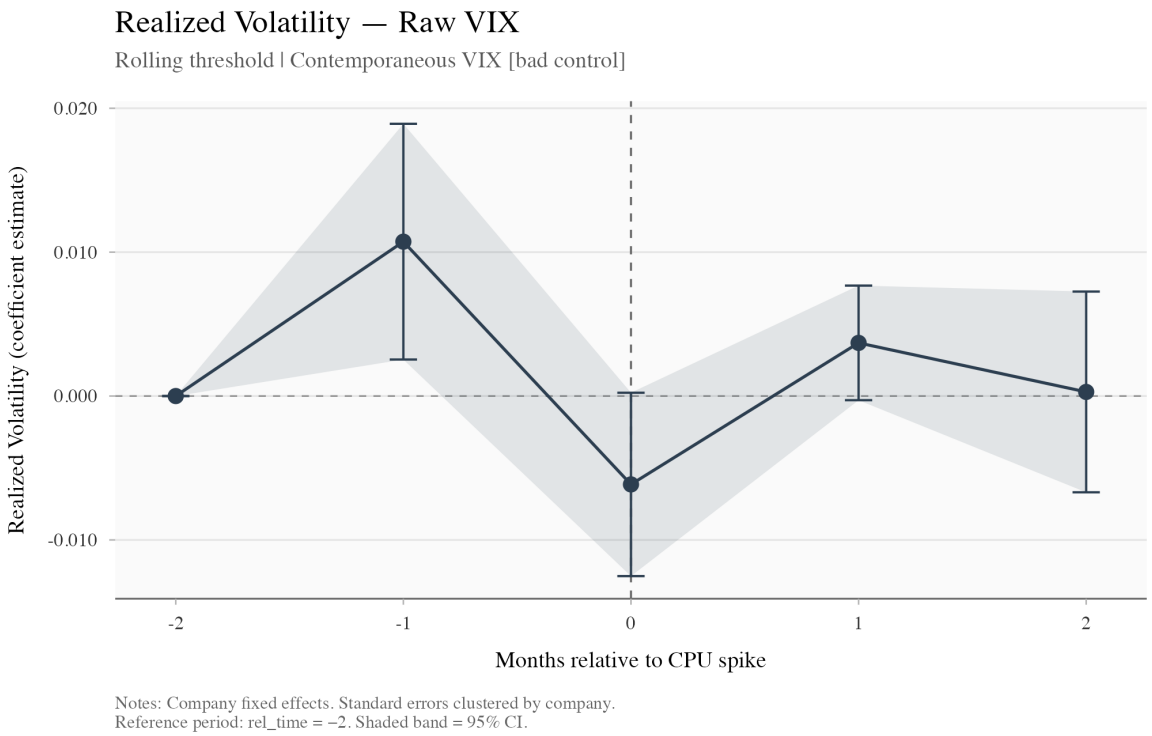


Figure 45: Realized volatility — Rolling threshold, Raw VIX.

Table 19: Realized volatility — Rolling threshold, VIX$(t-3)$. First lag with no overlap with any input month of CPU$(t{:}t-2)$.

Dependent Variable:	realized_vol
Model:	(1)
Variables	
rel_time-1	0.0039
	(0.0035)
rel_time0	-0.0152***
	(0.0027)
rel_time1	-0.0054***
	(0.0018)
rel_time2	-0.0071**
	(0.0033)
Fixed-effects	
company_name	Yes
Fit statistics	
Observations	750
R^2	0.37948
Within R^2	0.10936

Clustered (company_name) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

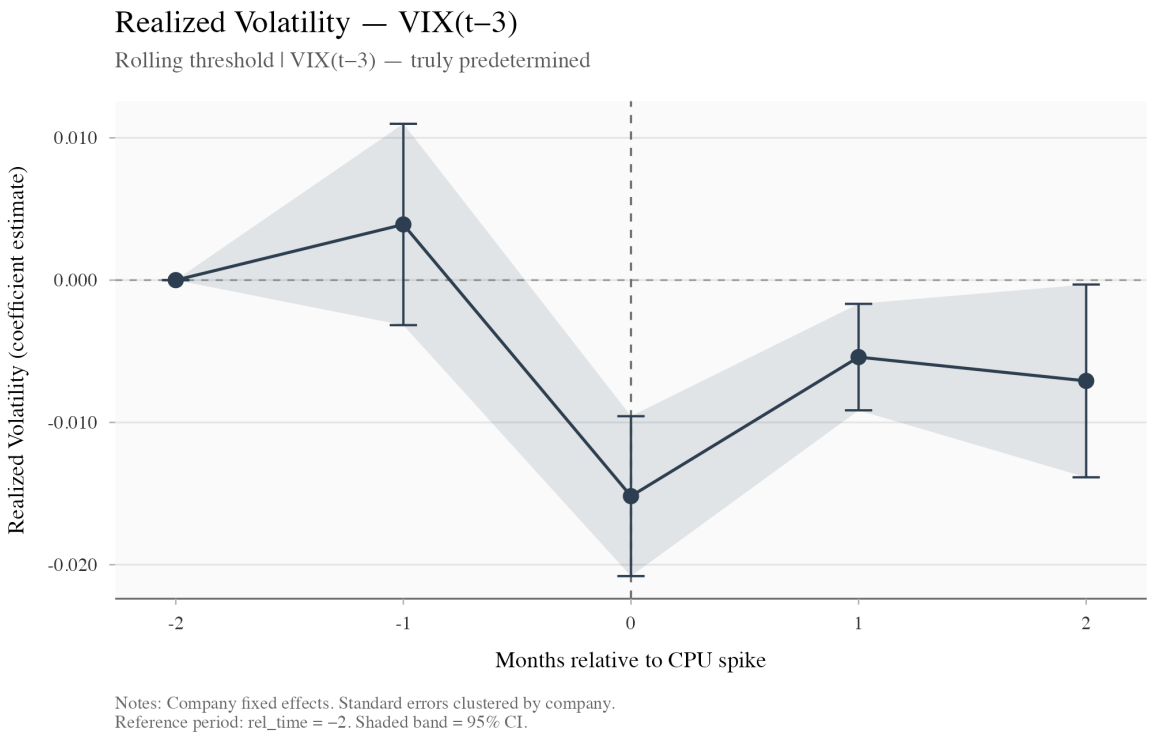


Figure 46: Realized volatility — Rolling threshold, VIX$(t-3)$.

Individual results — Baseline

9.6.2 *Local Maxima Spike Detection*

Table 20: Realized volatility — Local maxima: all three specifications. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Dependent Variable:	realized_vol		
	Baseline	Raw VIX	VIX(t-3)
Model:	(1)	(2)	(3)
Variables			
rel_time-1	0.0008	0.0066**	-0.0009
	(0.0027)	(0.0029)	(0.0027)
rel_time0	-0.0118***	-0.0041*	-0.0162***
	(0.0022)	(0.0023)	(0.0022)
rel_time1	-0.0044***	0.0021	-0.0075***
	(0.0012)	(0.0013)	(0.0013)
rel_time2	-0.0085***	-0.0002	-0.0080***
	(0.0024)	(0.0023)	(0.0024)
Fixed-effects			
company_name	Yes	Yes	Yes
Fit statistics			
Observations	1,050	1,050	1,050
R^2	0.29674	0.40214	0.35776
Within R^2	0.03864	0.18272	0.12205

Clustered (company_name) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 21: Realized volatility — Local maxima, Baseline.

Dependent Variable:	realized_vol
Model:	(1)
Variables	
rel_time-1	0.0008
	(0.0027)
rel_time0	-0.0118***
	(0.0022)
rel_time1	-0.0044***
	(0.0012)
rel_time2	-0.0085***
	(0.0024)
Fixed-effects	
company_name	Yes
Fit statistics	
Observations	1,050
R^2	0.29674
Within R^2	0.03864

Clustered (company_name) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

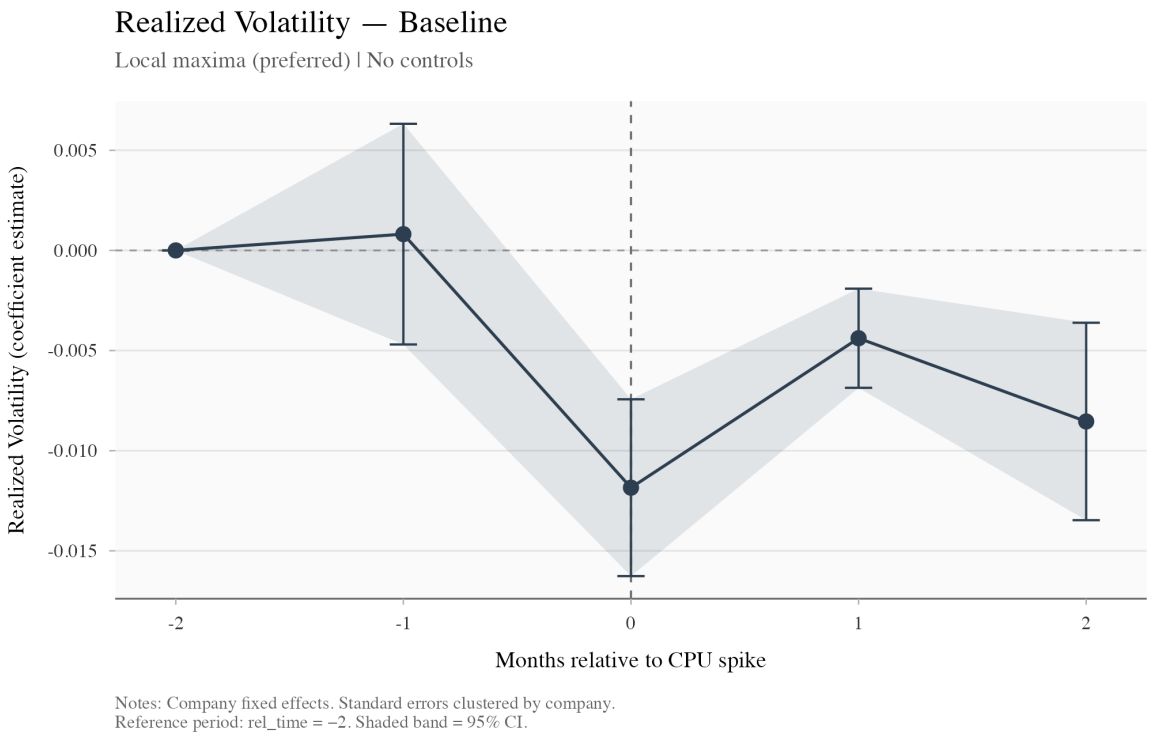


Figure 47: Realized volatility — Local maxima, Baseline.

Table 22: Realized volatility — Local maxima, Raw VIX.

Dependent Variable:	realized_vol
Model:	(1)
Variables	
rel_time-1	0.0066**
	(0.0029)
rel_time0	-0.0041*
	(0.0023)
rel_time1	0.0021
	(0.0013)
rel_time2	-0.0002
	(0.0023)
Fixed-effects	
company_name	Yes
Fit statistics	
Observations	1,050
R^2	0.40214
Within R^2	0.18272

Clustered (company_name) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

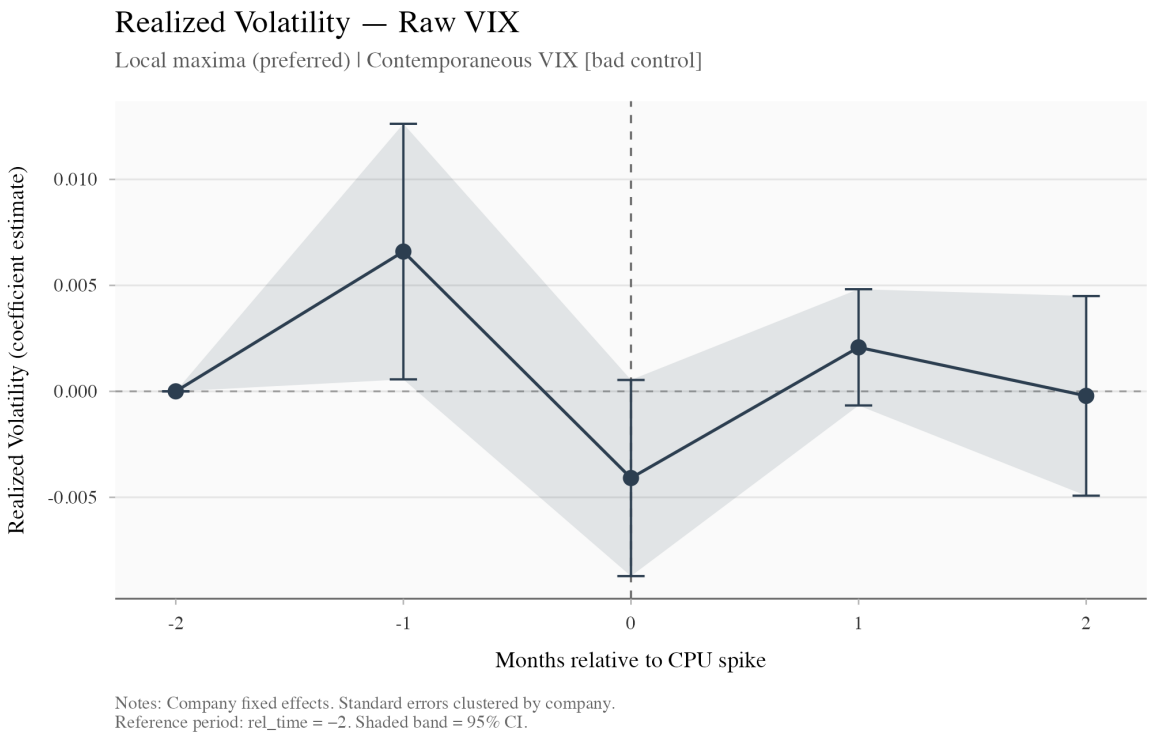


Figure 48: Realized volatility — Local maxima, Raw VIX.

Table 23: Realized volatility — Local maxima, VIX$(t-3)$.

Dependent Variable:	realized_vol
Model:	(1)
Variables	
rel_time-1	-0.0009
	(0.0027)
rel_time0	-0.0162***
	(0.0022)
rel_time1	-0.0075***
	(0.0013)
rel_time2	-0.0080***
	(0.0024)
Fixed-effects	
company_name	Yes
Fit statistics	
Observations	1,050
R^2	0.35776
Within R^2	0.12205

Clustered (company_name) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

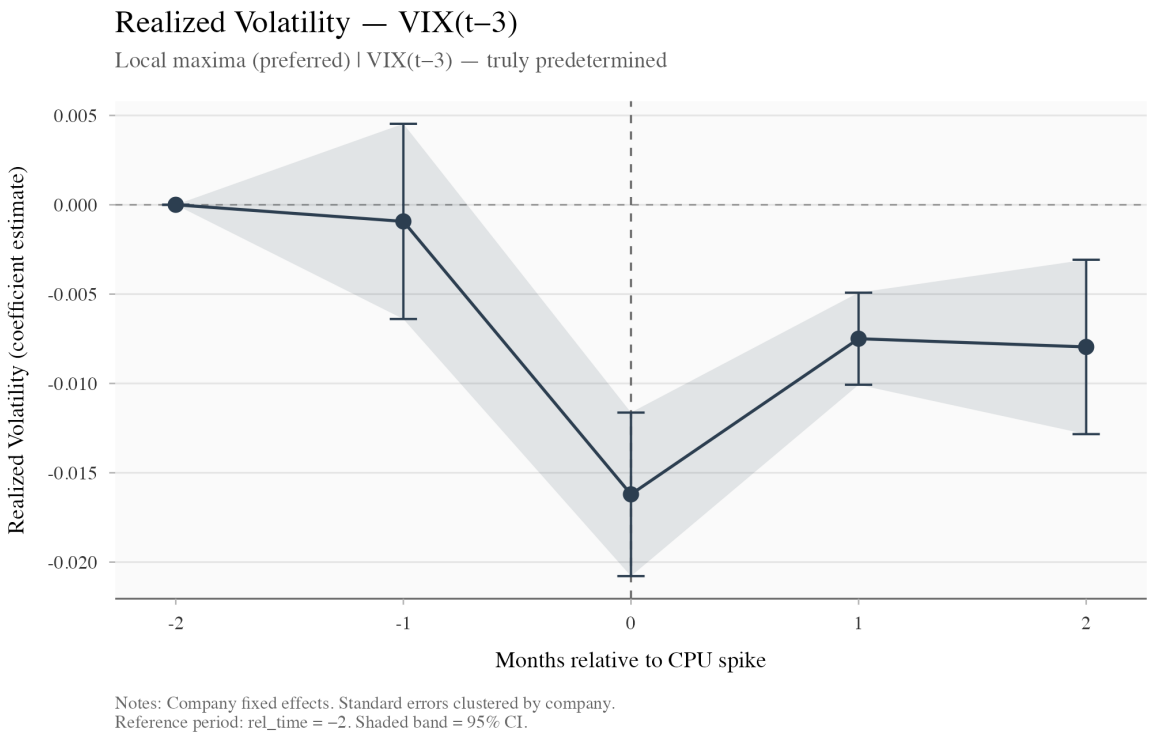


Figure 49: Realized volatility — Local maxima, VIX$(t-3)$.

9.7 Abnormal Trading Volume

Abnormal trading volume is defined as $\text{AbnormalVol}_{i,t} = \log V_{i,t} - \overline{\log V}_{i,t-12:t-1}$, where $V_{i,t}$ is mean daily volume in month t and $\overline{\log V}_{i,t-12:t-1}$ is the 12-month rolling mean of lagged log volume (see Section 5.3).

9.7.1 Rolling Threshold Spike Detection

Table 24: Abnormal trading volume — Rolling threshold: all three specifications. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Dependent Variable:	abnormal_vol		
Model:	Baseline (1)	Raw VIX (2)	VIX(t-3) (3)
Variables			
rel_time-1	0.0122	-0.0157	0.0191
	(0.0294)	(0.0316)	(0.0290)
rel_time0	-0.0363	-0.0716*	-0.0242
	(0.0316)	(0.0355)	(0.0309)
rel_time1	-0.0088	-0.0387	0.0131
	(0.0267)	(0.0312)	(0.0246)
rel_time2	-0.0614	-0.0991**	-0.0669*
	(0.0367)	(0.0420)	(0.0373)
Fixed-effects			
company_name	Yes	Yes	Yes
Fit statistics			
Observations	750	750	750
R^2	0.09791	0.12498	0.13196
Within R^2	0.01664	0.04616	0.05376

Clustered (company_name) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 25: Abnormal trading volume — Rolling threshold, Baseline.

Dependent Variable:	abnormal_vol
Model:	(1)
Variables	
rel_time-1	0.0122
	(0.0294)
rel_time0	-0.0363
	(0.0316)
rel_time1	-0.0088
	(0.0267)
rel_time2	-0.0614
	(0.0367)
Fixed-effects	
company_name	Yes
Fit statistics	
Observations	750
R^2	0.09791
Within R^2	0.01664

Clustered (company_name) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

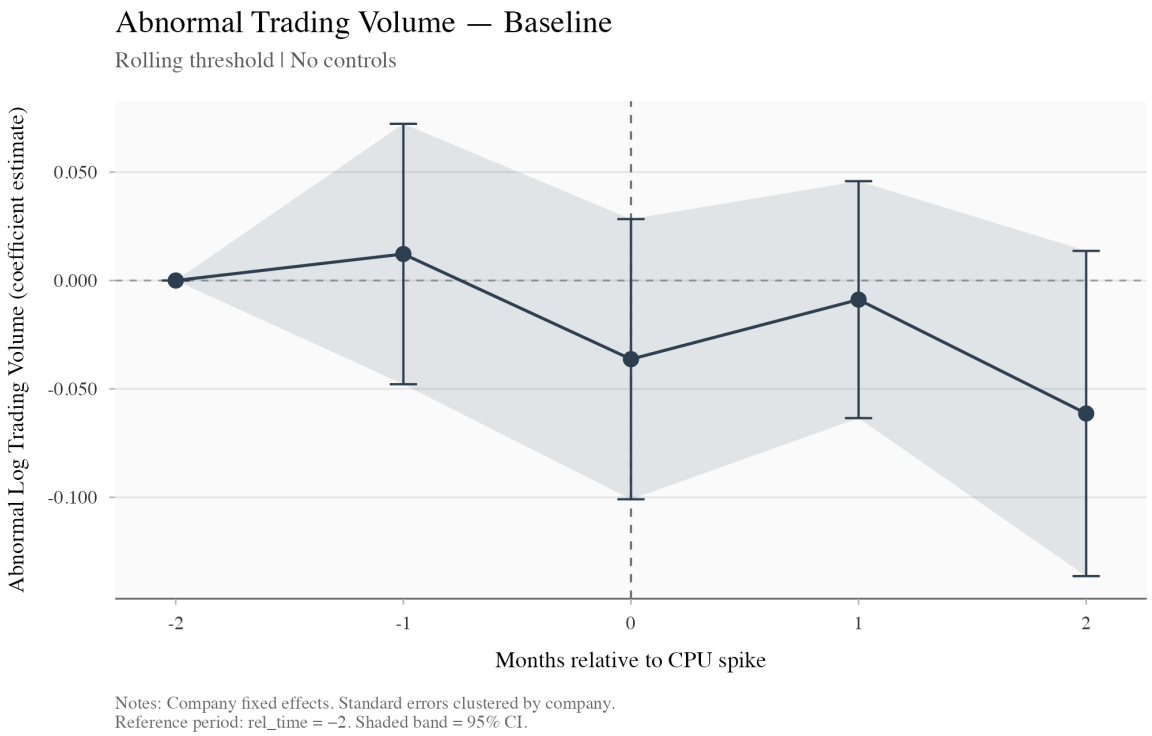


Figure 50: Abnormal trading volume — Rolling threshold, Baseline.

Table 26: Abnormal trading volume — Rolling threshold, Raw VIX.

Dependent Variable:	abnormal_vol
Model:	(1)
Variables	
rel_time-1	-0.0157
	(0.0316)
rel_time0	-0.0716*
	(0.0355)
rel_time1	-0.0387
	(0.0312)
rel_time2	-0.0991**
	(0.0420)
Fixed-effects	
company_name	Yes
Fit statistics	
Observations	750
R^2	0.12498
Within R^2	0.04616

Clustered (company_name) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

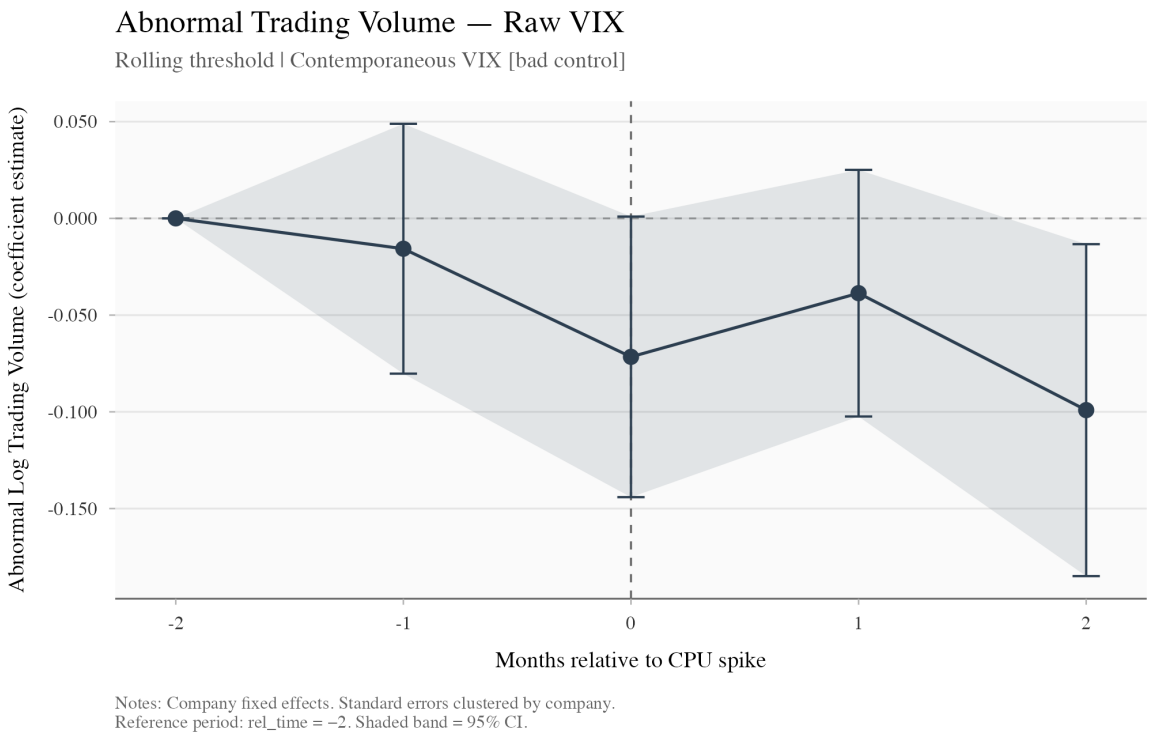


Figure 51: Abnormal trading volume — Rolling threshold, Raw VIX.

Table 27: Abnormal trading volume — Rolling threshold, VIX$(t-3)$.

Dependent Variable:	abnormal_vol
Model:	(1)
Variables	
rel_time-1	0.0191
	(0.0290)
rel_time0	-0.0242
	(0.0309)
rel_time1	0.0131
	(0.0246)
rel_time2	-0.0669*
	(0.0373)
Fixed-effects	
company_name	Yes
Fit statistics	
Observations	750
R^2	0.13196
Within R^2	0.05376

Clustered (company_name) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

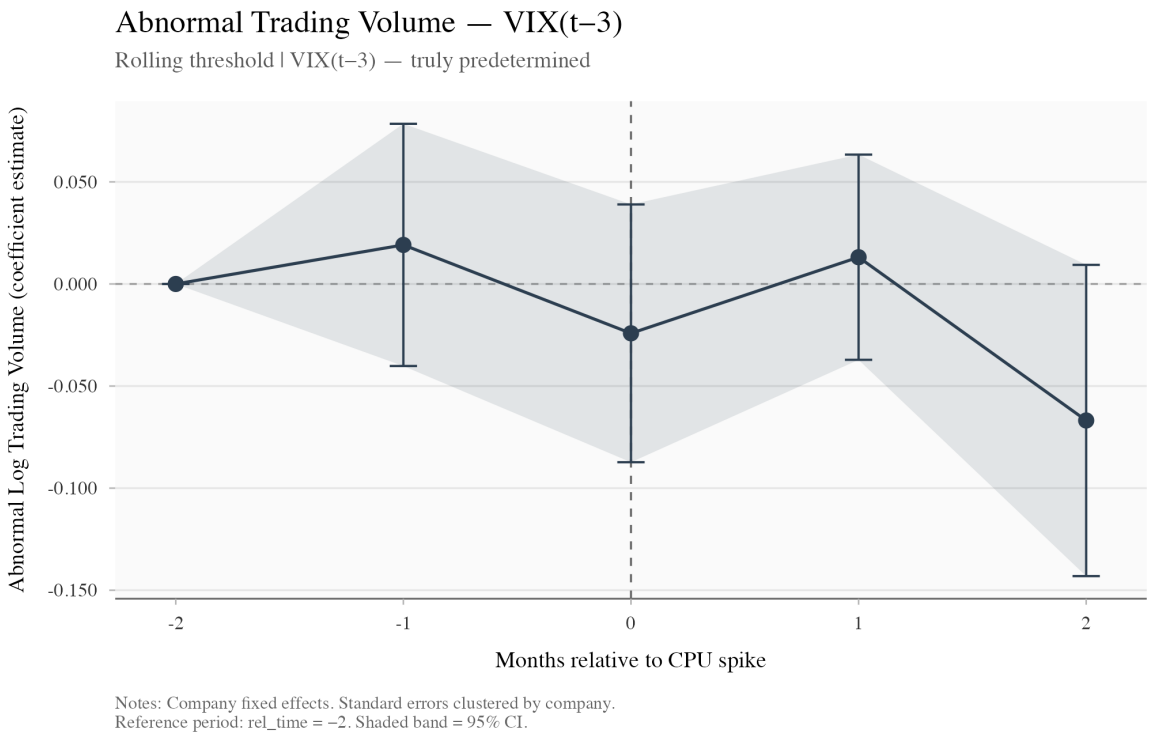


Figure 52: Abnormal trading volume — Rolling threshold, VIX$(t-3)$.

9.7.2 *Local Maxima Spike Detection*

Table 28: Abnormal trading volume — Local maxima: all three specifications. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Dependent Variable:	abnormal_vol		
	Baseline	Raw VIX	VIX(t-3)
Model:	(1)	(2)	(3)
Variables			
rel_time-1	0.0211	0.0117	0.0242
	(0.0223)	(0.0233)	(0.0224)
rel_time0	-0.0390**	-0.0517**	-0.0314*
	(0.0187)	(0.0222)	(0.0175)
rel_time1	-0.0355	-0.0461*	-0.0301
	(0.0215)	(0.0247)	(0.0196)
rel_time2	-0.0464*	-0.0600**	-0.0475*
	(0.0264)	(0.0293)	(0.0266)
Fixed-effects			
company_name	Yes	Yes	Yes
Fit statistics			
Observations	1,050	1,050	1,050
R^2	0.07789	0.08254	0.08102
Within R^2	0.01436	0.01933	0.01770

Clustered (company_name) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 29: Abnormal trading volume — Local maxima, Baseline.

Dependent Variable:	abnormal_vol
Model:	(1)
Variables	
rel_time-1	0.0211
	(0.0223)
rel_time0	-0.0390**
	(0.0187)
rel_time1	-0.0355
	(0.0215)
rel_time2	-0.0464*
	(0.0264)
Fixed-effects	
company_name	Yes
Fit statistics	
Observations	1,050
R^2	0.07789
Within R^2	0.01436

Clustered (company_name) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

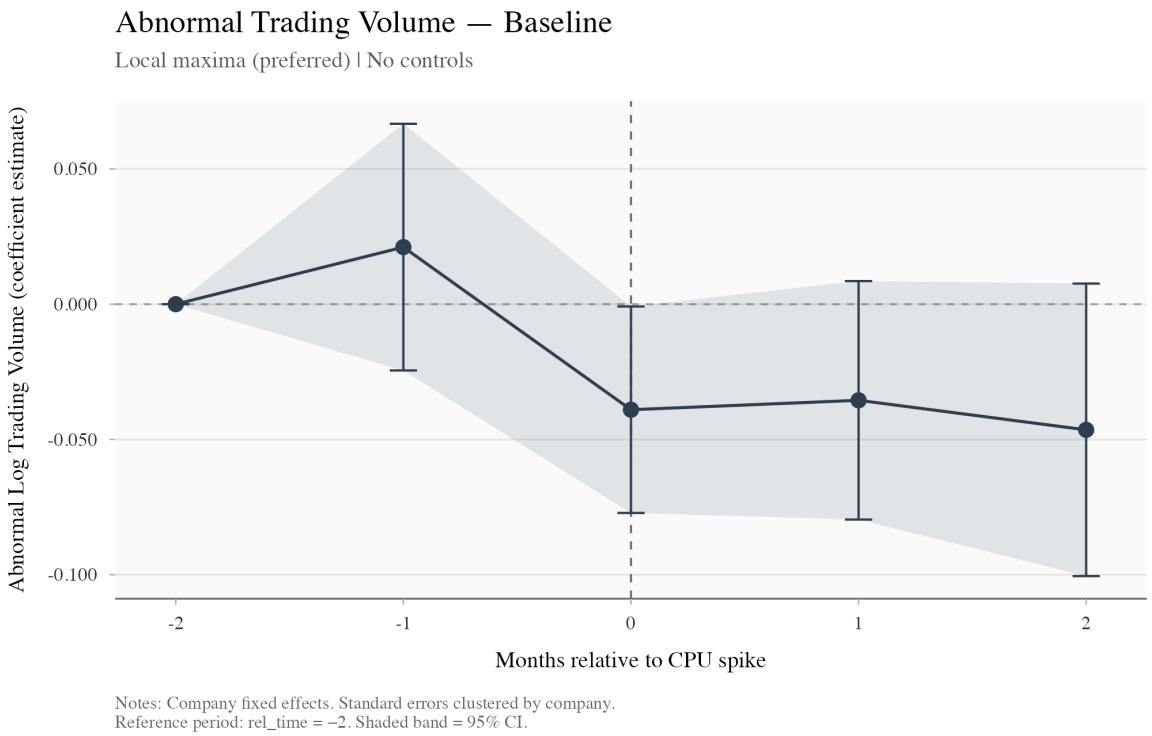


Figure 53: Abnormal trading volume — Local maxima, Baseline.

Table 30: Abnormal trading volume — Local maxima, Raw VIX.

Dependent Variable:	abnormal_vol
Model:	(1)
Variables	
rel_time-1	0.0117
	(0.0233)
rel_time0	-0.0517**
	(0.0222)
rel_time1	-0.0461*
	(0.0247)
rel_time2	-0.0600**
	(0.0293)
Fixed-effects	
company_name	Yes
Fit statistics	
Observations	1,050
R^2	0.08254
Within R^2	0.01933

Clustered (company_name) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

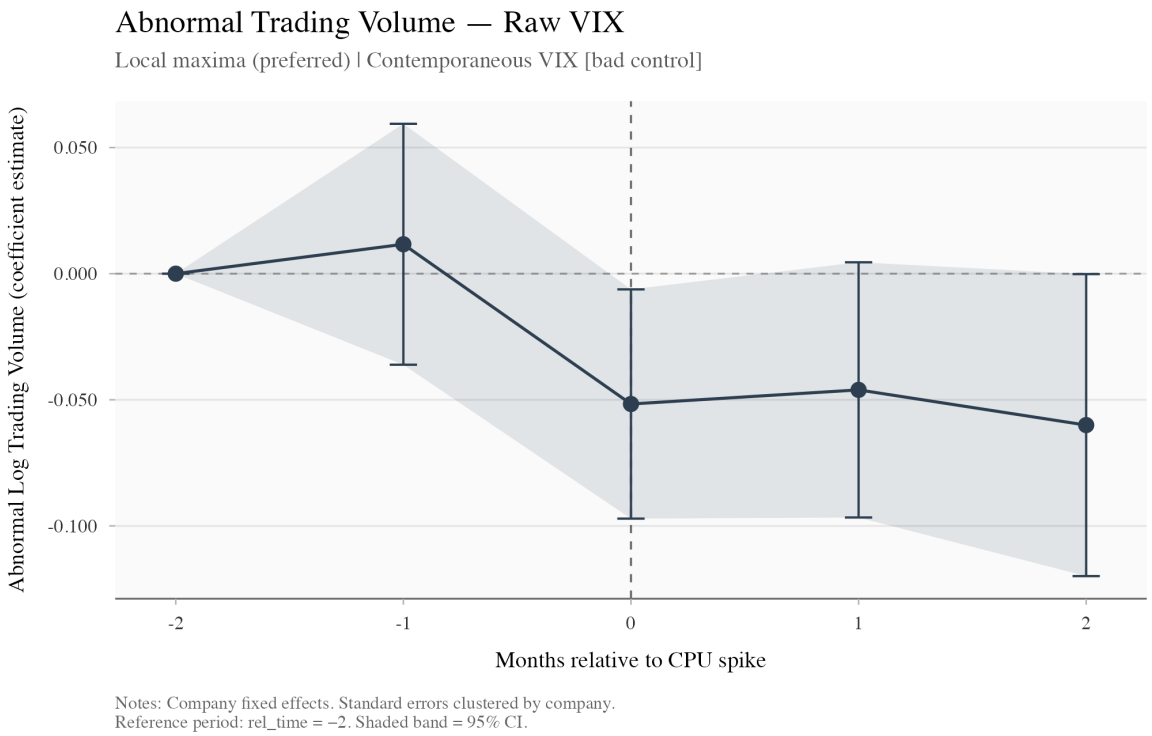


Figure 54: Abnormal trading volume — Local maxima, Raw VIX.

Table 31: Abnormal trading volume — Local maxima, VIX$(t-3)$.

Dependent Variable:	abnormal_vol
Model:	(1)
Variables	
rel_time-1	0.0242
	(0.0224)
rel_time0	-0.0314*
	(0.0175)
rel_time1	-0.0301
	(0.0196)
rel_time2	-0.0475*
	(0.0266)
Fixed-effects	
company_name	Yes
Fit statistics	
Observations	1,050
R^2	0.08102
Within R^2	0.01770

Clustered (company_name) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

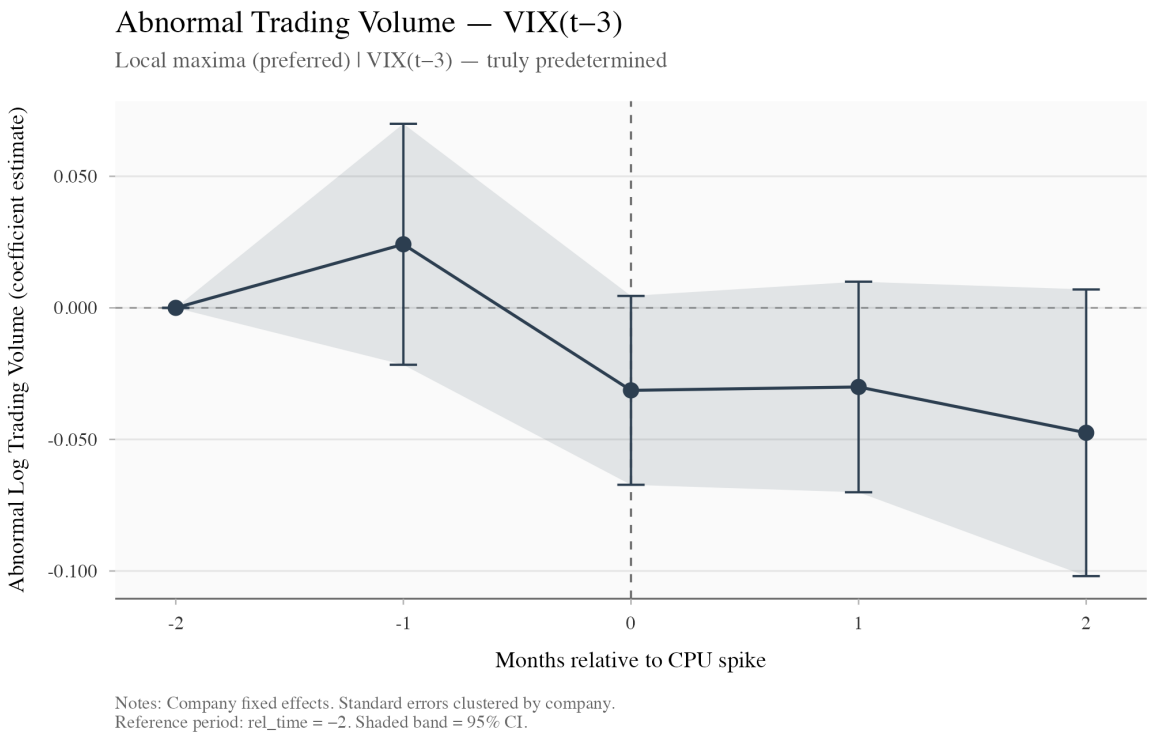


Figure 55: Abnormal trading volume — Local maxima, VIX$(t-3)$.

9.8 First-Difference Log Trading Volume

First-difference log volume is defined as $\Delta \log V_{i,t} = \log V_{i,t} - \log V_{i,t-1}$, where $V_{i,t}$ is mean daily volume in month t.

9.8.1 Rolling Threshold Spike Detection

Table 32: First-difference log volume — Rolling threshold: all three specifications. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Dependent Variable:	fd_log_volume		
	Baseline	Raw VIX	VIX(t-3)
Model:	(1)	(2)	(3)
Variables			
rel_time-1	0.0757	0.0800	0.0750
	(0.0501)	(0.0502)	(0.0502)
rel_time0	0.0166	0.0221	0.0153
	(0.0401)	(0.0395)	(0.0404)
rel_time1	0.0851***	0.0897***	0.0827***
	(0.0290)	(0.0283)	(0.0297)
rel_time2	0.0053	0.0112	0.0059
	(0.0481)	(0.0482)	(0.0481)
Fixed-effects			
company_name	Yes	Yes	Yes
Fit statistics			
Observations	750	750	750
R^2	0.03786	0.03838	0.03816
Within R^2	0.02310	0.02363	0.02341

Clustered (company_name) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 33: First-difference log volume — Rolling threshold, Baseline.

Dependent Variable:	fd_log_volume
Model:	(1)
Variables	
rel_time-1	0.0757
	(0.0501)
rel_time0	0.0166
	(0.0401)
rel_time1	0.0851***
	(0.0290)
rel_time2	0.0053
	(0.0481)
Fixed-effects	
company_name	Yes
Fit statistics	
Observations	750
R^2	0.03786
Within R^2	0.02310

Clustered (company_name) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

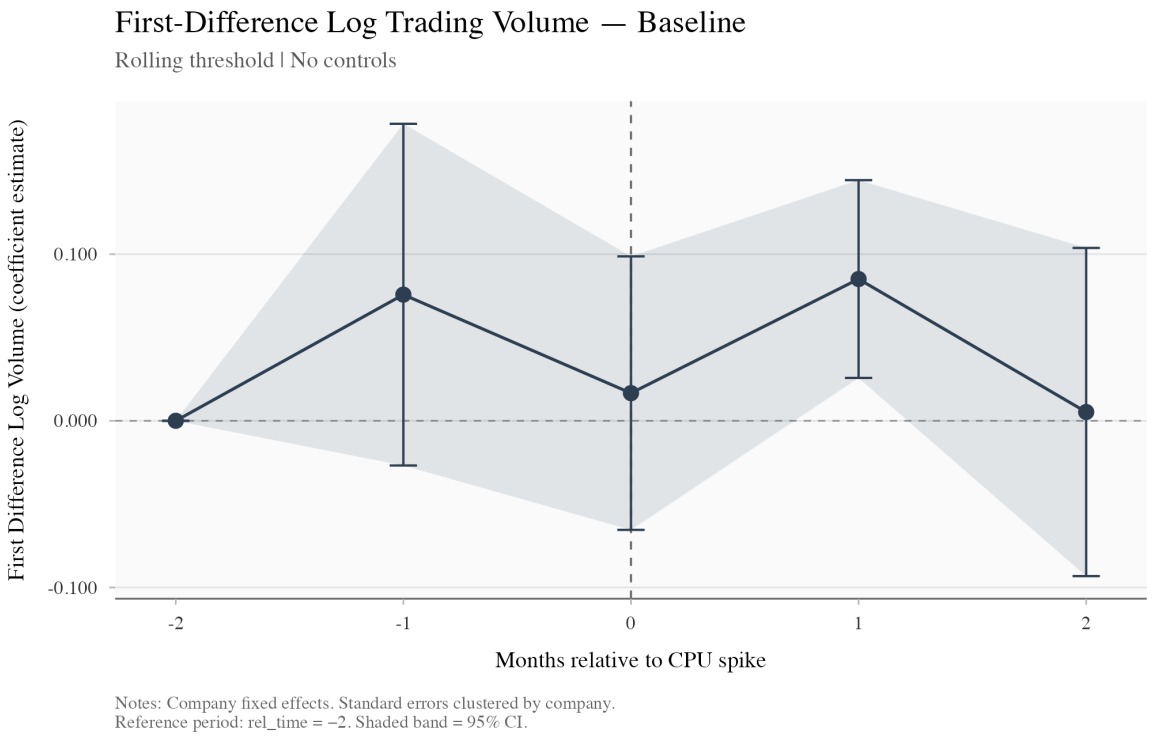


Figure 56: First-difference log volume — Rolling threshold, Baseline.

Table 34: First-difference log volume — Rolling threshold, Raw VIX.

Dependent Variable:	fd_log_volume
Model:	(1)
Variables	
rel_time-1	0.0800
	(0.0502)
rel_time0	0.0221
	(0.0395)
rel_time1	0.0897***
	(0.0283)
rel_time2	0.0112
	(0.0482)
Fixed-effects	
company_name	Yes
Fit statistics	
Observations	750
R^2	0.03838
Within R^2	0.02363

Clustered (company_name) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

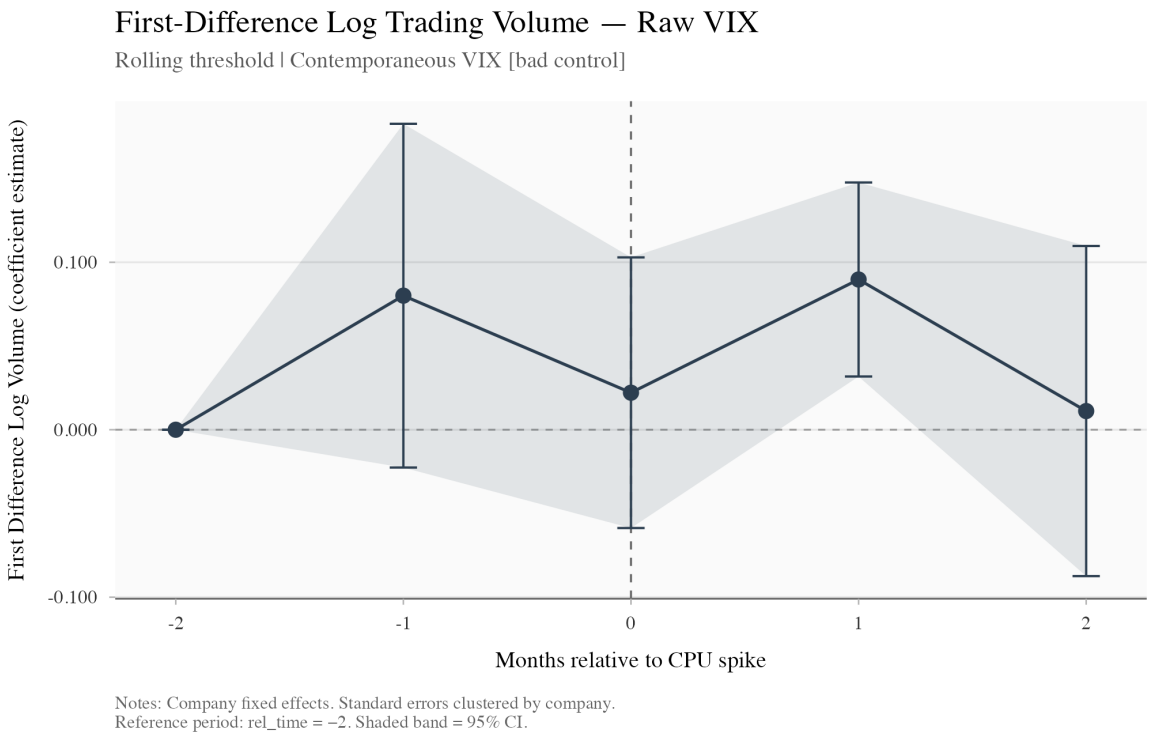


Figure 57: First-difference log volume — Rolling threshold, Raw VIX.

Table 35: First-difference log volume — Rolling threshold, VIX$(t-3)$.

Dependent Variable:	fd_log_volume
Model:	(1)
Variables	
rel_time-1	0.0750
	(0.0502)
rel_time0	0.0153
	(0.0404)
rel_time1	0.0827***
	(0.0297)
rel_time2	0.0059
	(0.0481)
Fixed-effects	
company_name	Yes
Fit statistics	
Observations	750
R^2	0.03816
Within R^2	0.02341

Clustered (company_name) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

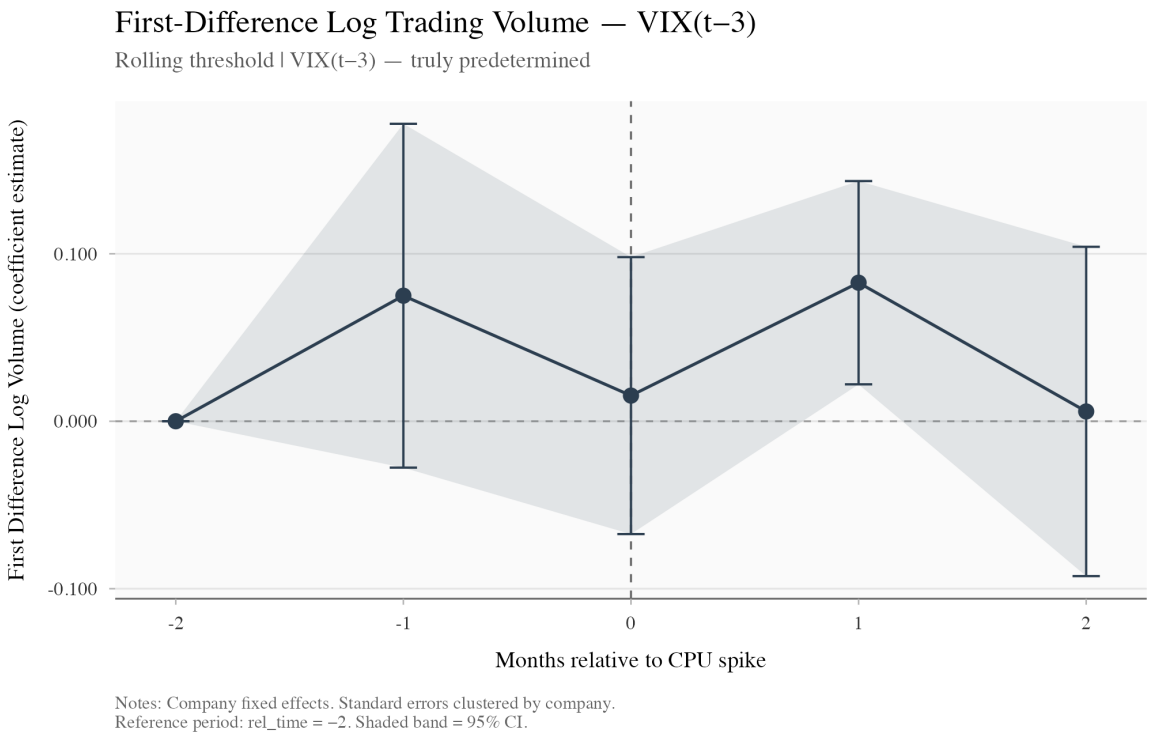


Figure 58: First-difference log volume — Rolling threshold, VIX$(t-3)$.

9.8.2 *Local Maxima Spike Detection*

Table 36: First-difference log volume — Local maxima: all three specifications. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Dependent Variable:	fd_log_volume		
	Baseline	Raw VIX	VIX(t-3)
Model:	(1)	(2)	(3)
Variables			
rel_time-1	0.1033***	0.1020***	0.1030***
	(0.0353)	(0.0347)	(0.0358)
rel_time0	0.0225	0.0207	0.0216
	(0.0250)	(0.0247)	(0.0258)
rel_time1	0.0809***	0.0794***	0.0803***
	(0.0247)	(0.0248)	(0.0245)
rel_time2	0.0674**	0.0654**	0.0675**
	(0.0309)	(0.0296)	(0.0307)
Fixed-effects			
company_name	Yes	Yes	Yes
Fit statistics			
Observations	1,050	1,050	1,050
R^2	0.03682	0.03691	0.03686
Within R^2	0.02557	0.02566	0.02561

Clustered (company_name) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 37: First-difference log volume — Local maxima, Baseline.

Dependent Variable:	fd_log_volume
Model:	(1)
Variables	
rel_time-1	0.1033***
	(0.0353)
rel_time0	0.0225
	(0.0250)
rel_time1	0.0809***
	(0.0247)
rel_time2	0.0674**
	(0.0309)
Fixed-effects	
company_name	Yes
Fit statistics	
Observations	1,050
R^2	0.03682
Within R^2	0.02557

Clustered (company_name) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

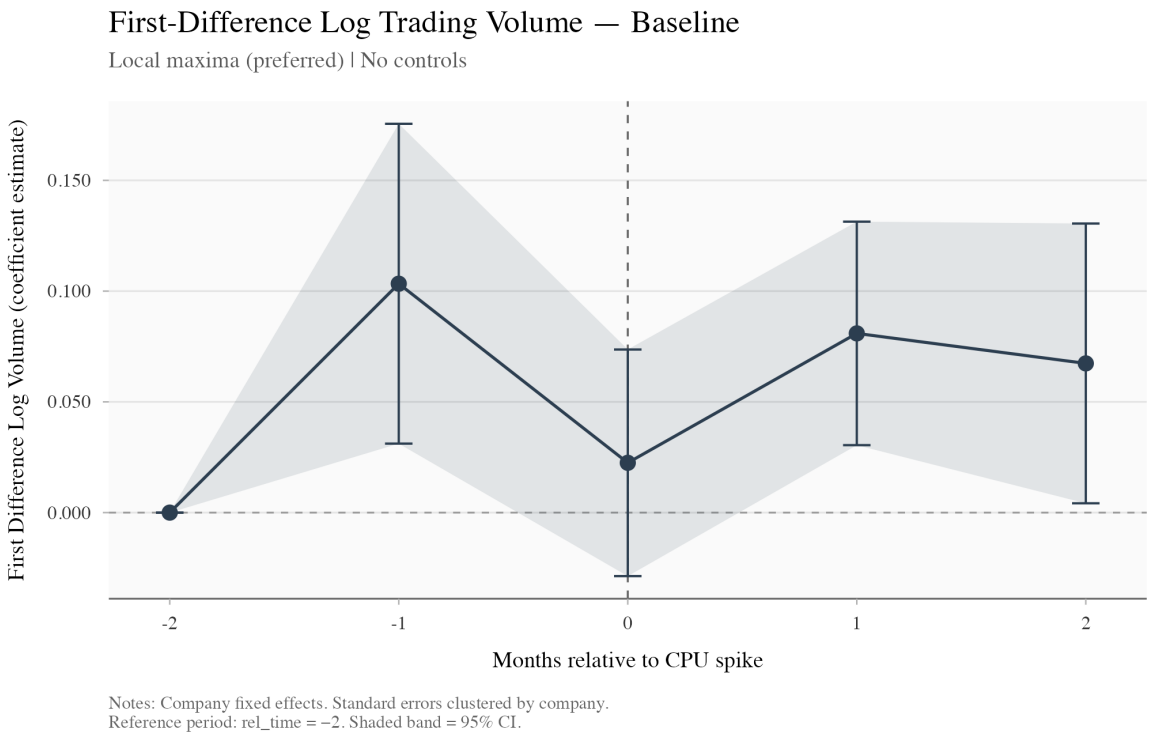


Figure 59: First-difference log volume — Local maxima, Baseline.

Table 38: First-difference log volume — Local maxima, Raw VIX.

Dependent Variable:	fd_log_volume
Model:	(1)
Variables	
rel_time-1	0.1020***
	(0.0347)
rel_time0	0.0207
	(0.0247)
rel_time1	0.0794***
	(0.0248)
rel_time2	0.0654**
	(0.0296)
Fixed-effects	
company_name	Yes
Fit statistics	
Observations	1,050
R^2	0.03691
Within R^2	0.02566

Clustered (company_name) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

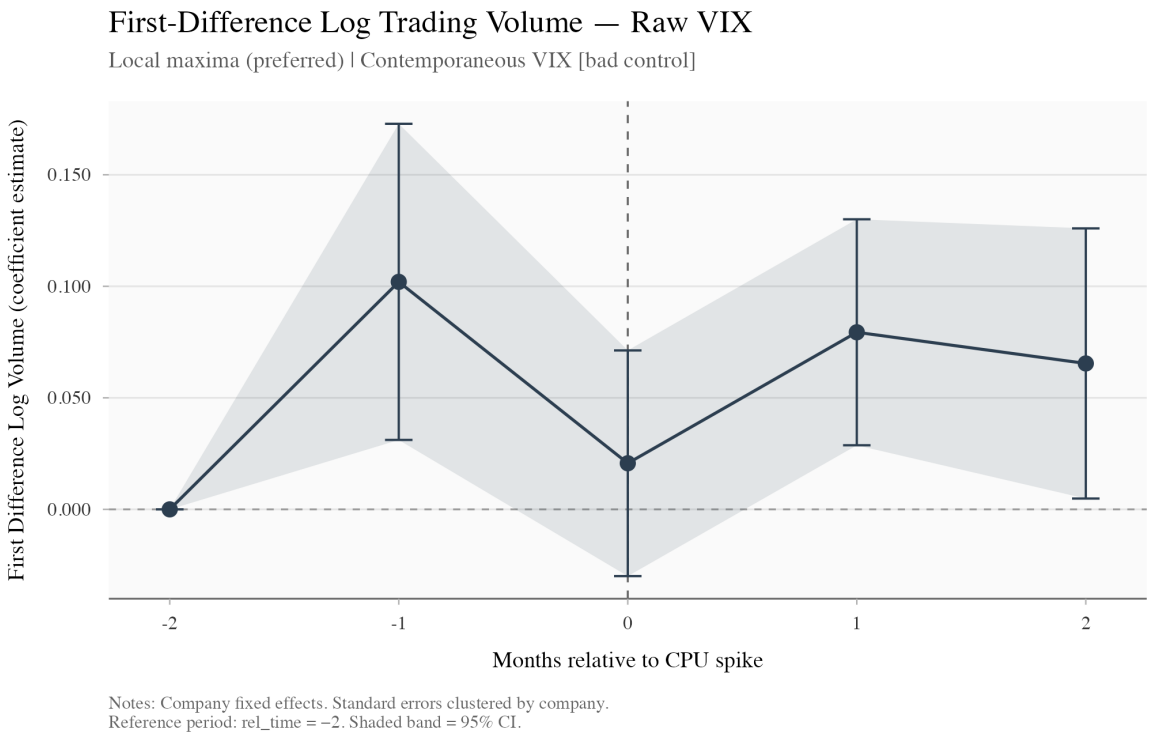


Figure 60: First-difference log volume — Local maxima, Raw VIX.

Table 39: First-difference log volume — Local maxima, VIX$(t-3)$.

Dependent Variable:	fd_log_volume
Model:	(1)
Variables	
rel_time-1	0.1030***
	(0.0358)
rel_time0	0.0216
	(0.0258)
rel_time1	0.0803***
	(0.0245)
rel_time2	0.0675**
	(0.0307)
Fixed-effects	
company_name	Yes
Fit statistics	
Observations	1,050
R^2	0.03686
Within R^2	0.02561

Clustered (company_name) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

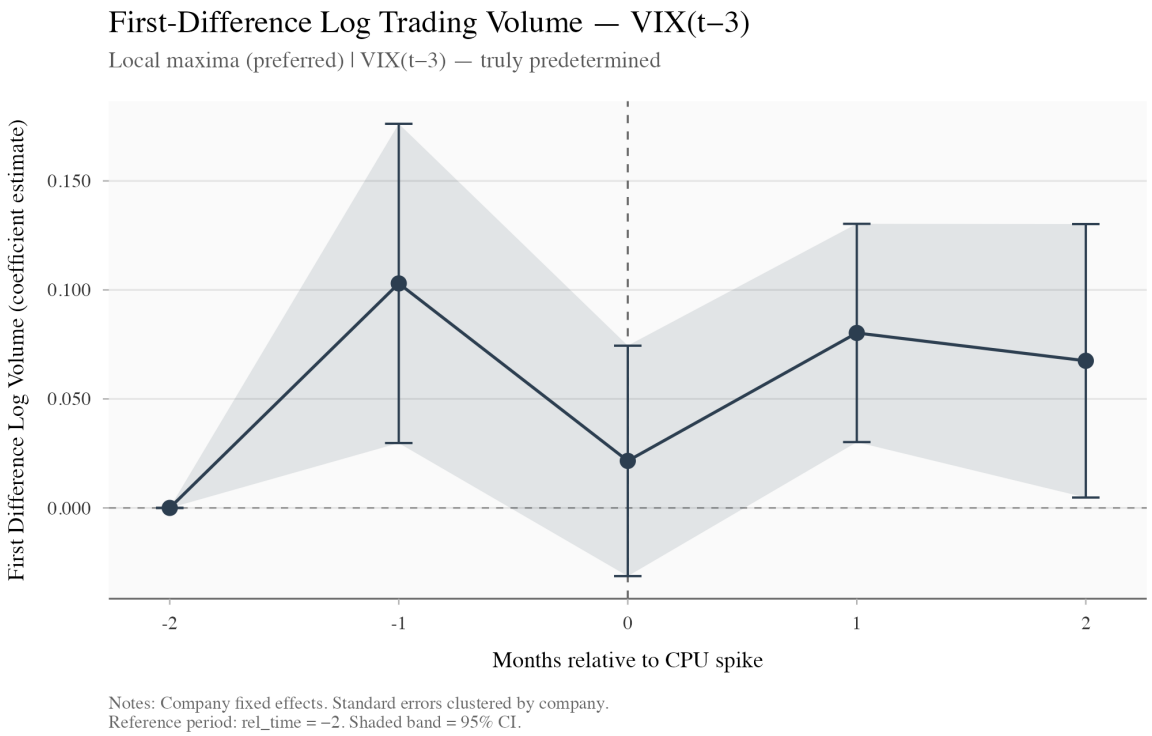


Figure 61: First-difference log volume — Local maxima, VIX$(t-3)$.

References

Angrist, J. D. and Pischke, J.-S. (2009). *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton University Press, Princeton, NJ.

Badertscher, B. A., Burks, J. J., and Easton, P. D. (2014). Financial reporting quality of u.s. private and public firms. *The Accounting Review*, 89(3):811–838.

Baker, A. C., Larcker, D. F., and Wang, C. C. Y. (2022). How much should we trust staggered difference-in-differences estimates? *Journal of Financial Economics*, 144(2):370–395.

Basaglia, P., Berestycki, C., Carattini, S., Dechezleprêtre, A., and Kruse, T. (2025). Climate policy uncertainty and firms' and investors' behavior. CESifo Working Paper 11782, CESifo.

Bernanke, B. S. (1983). Irreversibility, uncertainty, and cyclical investment. *The quarterly journal of economics*, 98(1):85–106.

Bezgin, M. S. and Gungor, S. (2025). Does climate policy uncertainty move with stock markets? evidence from advanced economies with the best climate change performance. *Financial Internet Quarterly*, 21(3):62–76.

Bloom, N. (2009). The impact of uncertainty shocks. *econometrica*, 77(3):623–685.

Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of econometrics*, 31(3):307–327.

Bonato, M., Cepni, O., Gupta, R., and Pierdzioch, C. (2022). Climate risks and state-level stock-market realized volatility. Technical Report 2022-46, University of Pretoria, Department of Economics.

Chen, Z., Zhang, L., and Weng, C. (2023). Does climate policy uncertainty affect chinese stock market volatility? *International Review of Economics & Finance*, 84:369–381.

Copeland, T. E. (1976). A model of asset trading under the assumption of sequential information arrival. *The Journal of Finance*, 31(4):1149–1168.

De Chaisemartin, C. and d'Haultfoeuille, X. (2020). Two-way fixed effects estimators with heterogeneous treatment effects. *American economic review*, 110(9):2964–2996.

Dixit, A. K. and Pindyck, R. S. (1994). *Investment under uncertainty*. Princeton university press.

Engle, R. F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of united kingdom inflation. *Econometrica: Journal of the econometric society*, pages 987–1007.

Fama, E. F. (1970). Efficient capital markets: A review of theory and empirical work. *The Journal of Finance*, 25(2):383–417.

Gavriilidis, K. (2021). Measuring climate policy uncertainty. *Available at SSRN 3847388*.

Gavriilidis, K., Känzig, D. R., Raghavan, R., and Stock, J. H. (2026). The macroeconomic effects of climate policy uncertainty. NBER Working Paper 34762, National Bureau of Economic Research, Cambridge, MA.

Gavriilidis, K., Känzig, D. R., and Stock, J. (2023). The macroeconomic effects of climate policy uncertainty. *Unpublished working paper*.

Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of econometrics*, 225(2):254–277.

Huang, W. (2026). The impact of climate policy uncertainty on corporate investment. *International Review of Financial Analysis*, 93:105118.

IPCC (2022). Climate change 2022: Impacts, adaptation and vulnerability.

Karpoff, J. M. (1987). The relation between price changes and trading volume: A survey. *Journal of Financial and Quantitative Analysis*, 22(1):109–126.

Lasisi, L., Omoke, P. C., and Salisu, A. A. (2024). Climate policy uncertainty and stock market volatility. *Asian Economics Letters*, 5(2).

Li, M., Han, X., and Li, Y. (2024). The impact of climate policy uncertainty on stock price synchronicity: Evidence from china. *Finance Research Letters*, 69:106166.

Lv, X. (2023). Climate policy uncertainty and stock market volatility. *Finance Research Letters*, 53:103605.

Moncomble, F. (2024). App_extractor. JavaScript software. First version released in 2023. Available from: https://fmoncomble.github.io/APP_extractor/.

Peng, Z. and Yang, W. (2024). Short selling and climate policy uncertainty. SSRN Working Paper, posted 13 December 2024.

Pennington, J., Socher, R., and Manning, C. D. (2014). Glove: Global vectors for word representation. In *Proceedings of the 2014 conference on empirical methods in natural language processing (EMNLP)*, pages 1532–1543.

Ren, X., Yan, H., and Gozgor, G. (2023). Climate policy uncertainty and idiosyncratic volatility: Evidence from the non-financial listed chinese firms. *Journal of Climate Finance*, 5:100026.

Shi, Y., Ho, K.-Y., and Liu, W.-M. (2016). Public information arrival and stock return volatility: Evidence from news sentiment and markov regime-switching approach. *International Review of Economics & Finance*, 42:291–312.

Sun, H., Xu, X., and Liu, Z. (2025). Climate policy uncertainty and executives' opportunistic stock selling. *Finance Research Letters*, 83:107706.

Treepongkaruna, S., Chan, K. F., and Malik, I. (2023a). Climate policy uncertainty and the cross-section of stock returns. *Finance Research Letters*, 55:103837.

Treepongkaruna, S., Chan, K. F., and Malik, I. (2023b). Climate policy uncertainty and the cross-section of stock returns. *Finance Research Letters*, 55:103837.

Vo, H., Phan, A., and Trinh, Q. D. (2025). Climate policy uncertainty: A catalyst for stock price crash? *Journal of Environmental Management*, 392:126629.

Wanidwaranan, P., Wongkantarakorn, J., and Padungsaksawasdi, C. (2025). Climate policy uncertainty and trading behavior: Evidence from aggregate herd behavior. *Energy Economics*, 149:108760.

Wing, C., Freedman, S. M., and Hollingsworth, A. (2024). Stacked difference-in-differences. *National Bureau of Economic Research (Working paper)*.

Yan, C., Zhou, Y., and Tan, Q. (2025). Climate policy uncertainty and stock market pricing efficiency: Evidence from china. *Economic Systems*, page 101358. In press, corrected proof.

Yang, X., Jiang, L., and Yang, Z. (2025). The impact of climate policy uncertainty on stock volatility: Evidence from china's eight high-carbon emission industries. *SSRN Electronic Journal*. Preprint, posted March 6, 2025.